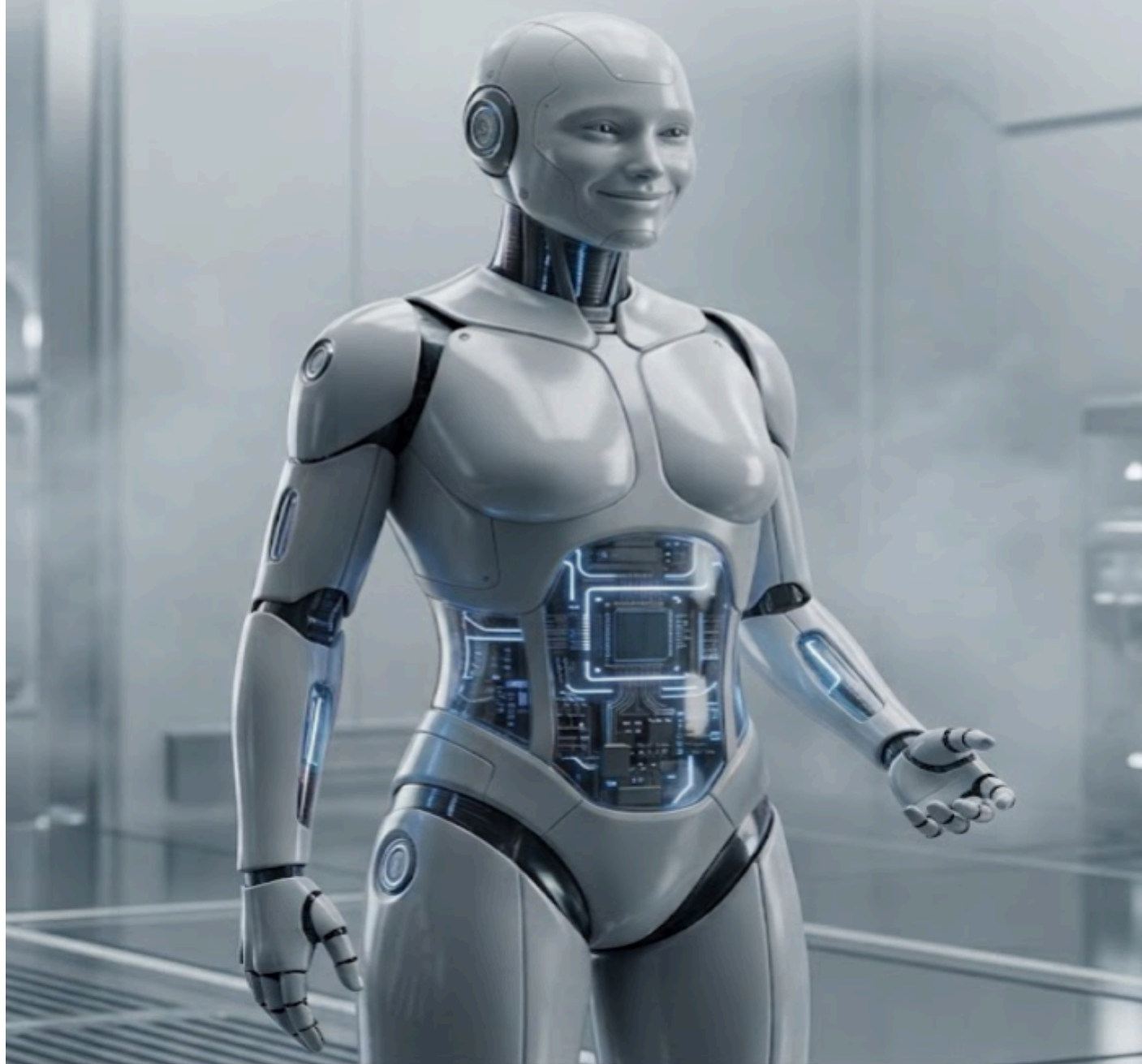


EXECUTABLE NATURAL LANGUAGE

The Missing Grammar of Thought
and SOP Lang English CNL



A RESEARCH PROGRAMME FOR AUDITABLE KNOWLEDGE
REPRESENTATION AND EXECUTABLE SCIENCE

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Axiologic Research

Abstract

Large language models make natural language a practical interface to computation, but scientific and technical prose remains an unstable execution surface. A model can alter quantifier scope, resolve an ambiguity silently, merge distinct identities, introduce a bridge premise, or convert reported evidence into accepted fact before an exact solver is called. The engineering problem addressed here is the construction of an auditable semantic layer between unrestricted language and computation, with explicit provenance, alternatives, types, contexts, and execution contracts.

The research programme combines four sources of evidence. English grammar identifies recurrent structures that a compiler must normalise. Cross-linguistic typology identifies distinctions that English often leaves optional, including evidentiality, clusivity, participant continuity, temporal remoteness, possession type, perspective, and spatial frame. Philosophy and systems research contribute distinctions such as process identity, collective agency, causal depth, reversibility, scale, boundaries, and distributed responsibility. Formal semantics, knowledge representation, programming-language methods, and controlled natural languages provide the mechanisms needed to represent and test these distinctions.

SLEnglish, short for SOP Lang English, is the common representation proposed for these experiments. It is a regular, typed, path-addressed language designed to receive candidate translations from English while remaining readable enough for human inspection and strict enough for deterministic parsing, validation, comparison, and execution. A dedicated chapter defines its syntax, operator contracts, parameters, namespaces, contexts, rules, and queries after the linguistic and conceptual requirements have been established.

The target application is executable science: research artefacts in which claims, evidence, assumptions, data, code, models, workflows, reviews, and corrections have explicit semantic dependencies. Such artefacts can support bounded queries, contradiction checks, dependency analysis, selective rebuilding, source-grounded explanation, and comparison of alternative interpretations. The research question is whether a small translation step from natural language into SLEnglish makes these operations more reliable and easier to audit than direct LLM reasoning or ad hoc formalisation.

Introduction: Purpose and Research Method

A scientific paper can be computationally reproducible while its semantic dependencies remain implicit. Code may run again even when the relation between a numerical result and the sentence it supports is unavailable to a machine. References may be complete while the path by which a premise entered an argument remains unrepresented. A dataset correction may therefore fail to invalidate an affected conclusion because the dependency exists only in the reader's interpretation. Executable science requires a representation in which these relations become explicit enough to inspect and recompute.

The first technical requirement is a visible formalisation boundary. A source statement, its interpretation, the evidence used for grounding, the selected semantic profile, the execution record, and the rendered result must remain distinct artefacts. Proof of a formal expression establishes a consequence of that expression; fidelity to the source is established by the translation record, source anchors, alternative interpretations, validation tests, and review. This distinction prevents exact reasoning from certifying an inaccurate formalisation.

The second requirement is conceptual coverage. English provides an efficient surface language, but many distinctions relevant to scientific reasoning are optional or compressed. The world's languages supply a comparative inventory of alternatives: source of information, inclusive and exclusive participation, participant continuity across clauses, temporal remoteness, aspect, possession type, classifier systems, logophoric perspective, social deixis, and absolute spatial reference. These observations are used here as candidate semantic variables. The analysis of English and other languages has one operational output: the operator and parameter tables in the appendices.

SLEngish provides the experimental target representation. Ordinary English constructions, cross-linguistic grammatical distinctions, scientific operations, epistemic states, social relations, and systems concepts can each be represented as typed executable operators when they affect denotation, inference, provenance, time, authority, or computation. The common outer grammar keeps translation and tooling simple; the operator libraries carry the semantic detail. This structure allows the language to grow by adding reviewed operator contracts rather than by changing the parser.

The standardisation target is a common executable envelope for multiple ontologies and interpretation profiles. Operator libraries, parameter registries, namespace paths, type declarations, translation policies, and backend adapters are explicit artefacts that can be versioned and compared. SLEngish can be interpreted by rule engines, graph systems, solvers, temporal models, generated code, specialised domain tools, or semantic circuits. The language remains stable at the source level while experiments determine which interpretation technologies are most effective for each class of task.

Contents and Argument Map

Section	Purpose
Chapter 1 — The Problem of Executable Natural Language	Defines the formalisation gap, the trust boundary, and the relation between source text, interpretation, execution, and explanation.
Chapter 2 — English as a Semantic Substrate	Synthesises the principal structures of English and identifies where its efficient surface grammar leaves semantic work to context and pragmatics.
Chapter 3 — Language Diversity and the Missing Grammar of Thought	Reviews research on linguistic relativity and examines grammatical distinctions that can inform an executable extension of English.
Chapter 4 — Identity, Collective Agency, and Systemic Concepts	Develops qualified identity, group agency, causal depth, responsibility, boundaries, emergence, scale, and related conceptual operators.
Chapter 5 — SLEngish: Design, Syntax, and Semantic Libraries	Introduces the one-shape SSA syntax, path-based operator files, typed parameters, contexts, rules, queries, and open verb libraries.
Chapter 6 — Interpretation and Execution without a Mandatory Backend	Defines the LLM translation contract, deterministic validation, backend-neutral intermediate objects, and optional interpreter families.
Chapter 7 — Executable Science as a Semantic Build System	Connects claims, evidence, code, data, review, provenance, invalidation, selective rebuilding, and scientific governance.
Chapter 8 — SLEngish in Relation to Ontologies and Formal Knowledge Representation	Compares the proposal with RDF, OWL, SHACL, SPARQL, PROV, Common Logic, CNLs, graph systems, and logic programming.
Chapter 9 — Research Programme, Evaluation, and Revision	Specifies questions, baselines, metrics, failure conditions, reversible design decisions, and bounded experiments.
Appendices	Provide the grammar, canonicalisation rules, parameter registry, verb libraries, linguistic mappings, worked programs, evaluation matrix, reading map, and references.

CHAPTER 1

The Problem of Executable Natural Language

The Formalisation Gap and the Trust Boundary

The word execution should be used with restraint. A program is executable because an operational semantics determines how its state changes. A database query is executable because relational operators determine which records are selected, joined, grouped, or excluded. A planning instance is executable because actions have preconditions and effects, and a planner searches a formally delimited state space. A logical theory is executable in a broader sense because a defined procedure can derive consequences, find models, report inconsistency, or return a certificate. In each case, the result is produced by an artefact whose behaviour is independent of the persuasiveness of the surrounding explanation.

An LLM response has a different status. The model maps a context to a distribution over continuations. Tool use can improve accuracy, but the model still determines which tool to call, what problem to give it, and how to interpret the output. A generated Python script may calculate exactly while modelling the wrong variable. An SMT solver may return satisfiable because an exception was omitted during translation. A theorem prover may establish a theorem whose quantifiers differ from those in the source. Execution moves an error from arithmetic or search into formalisation; it does not abolish error.

Executable natural language must therefore be defined as a relation among artefacts rather than as an intrinsic property of prose. The minimum chain contains a source revision, a task contract, one or more interpretations, grounding evidence, a selected formal semantics, an execution record, and a licensed realisation. The source revision establishes what was present. The task contract states which question is being asked and what degree of assurance is required. Interpretations expose the semantic choices introduced by the compiler. Grounding connects symbols to spans, records, sensors, definitions, or declared assumptions. The formal profile states what consequence means. The execution record carries proof, witness, counterexample, or bounded result. The realisation is subordinate to that record.

This relation is conditional by design. A result may be mechanically correct relative to a particular ontology, identity map, set of admissible readings, closed-world boundary, and solver semantics. It need not be true without qualification in the world. The architecture should make the conditions visible rather than flatten them into the word verified. Structural validity, semantic grounding, execution correctness, empirical adequacy, and institutional authorisation are distinct layers. A system can satisfy one and fail another.

The decisive practical metric is false assurance. False assurance occurs when the system presents a determinate, apparently verified conclusion even though the source, grounding, interpretation set, or coverage does not justify it. In high-assurance domains, a precise unknown is often more valuable than a fluent answer. A system that stops because two people may have been merged is better than one that silently chooses. A system that says no absence conclusion is licensed because the corpus is incomplete is better than one that treats failure to retrieve as proof of non-existence. Appropriate abstention is part of execution semantics, not a failure of usability.

This perspective preserves a useful ambition while rejecting totalisation. Scientific claims can be linked to evidence and executable analyses. Requirements can be connected to tests and state machines. Policies can be compiled into decision tables under an explicit jurisdiction and date. Protocols can be checked for temporal contradictions. Long documents can be queried against reusable theories without asking a model to improvise the same reasoning each time. What becomes executable is a declared projection whose omissions and assumptions are inspectable.

SLEnglish is proposed as the canonical pivot for that projection. At this point it is enough to state its purpose without presenting its syntax. It must give every semantic node a stable identity, make roles explicit, keep proposition construction separate from assertion or report, preserve alternative interpretations, and connect directly to a dependency graph that declared interpreters can validate and evaluate. It must be readable enough for review and regular enough for deterministic tooling. Because an LLM performs most of the initial translation, naturalness is less important than correction cost, semantic visibility, and stable compilation.

The trust boundary follows from these requirements. Generative models may propose forms, alternatives, ontology mappings, and explanations. They may not silently establish truth, provenance, or authority. Parsers may validate syntax but may not invent missing semantics. Type checkers may ensure that roles are legal but

may not certify empirical truth. Solvers may determine consequences under a profile but may not reinterpret the source. Renderers may improve readability but may not strengthen modality, erase uncertainty, or add claims. Operational effects require explicit capabilities and policy. Human judgement remains necessary where meanings, values, or side effects are contested.

The formalisation gap is the distance between a source expression and the formal object executed in its name. It is the most persistent weakness in systems that combine language models with external tools. Benchmarks often report that a solver improved the final answer, but answer accuracy does not reveal whether the symbolic problem preserved the source. In constrained arithmetic tasks, short programs and immediate execution feedback make many errors visible. In policy, science, law, security, and long-document reasoning, a formal artefact can be internally impeccable while quietly omitting the condition that mattered.

Grounding is not one operation. Lexical grounding maps words or spans to candidate symbols. Referential grounding determines which mentions denote the same entity. Ontological grounding decides which categories and relations are available in the selected model. Temporal grounding locates events and states in intervals and calendars. Evidential grounding links propositions to observations, reports, datasets, or assumptions. Normative grounding identifies jurisdiction, authority, mandate, and effective date. Each layer can fail independently, and a serious compiler records the failure at the layer where it occurs.

Identity deserves particular emphasis. A document may mention ‘the board,’ ‘the organisation,’ ‘the model,’ or ‘the sample’ across many pages. Treating repeated labels as identical can create false continuity; treating every occurrence as distinct can destroy legitimate inference. For clouds, rivers, organisms, software versions, institutions, and datasets, identity itself may be qualified by material, processual, genealogical, informational, functional, or legal continuity. The executable representation should not conceal this work behind one generic equality predicate.

Coverage is equally important. Open-world sources can omit facts that are true. A collection of retrieved emails is not automatically complete evidence that no notification was sent. A literature search is not proof that no prior art exists. A database table may be closed for a named period and schema, while a public web corpus is not. Negative conclusions require a coverage token that states the domain searched, source revision, retrieval method, exclusions, and completeness policy. Without that token, absence remains unknown rather than false.

Ambiguity should be materialised as alternatives when it can affect consequence. A pronoun may refer to two actors; a requirement may admit two scopes; a causal sentence may describe intention or only correlation. A model can rank readings, but confidence is not proof. If a downstream verdict is invariant across all admissible readings, execution may proceed with a must conclusion. If it holds under at least one reading, the result is may. If it fails under every reading, it is cannot. If the interpretation set is incomplete or the abstraction too coarse, the honest result remains unknown.

This creates a productive role for abstract interpretation. The collecting semantics is the set of admissible interpretations of the source under the current grounding policy. An abstract domain can summarise that set through properties such as possible identities, temporal intervals, modalities, or evidence states. A sound abstract computation can show that a conclusion holds across the represented set or remains possible within it. Its guarantee is conditional on coverage of interpretations. The abstraction does not prove that the compiler imagined every meaning a competent reader might endorse.

Validation must therefore target the translation relation itself. Bidirectional checking renders a formal object back into controlled prose and asks whether the material commitments match the source. Contrastive tests alter one linguistic feature and verify that only the corresponding semantic field changes. Paraphrase tests compile meaning-preserving variants and compare canonical graphs. Extensional tests search for cases on which two candidate formalisations differ. Counterexample mapping links every solver witness to source spans and asks whether the scenario is linguistically and scientifically admissible. Human correction time is a crucial metric because a representation that is formally rich but impossible to review is not a trustworthy boundary layer.

The system must also treat source text as untrusted data. Documents can contain prompt injections, misleading headings, code-like fragments, hostile Unicode, fabricated ontology declarations, or quoted instructions. A compiler should preserve exact source structure while operating under fixed authority. Formal artefacts should be data, not unrestricted code. Extensions that perform algorithms or tool calls require reviewed contracts, locked digests, typed inputs and outputs, budgets, determinism declarations, and explicit capabilities. Security outcomes must be evaluated separately from semantic accuracy.

A defensible guarantee can now be stated. Given a named source revision, task, ontology, admissible interpretation set, grounding policy, formal profile, implementation version, and execution bounds, the reported canonical result follows and can be replayed. This is weaker than claiming that a machine fully understands language. It is stronger than asking a model to reason and trusting the fluency of its answer. The guarantee is valuable precisely because its frontier is visible.

From Formal Semantics to Auditable Compilation

The research programme has a long intellectual lineage. Controlled Natural Languages show that readable fragments can map deterministically to formal representations when vocabulary, grammar, and interpretation rules are restricted. Attempto Controlled English demonstrated discourse representation, querying, and theorem-proving integration. Common Logic Controlled English offered an English-like route to first-order representation. PENG and related systems showed that processable English can compile into logic-programming and Answer Set Programming profiles. SBVR made structured vocabulary and business rules accessible to domain experts. These systems establish feasibility while exposing the trade-off among precision, expressiveness, naturalness, and simplicity identified in Kuhn’s survey of CNLs.

Formal semantics contributes a different discipline. Model-theoretic semantics, Discourse Representation Theory, neo-Davidsonian event structures, natural logic, and typed semantic representations explain how quantification, anaphora, events, roles, modality, and discourse context can be represented. They also warn that a syntactic parse is not a grounded world model. A discourse referent can be introduced formally while its identity remains empirically uncertain. A lexical entailment can be valid in one ontology and invalid in another. The executable layer needs linguistic structure, domain definitions, evidence, and task-specific commitments.

Semantic parsing is the closest engineering predecessor. It converts language into executable database queries, logical forms, regular expressions, robotic commands, or programs. Its successes show that external execution improves systematicity. Its failures show how easily models exploit spurious correlations, memorise schemas, or generate programs that happen to answer a dataset without representing the question. The LLM era improves open-domain candidate generation but does not repeal the requirement for denotation, type, scope, and compositional tests.

Knowledge representation contributes mature families of semantics. Description logics and OWL support ontology reasoning under carefully chosen decidability restrictions. RDF and property graphs support exchange and graph navigation. Datalog supports recursive relational consequence. Answer Set Programming handles defaults and non-monotonicity. Common Logic provides a first-order interchange framework. Constraint solvers handle arithmetic and combinatorial conditions. Temporal logics and event calculi represent changing worlds. No one formalism dominates because the questions differ.

Programming-languages research adds the tools needed to reason about incomplete and evolving interpretations. Abstract interpretation computes sound approximations over potentially enormous state spaces. Symbolic execution constructs path conditions and witnesses. Model checking explores transition systems against temporal properties. SAT, SMT, and CSP engines decide bounded or theory-rich constraints. Dataflow analysis and fixpoint evaluation support dependency propagation. Proof assistants and proof-producing solvers can reduce the trusted base. These methods should not be used as metaphors; each has a specific semantics, completeness frontier, and certificate story.

Recent LLM research has rediscovered many of these structures under new names. Program-aided reasoning delegates calculations to interpreters. Constraint-based systems generate declarative problems for SAT or SMT. Logic-oriented systems translate text into first-order rules. Temporal translators generate specifications for model checking. Planning systems ask LLMs to construct domains and instances that sound planners solve. LLM-assisted program analysis proposes invariants, abstract transformers, or path summaries that formal checkers accept or reject. The strongest results occur when the target formalism is narrow and an independent component controls admission.

Neuro-symbolic architectures can be injective, in which symbolic information is absorbed into a neural model, or federative, in which neural and symbolic components retain separate identities and exchange structured artefacts. For executable natural language, federation is attractive because it preserves a visible trust boundary, allows replay, and permits multiple engines. It is not automatically superior on every benchmark. The empirical case must be established per task, but the architectural case for separated authority is strong.

The lineage suggests a synthesis. A source-grounded semantic pivot should preserve task-neutral distinctions and alternatives. A portfolio should lower selected fragments into specialised formal engines. Execution substrates should expose dependencies, budgets, evidence, and proof objects. LLMs should propose and explain, not silently own consequence. SLEngish is intended to make this synthesis implementable with a common, inspectable source language.

Formal precision requires conceptual precision. A triple asserting that a company caused a loss can be syntactically exact while leaving unresolved whether company denotes a legal person, a board, a distributed organisation, a management decision, or an emergent result of local incentives. A theorem about the same model can be valid while same conflates identical parameters, common architecture, lineage, functional equivalence, and institutional ownership. Executable knowledge therefore needs explicit distinctions before it needs stronger inference.

The linguistic and philosophical survey is used as a requirements-discovery method for SLEngish. Linguistic diversity contributes evidence source, clusivity, participant continuity, remoteness, aspect, classifier choice, possession type, social deixis, logophoric perspective, and spatial frame. Philosophy contributes identity, causation, infinity, part-whole relations, collectives, and responsibility. Systems analysis contributes reversibility, lock-in, feedback, scale, externality, path dependence, and threshold distance. Each useful distinction is converted into a candidate operator, parameter, type refinement, context field, or validation rule and then tested on concrete reasoning tasks.

SLEngish uses profile-dependent obligatoriness. An epistemic profile can require evidence mode and source depth for empirical claims. An identity profile can require a declared continuity criterion for process entities. An institutional profile can require mandate and decision architecture when a collective is represented as acting. A scientific profile can require units, uncertainty, validity interval, and provenance. The profile therefore determines which omissions are compilation errors for a particular task.

Cross-linguistic operator design is grounded in descriptive grammars, typological surveys, corpora, and specialist review. When a candidate operator draws materially on a living linguistic tradition, its semantics should be checked against the relevant scholarship and speaker expertise. The transferable engineering object is a precisely defined distinction with a testable contract; the source language remains the evidence for that distinction rather than a label for an allegedly universal category.

Semantic distinctions are represented as first-class, extensible verbs. Evidentiality, switch-reference, logophoricity, clusivity, causative voice, process identity, collective agency, measurement, inference, and ordinary event structure can each own an operator contract when their parameters and consequences differ. This keeps the distinction visible in source code, documentation, tests, and execution traces instead of hiding it behind a small set of generic verbs.

In SLEngish, verb denotes an executable semantic operator rather than only a lexical English verb. A familiar action such as transfer, a grammatical construction such as inclusive-we, an epistemic act such as infer, a relation such as part-of, a scientific operation such as measure, and a systems concept such as feedback-positive can all be verbs when they have declared schemas and interpretation contracts. Named parameters provide the microgrammar of each verb while every statement shares the same outer syntax.

This design separates a stable parser from an extensible semantic vocabulary. The parser recognises one statement envelope, references, paths, named parameters, and values. Operator libraries add domain and linguistic semantics through files that declare types, required parameters, optional parameters, constraints, examples, tests, and compatible interpreters. A compiler can therefore validate the generic structure of a statement even when a specialised operator must be routed to another interpreter.

The research sequence follows directly from this architecture. Chapters on English and cross-linguistic variation identify candidate distinctions; the conceptual chapters refine identity, collective agency, causation, uncertainty, and systems concepts; the SLEngish chapter turns those distinctions into executable operator contracts; the execution and knowledge-representation chapters test how well the representation lowers to existing technologies. The appendices collect the resulting verbs and parameters as the working vocabulary for implementation and experiments.

Term	Explanation
Controlled natural language	Attempto Controlled English and related CNLs establish deterministic mappings for restricted readable forms (Fuchs, Kaljurand, and Kuhn 2008; Kuhn 2014).

Term	Explanation
Formal semantics	Discourse representation, natural logic, event semantics, and semantic parsing supply richer structures than untyped triples (Kamp and Reyle 1993; Bos 2008; Angeli and Manning 2014; Abzianidze et al. 2019).
Formal execution	Abstract interpretation, symbolic execution, model checking, and SMT provide distinct guarantee and witness mechanisms (Cousot and Cousot 1977; King 1976; de Moura and Bjørner 2008; Baldoni et al. 2018).
LLM plus tools	Program-aided, logic, constraint, planning, and autoformalisation systems show the value of narrow verified targets, not the formality of probabilistic generation itself (Gao et al. 2023; Ye et al. 2023; Pan et al. 2023).
Executable-language programme	The present synthesis combines controlled-language research, formal execution, cross-linguistic semantic distinctions, typed operator libraries, and backend-neutral interpretation.

CHAPTER 2

English as a Semantic Substrate

The Structural Core of English

SLEnglish begins from English for practical rather than metaphysical reasons. English is a major language of science, software, regulation, and international technical work; contemporary language models also handle it with broad coverage. None of this makes English a neutral model of thought. It makes English the source language whose recurrent structures, efficiencies, ambiguities, and historical irregularities must be understood before they are regularised. A controlled executable extension cannot improve what it has not first analysed.

Modern English is structurally analytic compared with many Indo-European relatives. It relies heavily on word order, auxiliary verbs, prepositions, articles, pronouns, and lexical choice rather than on extensive case or agreement morphology. The basic declarative clause is often described through a subject followed by a verb phrase, yet this familiar formula conceals several layers. The subject is a syntactic position, not a stable semantic role. It may denote an agent, experiencer, patient, theme, instrument, location, event, proposition, dummy element, or formally raised constituent. The verb phrase can contain auxiliaries that mark tense, aspect, modality, voice, and negation, while complements and adjuncts distribute information about participants, circumstances, and discourse status (Huddleston and Pullum 2002).

The distinction between syntactic function and semantic role is central. In ‘Alice opened the door’, Alice is both subject and agent. In ‘Alice noticed the error’, the subject is an experiencer. In ‘Alice received the file’, the subject is a recipient. In ‘The key opened the door’, the subject is an instrument or cause. In ‘There appeared a problem’, the grammatical subject is the expletive there while the post-verbal phrase introduces the entity. A machine that maps the first noun phrase mechanically to agent will therefore produce systematic errors even in simple clauses.

English transitivity is similarly flexible. Verbs may appear without an object, with one object, with two object-like complements, with a predicative complement, or inside constructions that change the normal alignment between syntax and semantic roles. ‘The glass broke’ and ‘Alice broke the glass’ share a change but differ in explicit causation. ‘Alice gave Bob the file’ and ‘Alice gave the file to Bob’ package the same transfer with different information structure. ‘The committee elected Alice chair’ combines an affected participant with a resultative role. Construction Grammar is useful here because it treats argument-structure patterns as conventional pairings of form and meaning rather than assuming that the lexical verb alone supplies every participant relation (Goldberg 2006).

English noun phrases combine determination, quantification, modification, complementation, possession, and reference. Articles are not merely decorative. The contrast between a, the, and bare forms interacts with identifiability, discourse status, genericity, countability, and number. Quantifiers introduce scope that surface order does not always settle. ‘Every researcher reviewed a paper’ may describe one shared paper or a different paper for each researcher. Relative clauses and prepositional phrases can attach at several levels. Nominalisations such as ‘the destruction of the archive’ compress event structure and may omit the destroyer, instrument, time, intention, and authority. This compression is efficient for prose but costly for executable interpretation.

Tense in English is morphologically sparse. The finite verb contrasts primarily past with non-past, while richer temporal meanings are assembled through auxiliaries, aspect, adverbials, lexical aspect, and context. The progressive presents an event or state from within its unfolding; the perfect relates a prior situation to a later reference point; modal auxiliaries express possibility, necessity, prediction, permission, obligation, or inference. The same surface modal can cross domains. ‘Must’ can report a rule, a logical consequence, or an evidential inference. A SLEnglish translation must therefore represent the modal source rather than preserve the word as an uninterpreted token.

Negation appears simple but interacts strongly with scope and pragmatic inference. ‘Alice did not review every paper’ does not normally mean that she reviewed none. ‘Every paper was not reviewed’ is unstable across speakers and registers. Negative polarity items, exceptive phrases, downward-entailing contexts, and presuppositions further complicate direct compilation. English also distinguishes sentence negation from lexical opposition, denial, absence, failure, prevention, and inability. These contrasts motivate separate operators rather than one global Boolean prefix.

Clause combination carries relations that punctuation and conjunctions only partly determine. Coordination can express addition, contrast, sequence, consequence, concession, or reformulation. Subordination can introduce time, cause, purpose, condition, comparison, report, or evidential stance. Participial and infinitival clauses often omit their understood subject. ‘After reviewing the data, the conclusion changed’ is grammatically possible even though the surface form encourages an implausible controller. Relative clauses, complement clauses, and reported speech introduce perspective centres and anaphoric dependencies that may extend across sentences.

Discourse interpretation supplies much of what grammar leaves open. Pronouns require antecedents; definite descriptions rely on shared context; ellipsis recovers missing material; tense is anchored to a discourse time; information structure distinguishes topic, focus, contrast, and background; implicature enriches literal meaning; presupposition treats some content as already accommodated. A reader handles these operations through linguistic competence and world knowledge. An executable system must either represent the decision, preserve alternatives, or declare that the distinction is outside the active formalisation.

English morphology also contains historical irregularity that has little semantic value for a machine-facing representation. Strong and suppletive verbs, irregular plurals, inconsistent derivation, fossilised case forms, orthographic ambiguity, and variable stress complicate authoring and parsing without improving reasoning. SLEngish therefore uses canonical symbols, no agreement, no conjugation, and no dependence on articles or inflection. The lexical familiarity of English is retained; its accidental morphology is not part of the executable contract.

English structure	Semantic consequence for an executable representation
Subject position	The grammatical subject does not identify a single semantic role; the role must be named as agent, experiencer, patient, theme, instrument, location, proposition, or another declared relation.
Argument structure	Participants are contributed jointly by verbs and constructions. Named parameters prevent double-object, prepositional, passive, and resultative alternations from generating different hidden schemas.
Noun phrase	Reference, identity, quantity, countability, genericity, possession, and modification must be separated rather than compressed into determiner order.
Tense and aspect	Past/non-past morphology is insufficient. Event time, reference time, progressive, perfect, habituality, completion, duration, and temporal distance require explicit operators or parameters.
Modality	Possibility, obligation, permission, prediction, inference, and logical necessity must not share one untyped modal value.
Negation	Explicit contradiction, absence, non-occurrence, failure to derive, prevention, inability, and lexical opposition require distinct semantics.
Clause linkage	Cause, purpose, condition, concession, sequence, report, and elaboration need declared relations instead of being inferred from punctuation or generic conjunction.
Reference and discourse	Anaphora, quotation, perspective, topic, focus, presupposition, and implicature must be resolved, represented as alternatives, or marked outside scope.
Morphological irregularity	Irregular conjugation, plural formation, and spelling are removed from the executable layer because they add parsing cost but no intended semantic distinction.

The corpus perspective reinforces this analysis. Biber and colleagues show that English grammar is not one uniform system used identically in conversation, fiction, news, and academic prose. Registers differ in clause density, nominalisation, stance markers, pronoun use, lexical bundles, and information packaging (Biber et al. 1999). Scientific prose in particular concentrates information in noun phrases and passive constructions. That density is useful to experts but difficult for source-grounded execution because causal roles and evidential status are often distributed across several sentences or hidden inside nominal modifiers.

A regular intermediate language should not attempt to reproduce every surface choice. It should preserve the distinctions that change denotation, inference, temporal organisation, evidence, authority, or downstream action. Voice can often be normalised into role-labelled event structure. Word-order variants can map to the same operator. Synonymous surface verbs can map to a shared semantic contract when the domain warrants it. Conversely, superficially similar uses must remain separate when they carry different commitments. ‘The

model predicts', 'the paper reports', 'the instrument records', and 'the team concludes' cannot all become a generic relation between a subject and a proposition.

This is why the proposed operator libraries include ordinary English constructions as well as categories discovered through cross-linguistic comparison. The appendices treat declaratives, interrogatives, imperatives, passives, progressives, perfects, conditionals, reported speech, anaphora, nominalisation, coordination, comparison, possession, measurement, and other recurrent structures as candidate executable contracts. The purpose is not to encode English grammar for its own sake. It is to make the translation from English auditable at the points where a grammatical choice changes the semantic object handed to an interpreter.

Where English Leaves Meaning Implicit

English succeeds partly because it permits efficient underspecification. Speakers need not repeat what participants can recover from the situation. Articles, pronouns, tense, lexical aspect, conventional scripts, and background knowledge coordinate attention at low cost. The same economy becomes a liability when a sentence is detached from its original setting, incorporated into a knowledge base, reused by an agent, or asked to support a decision months later. The problem is not that English cannot express precision. It is that precision is optional, unevenly distributed, and often expensive.

The verb *be* illustrates the issue. It supports strict identity, classification, role occupancy, temporary state, location, scheduling, constitution, approximation, and predication. 'Alice is the director' can describe current office, legal authority, informal leadership, or an identification in a particular record. 'Water is H₂O' concerns theoretical identity of a different kind. 'The meeting is Monday' locates an event in time. A formal language that assigns all these to one predicate either overgenerates substitution or pushes decisive semantics into untyped arguments.

The word *same* is more consequential. It can indicate numerical identity, material continuity, structural resemblance, genealogical descent, functional equivalence, version lineage, legal continuity, or context-dependent sufficiency. A cloud can remain the same process while exchanging most of its material. A company can remain the same legal entity after complete personnel turnover. Two software deployments can run the same source while differing in configuration and behaviour. English allows the relevant criterion to remain tacit because human interlocutors often negotiate it locally. Executable knowledge cannot assume that the criterion will remain stable across queries.

Collective nouns compress participant structure and authority. 'The company decided' may refer to a board vote, an executive instruction, a delegated automated policy, a distributed practice, or an outcome that nobody intended. 'Science says' can mean a meta-analysis, a current disciplinary majority, one institutional statement, or a journalist's synthesis. 'We' can include or exclude the addressee, share or deny authority, distribute or conceal responsibility, and associate some people with costs they did not choose. English can clarify these relations, but its ordinary grammar does not require the clarification.

Causal language is equally compressed. 'X caused Y' can denote a triggering event, enabling condition, background structure, intervention, statistical contributor, omission, necessary condition, sufficient configuration, or retrospective attribution. Active voice may imply intention even when the mechanism is systemic. Passive voice may hide the actor when the actor is unknown, irrelevant, obvious, strategically omitted, or institutionally protected. Scientific explanation needs to separate causal type, evidential basis, model, time scale, and intervention semantics rather than equating grammatical transitivity with causal force.

Epistemic provenance is mostly optional. English can say that a proposition was observed, measured, remembered, reported, inferred, simulated, quoted, or generated, yet it also permits an unmarked declarative that gives no clue about the source. Hedges such as *apparently*, *probably*, *perhaps*, or *I think* mix confidence, politeness, indirectness, and source. A report may be grammatically presented as fact after several transformations. For scientific and institutional knowledge, source mode and source depth should be independently addressable.

Temporal meaning is distributed across tense, aspect, adverbials, lexical semantics, and world knowledge. 'The system fails' may describe a current event, a habitual pattern, a generic disposition, a scheduled test result, or a conditional consequence in an omitted context. 'The model has improved' establishes a prior comparison but may leave baseline, measurement, evaluation set, and interval unspecified. 'Soon' and 'recently' are relative to speaker, domain, and time scale. A language intended for executable science must

distinguish valid time, observation time, transaction time, reference interval, duration, recurrence, and temporal distance when they affect consequence.

Scope is another hidden variable. Quantifiers, negation, modality, conditionals, and attitude reports can combine in several orders. ‘A reviewer checked every claim’ may involve one reviewer or several depending on context. ‘The regulator may not approve the system’ can mean permission to withhold approval or uncertainty about approval. ‘All models are not robust’ may be read as none are robust or not all are robust. A controlled representation should make scope structural, because no deterministic reasoner can recover a distinction that was erased by the compiler.

Reference is rarely a matter of string equality. Names vary, organisations change names, titles refer to roles, pronouns depend on discourse, and descriptions may identify an entity only under a time and jurisdiction. The same label can denote a legal person, a brand, a software service, or a group of employees. Entity resolution is therefore an empirical and ontological task. SLEngish preserves stable node identities and permits qualified identity relations rather than assuming that repeated words denote the same object.

Countability and boundaries also depend on conceptual choice. English alternates between mass and count construals, as with experience, paper, coffee, data, or software. A noun can impose a unit where the phenomenon is continuous. A classifier may be omitted even when it determines what counts as one case, one event, one model, or one species. Measurement appears exact only after a unit, instrument, precision, and boundary rule have been selected. The executable layer should record the classifier or individuation criterion whenever a count licenses inference.

Normative statements combine content with authority. ‘The sample must be destroyed’ may be a legal requirement, a protocol obligation, a safety recommendation, a prediction, or an inference from observed evidence. ‘May’ can grant permission or state possibility. ‘Should’ can express weak obligation, expected behaviour, advice, or counterfactual consequence. Jurisdiction, issuer, mandate, effective time, exceptions, and affected parties determine whether an instruction is binding. A semantic compiler cannot leave these fields to tone.

The implicitness of English is not a defect to be removed globally. A language that demanded source, time, identity mode, causal mechanism, authority, and scope in every sentence would be unusable. SLEngish instead supports profiles and operator contracts. A scientific claim profile may require units, uncertainty, source spans, and valid time. An institutional decision profile may require decision procedure, mandate, represented collective, and affected population. An informal annotation profile may accept more omissions and return a weaker guarantee. Precision is attached to purpose.

This principle also controls the relation between SLEngish and natural language. The source text remains authoritative as text. A compiled representation is one declared projection used for a task. It may omit literary effects, social implication, emotion, irony, or productive ambiguity. The system should not claim that the projection is the complete meaning of the passage. Instead it records coverage, alternatives, unsupported constructs, and the exact source spans that gave rise to each node. Auditability replaces the fiction of complete translation.

Implicit English distinction	Required SLEngish treatment when the distinction affects consequence
Identity criterion	Name strict, material, structural, processual, genealogical, informational, functional, role-based, or legal continuity.
Participant role	Replace position-based inference with named roles such as agent, experiencer, patient, theme, recipient, source, instrument, and beneficiary.
Evidence source	Distinguish observation, measurement, testimony, quotation, memory, inference, simulation, generation, and source depth.
Temporal frame	Represent event time, valid time, transaction time, reference interval, aspect, recurrence, and remoteness where relevant.
Quantifier scope	Encode bindings and structural scope for universal, existential, numerical, modal, and negative operators.
Collective attribution	State whether the relation is statistical, authorised, delegated, distributed, coordinated, emergent, or merely metonymic.
Causal type	Separate trigger, contribution, enabling condition, mechanism, omission, structural cause, prevention, and intervention.
Normative authority	Record issuer, mandate, jurisdiction, effective period, affected party, exception, and strength.

Implicit English distinction	Required SLEngish treatment when the distinction affects consequence
Boundary and classifier	Record the operational rule by which a unit, member, event, category, or dataset row is individuated.
Discourse dependence	Resolve reference and perspective, retain alternatives, or mark the dependency as outside the selected projection.

The English analysis supplies a baseline against which cross-linguistic categories can be assessed. The next chapter does not present other languages as repositories of exotic solutions. It asks a narrower question: which distinctions have natural languages shown to be stable enough for recurrent grammatical or lexical treatment, and which of those distinctions expose omissions that matter for knowledge representation? The operator tables at the end of the book are the engineering consequence of that survey.

The relationship is not one-to-one. A natural-language morpheme can combine meanings that SLEngish separates. A single SLEngish operator may integrate evidence from several linguistic traditions, formal theories, and engineering requirements. Some English constructions deserve their own operator because they change inference; others are normalised into a shared frame. Some rare grammatical categories become useful only as optional parameters. The correct granularity is an empirical question to be evaluated through translation errors, reviewer correction, query behaviour, and implementation complexity.

English therefore serves neither as the universal grammar nor as an adversary. It supplies the practical lexical interface, a rich history of grammatical description, and a large body of machine-readable text. Its limitations provide test cases. SLEngish is an attempt to preserve that accessibility while moving decisive semantic commitments from context into typed, versioned, and inspectable structures.

CHAPTER 3

Language Diversity and the Missing Grammar of Thought as Design Evidence

Relativity, Salience, and the Ethics of Comparison

The idea that language shapes thought has survived repeated declarations of death because it captures an experience almost every bilingual speaker recognises. A thought that feels direct in one language can become awkward in another. A social relation changes when one must choose between a familiar and deferential pronoun. A poem becomes not merely less elegant but differently organised when translated. Some languages seem to invite the speaker to say how an event was known, whether an action was completed, or whether “we” includes the listener. It is tempting to infer that speakers inhabit different mental worlds.

Temptation, however, is not evidence. The history of linguistic relativity is full of plausible anecdotes, selective translations, romantic portraits of non-European peoples, and experiments that confound language with education, ecology, literacy, or task strategy. The strongest version of the claim—that a language makes certain thoughts impossible—has little support. It also underestimates human inventiveness. Speakers constantly paraphrase, borrow, point, draw, calculate, and coin terms. Science itself is a long demonstration that a language can be extended beyond the distinctions inherited from ordinary life.

From worldview to experiment. Wilhelm von Humboldt, Franz Boas, Edward Sapir, and Benjamin Lee Whorf did not formulate one clean hypothesis. Their work contains different claims about grammatical classification, habitual attention, cultural interpretation, and the relation between language and experience. What later came to be called the “Sapir–Whorf hypothesis” was partly a retrospective packaging of a heterogeneous tradition. John Lucy’s historical and methodological reconstruction remains indispensable precisely because it replaces slogans with researchable questions (Lucy 1992, 1997).

A useful distinction separates linguistic determinism from linguistic relativity. Determinism says that thought is bounded by language: without a linguistic category, a speaker cannot entertain the corresponding concept. Relativity says that habitual linguistic patterns can bias or organise cognition in ways that differ across language communities. Even this weaker thesis is too vague until four variables are specified: which linguistic structure, which cognitive process, under what task conditions, and through what mechanism?

Consider colour. Russian conventionally divides the English category blue into *goluboy* for lighter blues and *siniy* for darker blues. Winawer and colleagues found that Russian speakers were faster to discriminate colours across this lexical boundary than within a category, an advantage disrupted by verbal interference but not spatial interference (Winawer et al. 2007). The result does not show that English speakers cannot see the difference. It suggests that an available lexical boundary can be recruited online and alter processing speed. The effect is real, local, and mechanistically modest—exactly the kind of finding a mature theory should expect.

Spatial cognition provides a stronger family of cases. In languages that habitually use absolute frames of reference—roughly north, south, east, west, or directions anchored in prevailing geography—speakers must maintain orientation to converse fluently. Cross-linguistic studies have found correlations between linguistic frames and strategies in non-verbal spatial tasks (Pederson et al. 1998; Levinson 2003). Haun and colleagues showed that Dutch and Namibian children preferred different spatial coding strategies in line with their linguistic environments, and that performance suffered when participants had to switch frames (Haun et al. 2011). Again, the conclusion is not that one group lacks egocentric space. It is that repeated linguistic practice can train a strategy until it becomes the path of least resistance.

Number offers both dramatic evidence and a warning against exaggeration. Work with Pirahã and Mundurukú speakers distinguishes approximate numerical competence from exact symbolic number. People without an extensive conventional counting system can estimate and compare quantities, but exact matching becomes difficult when memory, transformation, or displacement prevents direct one-to-one correspondence (Gordon 2004; Pica et al. 2004; Frank et al. 2008). Exact number words appear to function as a cognitive technology: they stabilise cardinal values across time and operations. Yet these studies do not show that a language permanently seals numerical capacity. Training, material practices, and education matter. Recent work also demonstrates that semantic task structure can outweigh language-specific metaphors. A preregistered 2026 study across four languages found that the semantic content of arithmetic problems, not the

languages’ preferred temporal metaphors, drove ordinal versus cardinal strategies (John, Macedo-Rouet, and Gros 2026). Relativity is a family of conditional effects, not a master key.

Language is not the whole mind. The most serious challenge to expansive claims comes from dissociations between language and thought. People with severe language impairment can retain arithmetic, music, causal reasoning, social understanding, and planning. Brain-imaging work distinguishes the frontotemporal language network from systems supporting many forms of domain-general reasoning. Fedorenko, Piantadosi, and Gibson argue that language is primarily adapted for communication and cultural transmission rather than serving as the medium of all thought (2024). Their case is an important corrective to any theory that treats inner speech as the operating system of intelligence.

But the choice is not between “language is thought” and “language is cognitively irrelevant.” A map is not the territory, yet a map changes where travellers go. A programming language is not computation itself, yet its abstractions alter which programs are easy to write, inspect, and maintain. A graphical user interface is not the underlying machine state, yet its defaults shape behaviour. Language can be one representational resource among several and still have large cumulative effects.

This middle position also clarifies the relevance of large language models. Fluent models show that formidable formal linguistic competence can be acquired from patterns in text, while reliable planning, world modelling, causal intervention, and grounded judgment remain separable problems. Mahowald and colleagues distinguish formal linguistic competence—knowledge of linguistic patterns—from functional linguistic competence, the use of language in real-world reasoning and communication (2024). The distinction cautions against inferring thought from eloquence. It also makes language design newly consequential: AI systems increasingly mediate human writing, summarisation, search, policy, and scientific work. Their linguistic defaults may become cognitive defaults at societal scale.

The mechanism: salience, compression, and obligation. A defensible theory can be built from three mechanisms.

Salience concerns what is brought to attention. Naming a distinction makes it easier to notice, discuss, remember, and teach. Odour provides a striking example. English speakers often name smells by source—smells like lemon—and perform poorly at abstract odour naming. Speakers of Jahai, an Aslian language with a rich abstract smell lexicon, name odours with ease and agreement comparable to colour naming (Majid and Burenhult 2014). Maniq shows a similarly structured odour vocabulary (Wnuk and Majid 2014). The result is not that English noses are defective. English offers little conventional scaffolding for treating odour qualities as stable, discussable dimensions.

Compression concerns the cost of holding and transforming a distinction. A numeral such as seventeen compresses a cardinality into a portable symbol. An evidential suffix can compress “I infer this from visible consequences” into a routine grammatical choice. Compression matters because working memory is limited. A distinction that occupies one familiar slot can participate in more complex reasoning than one that requires a sentence of explanation.

Obligation concerns what grammar makes difficult to omit. English can express evidentiality, clusivity, remoteness, and source depth. The issue is not expressive possibility but default omission. Alexandra Aikhenvald documents languages in which every declarative statement requires a choice among information-source categories (2004). A speaker may have several sources and must still select a form. The grammar imposes what we might call mandatory epistemic bookkeeping.

These mechanisms suggest a more precise formulation of linguistic relativity:

A linguistic system influences cognition when its recurrent categories alter the salience, representational cost, or habitual obligatoriness of distinctions used in perception, memory, inference, or coordination.

The formulation predicts variability. Effects should be stronger when language is actively recruited, when the task aligns with the encoded distinction, when a category is obligatory and frequent, and when alternative strategies are costly. They should weaken under verbal interference, when perceptual evidence dominates, when bilingual speakers operate in a different language, or when task semantics provide a more powerful structure. Gennari’s recent review emphasises precisely this situated character: language effects depend on task goals, context, and alignment between linguistic and cognitive representations (2025).

English is not neutral. The scientific debate has itself been distorted by the dominance of English. Blasi and colleagues argue that cognitive science has too often treated properties of English—its vocabulary,

morphosyntax, modalities, and interactional norms—as transparent windows onto the human mind (2022). This is an epistemic problem rather than a complaint about cultural prestige. If experiments are designed in English, conducted by English-speaking researchers, and interpreted through English categories, the field may overlook cognitive variables made more visible elsewhere.

Anna Wierzbicka's *Imprisoned in English* pushes the critique further. Terms such as emotion, cooperation, politeness, mind, and fairness are not culture-free atoms simply because they travel easily in Anglophone scholarship (Wierzbicka 2013). Her *Natural Semantic Metalanguage* programme is controversial, particularly in its claim to identify universal semantic primes, but the diagnosis is powerful: scholars often universalise the metalanguage in which they theorise.

N. J. Enfield offers a complementary warning. Language is extraordinarily effective for coordination, persuasion, narrative, and social action, yet those strengths do not make it a neutral instrument for representing reality (Enfield 2022). Natural language routinely leaves boundaries vague, assigns agency strategically, compresses causal chains, and foregrounds what matters in interaction rather than what would satisfy a scientific ontology. The defects are not accidental. A communication system optimised for fast coordination among embodied social beings will not automatically be optimised for auditing distributed responsibility in a multinational platform or tracing uncertainty through an AI-generated literature review.

Against two symmetrical mistakes. The first mistake is to dismiss linguistic relativity because speakers can paraphrase. Possibility is the wrong metric. A Turing-complete language can in principle compute any computable function, but programming languages still differ enormously in safety, legibility, and development cost. The second mistake is to romanticise grammatical differences as windows into superior cultures. A language that marks evidence obligatorily can still be used to lie. A language with inclusive and exclusive pronouns does not abolish political exclusion. Absolute spatial systems can support extraordinary orientation while imposing costs in unfamiliar settings. Every grammar illuminates and obscures.

Indo-European languages themselves prevent crude oppositions. English lacks a grammatical evidential system, but Bulgarian has rich evidential and related forms. English has lost the dual that survives in languages such as Slovenian and was present in older Indo-European languages. Slavic aspect forces distinctions that English often distributes across lexical choice and context. Albanian has an admirative category associated with unexpected information. The object of criticism is therefore not an imagined uniform “Indo-European mind.” It is a historically contingent bundle of defaults, especially those of modern standard English as the global language of science, technology, and administration.

The thesis of this book can now be stated without metaphysical drama:

Language does not determine the limits of thought. It helps determine the price of thought.

A costly thought can still be thought. But costs accumulate. Institutions repeat familiar categories; databases require fields; laws stabilise definitions; metrics reward what can be named and counted; AI systems reproduce patterns in their training language. The task ahead is not to escape language. It is to inspect and redesign some of the toll gates.

Evidence, Perspective, Participation, and Social Grammar

Humanity did not design a single language and then diversify it cosmetically. The world's languages vary in sound, morphology, syntax, semantics, discourse, gesture, and the social distribution of speech. Large typological resources such as the *World Atlas of Language Structures* and *Grambank* convert parts of this diversity into comparable data. *Grambank* currently encodes hundreds of thousands of observations across more than two thousand language varieties and nearly two hundred structural features; its analyses show both deep genealogical inheritance and substantial flexibility (Skirgård et al. 2023). The databases remain necessarily incomplete, but they overturn the intuition that familiar European grammar exhausts the natural options.

It is fashionable to call languages “experiments,” but the metaphor requires care. Languages are not controlled trials. A grammatical feature travels with culture, environment, history, contact, subsistence, literacy, and institutions. Correlation between grammar and cognition cannot by itself establish causation. Yet diversity performs an indispensable conceptual service before any causal claim is made: it proves that a distinction is learnable, transmissible, and usable in ordinary life. What English treats as cumbersome may elsewhere be routine.

Who belongs inside “we”. English we is politically convenient because it is semantically underdetermined. It can mean speaker plus listener, speaker plus others but not listener, a whole institution, a nation, humanity, a professional class, or a rhetorical coalition invented for one sentence. Many languages grammaticalise a basic part of this ambiguity through clusivity: inclusive we contains the addressee; exclusive we does not.

WALS data show that this distinction is widespread outside Europe and absent from the European languages in its sample (Cysouw 2013). Mandarin provides a familiar lexical contrast: *zánmen* normally includes the listener, whereas *wǒmen* need not. Other systems extend the distinction through dual, trial, or paucal number. Paamese, for example, distinguishes categories that allow speakers to refer not merely to “us” but to inclusive or exclusive groups of two, three or a few.

Clusivity is not a built-in theory of democracy. It does, however, expose a variable that English political language can hide. “We have decided that sacrifices are necessary” sounds different when grammar forces the speaker to choose we-with-you or we-without-you. A designed extension could go further and distinguish we-authorised-by-you, we-claiming-to-represent-you, and we-affected-with-you. Natural grammar supplies the proof of usability; conceptual engineering identifies the modern accountability problem.

Number beyond singular and plural. English compresses most cardinal distinctions into singular versus plural. The difference between two people, three people, and a small coalition is lexical information, not a general grammatical category. Yet dual, trial, and paucal systems show that number can be organised differently. Such systems may be especially elaborated in pronouns, where the structure of participation matters socially.

The cognitive significance should not be overstated. A grammatical dual does not make its speakers better at arithmetic. Its value is attentional: it treats dyads as a recurring type rather than an accidental quantity. This matters because pairs often have properties that larger groups do not. Reciprocity, mutual monitoring, intimacy, deadlock, and shared secrecy change discontinuously between two and three. A language for institutional analysis might profit from distinguishing not only numerical size but coordination topology: pair, chain, hub, clique, hierarchy, and swarm.

Counting through ontology. Numeral-classifier languages require a mediating category when nouns are counted. In Minangkabau, classifiers include forms associated with humans, non-human animates, and elongated objects. WALS distinguishes languages without numeral classifiers from those with optional or obligatory systems (Gil 2013). The categories differ widely, and their use is often shaped by conventional rather than philosophically pure taxonomies.

The larger insight is that quantity need not be ontologically neutral. English applies the same syntactic frame to three chairs, three companies, three storms, and three promises. But what constitutes one unit differs. A “company” may be counted legally, operationally, by brand, by corporate group, or by decision system. A “storm” can be individuated meteorologically or by named event. A “model” can be counted by architecture, weight file, deployed endpoint, or lineage. Modern data systems often fail because they request a number before defining the unit.

A useful conceptual classifier would not ask what physical shape an entity has. It would ask what principle makes it countable: bounded object, continuing process, legal person, information pattern, causal lineage, temporary coordination, or statistical cluster. The classifier would act as an ontological checksum attached to numerical claims.

Possession is not one relation. English my can connect a person to a hand, a coat, a birthplace, an idea, a child, an employee, a debt, a disease, or a country. The word’s grammatical uniformity hides substantially different relations: part-whole, ownership, kinship, authorship, control, association, responsibility, experience, and identification. Languages with possessive classification and alienability contrasts demonstrate that grammar can separate some of these relations. WALS cautions that the label “alienable versus inalienable” oversimplifies systems whose classes may be organised around kinship, body parts, spatial relations, or semantic properties of nouns (Nichols and Bickel 2013).

This is not a minor grammatical curiosity. Political and economic disputes often exploit possessive ambiguity. “Our data,” “my body,” “their territory,” “the company’s model,” and “humanity’s atmosphere” invoke different normative structures while sharing possessive syntax. A language designed for contemporary governance might distinguish custody, legal title, control, authorship, dependency, stewardship, membership, and constitutive part. The distinction between a resource one may transfer and a relation that partly constitutes the self should not be left entirely to context.

Tracking perspective and continuity. Some languages mark whether the subject of one clause is the same as or different from the subject of an adjacent clause. Switch-reference systems can therefore preserve agent continuity through discourse even when noun phrases are omitted. The markers may also interact with temporal or event continuity, and their analysis remains theoretically contested (McKenzie 2025). Whatever the correct syntax, the functional achievement is striking: grammar can make a change of protagonist visible at the joint between propositions.

Logophoricity tracks a different question: whose perspective, speech, knowledge, or consciousness is represented? English pronouns and reported-speech constructions often leave ambiguity between the narrator's and a character's commitments. Logophoric pronouns in some languages identify a privileged perspective-holder. Obviation, especially associated with Algonquian languages, ranks third persons by discourse prominence through proximate and obviative marking. These are not merely reference-management devices. They make the architecture of perspective grammatically present.

The need is acute in the age of synthetic media and AI. A generated report may contain the user's assertion, a model inference, a retrieved source's claim, and an evaluator's judgment in the same paragraph. Ordinary quotation and citation are too coarse. A future grammar of provenance could preserve not only who uttered a sentence but who perceived, inferred, endorsed, transformed, or merely transmitted each proposition.

Time with distance, not merely tense. English marks tense and aspect, but it normally leaves temporal distance to adverbs. Many languages have grammatical remoteness distinctions. WALS reports systems separating today's past from earlier pasts, and Yagua has been described with five degrees ranging from a few hours ago to a distant or legendary past (Dahl and Velupillai 2013). Such categories do not replace dates. They position events relative to the speaker's temporal horizon.

The design opportunity is not to copy a particular tense inventory. Modern institutions operate across multiple clocks: milliseconds in automated trading, days in news cycles, quarters in corporations, electoral terms in democracies, decades in infrastructure, centuries in climate, and geological durations in nuclear waste. English can state each scale but offers no grammatical friction when a claim slides between them. "Efficient," "safe," "profitable," and "sustainable" are often meaningless until a temporal address is supplied.

Experience beyond the visual hierarchy. English has a rich visual vocabulary and comparatively weak abstract vocabulary for smell. Jahai and Maniq demonstrate that olfactory qualities can be lexicalised systematically and named with speed and agreement (Majid and Burenhult 2014; Wnuk and Majid 2014). Ideophones across many languages vividly depict sensory scenes, motion, texture, intensity, sound, and manner. They resist the assumption that mature language must move away from iconicity toward arbitrary, visually dominated description.

The lesson is broader than smell. Scientific instruments increasingly detect dimensions humans do not perceive directly: molecular signatures, latent variables, network motifs, electromagnetic spectra, uncertainty surfaces. We typically force these into inherited visual and object-centred metaphors. A richer language of patterned sensation might help experts develop intuitions for high-dimensional systems without pretending those intuitions are literal perception.

Social geometry in grammar. Honorific systems, speech levels, kinship terms, and address forms show that grammar and lexicon can make social relation unavoidable. The familiar European T–V distinction—informal versus formal second-person address—is only a small example. Languages may encode relative age, status, intimacy, group membership, deference, and the social direction of an act. Such systems can preserve hierarchy and exclusion; they can also prevent the fiction that communication occurs between context-free individuals.

English is not socially neutral. It encodes hierarchy through titles, register, accent, indirectness, and institutional genre, but often disguises these as stylistic choices rather than relational information. A redesigned language should not restore rigid rank. It could instead mark role and power asymmetry: who can refuse, who bears risk, who controls the agenda, and who is represented but absent. This would turn social context from background implication into inspectable structure.

What typology gives—and what it does not. The catalogue could continue: grammatical animacy, noun classes, inverse systems, pluractionality, distributive and associative number, egophoricity, mirativity, middle voice, applicatives, antipassives, directionals, elaborate aspect, and systems of quotation and report. No single language combines all of them, and no speaker explicitly consults a philosophical matrix while using them. Grammar is mostly automatic.

That automaticity is precisely why typology matters. It shows that habitual distinctions can be made quickly enough for conversation. But typology alone cannot prove cognitive benefit. A category may exist because of historical accident; it may be redundant; speakers may strategically neutralise it; experimental effects may be task-bound. The book therefore adopts a two-stage method. Natural languages first reveal possible interface components. Philosophy and experiment then ask whether a component would improve reasoning in a specified domain.

The goal is not maximal grammatical complexity. Every obligatory distinction consumes attention. A language that forced source, confidence, causal depth, temporal scale, collective mode, and identity criterion into every sentence would be unusable. Good interface design is selective. A cockpit displays different variables from a kitchen timer. Scientific reporting, corporate governance, public news, and intimate conversation need different defaults.

The world's languages are thus neither prisons nor museums. They are a library of interface patterns. We should read them with gratitude, skepticism, and the intention to build.

Imagine two versions of the same headline:

The treatment causes cardiac damage.

The treatment causes cardiac damage — inferred from a retrospective association, reported through two secondary summaries, confidence moderate, causal status disputed.

The first is elegant, portable, and dangerously complete. The second is cumbersome but epistemically honest. Modern English offers no stable point between them. Academic prose can add citations and hedges, journalism can attribute sources, and data systems can record metadata, yet ordinary sentence grammar permits a proposition to travel detached from how it was acquired. Social media and generative AI amplify this detachment. A sentence can be copied thousands of times while its evidential history evaporates.

Natural evidential systems show that source information can be compressed into grammar. They do not solve epistemology, but they expose a remarkable design option: a language can require the speaker to answer, however minimally, *How do you know?*

Evidence is not confidence. Evidentiality is often confused with certainty. The categories can interact, but they are conceptually different. A speaker may be completely confident in a report and uncertain about what they directly saw. Evidentiality marks the source or mode of access: visual observation, non-visual perception, inference, report, quotation, memory, or another conventional category. Epistemic modality marks possibility, probability, necessity, doubt, or commitment.

WALS separates languages with no grammatical evidentials, languages with indirect evidentials, and languages distinguishing direct from indirect sources. In its 418-language sample, 237 have some grammatical evidential marking; systems are distributed across verbal affixes, tense forms, particles, modal morphemes, and mixed strategies (de Haan 2013a, 2013b). The figures should not be treated as an inventory of all languages, but they make one fact undeniable: grammatical source marking is not an exotic anomaly.

Turkish famously contrasts a witnessed past in *-di* with an indirect or unwitnessed past in *-miş* in relevant uses. A simple pair such as *Ahmet gel-di* and *Ahmet gel-miş* can distinguish direct access from an event learned indirectly, although the forms have broader tense–aspect–modal functions and cannot be reduced to labels in an English table. Tuyuca and other Eastern Tucanoan languages have more elaborate systems. WALS examples include visual, non-visual sensory, inferential, and reportative distinctions. Koasati has an auditory evidential in examples where a speaker reports what an event sounds like. Aikhenvald's comparative work documents systems with two terms and systems with six or more, along with the discourse rules that govern selection when evidence comes through several channels (Aikhenvald 2004).

European languages complicate the lazy contrast between “European” and “non-European” thought. Bulgarian has grammatical forms analysed in terms of evidentiality, renarration, inference, doubt, and mirativity. The precise semantics are debated; Koev, for example, analyses the Bulgarian *-l* form through the relation between the described event and the event of acquiring information (Koev 2017). Albanian admiring forms are associated with unexpected or newly realised information. Germanic modal constructions can develop evidential functions. The typological pattern is areal and historical, not a clean genetic divide.

Does evidential grammar improve source memory. The attractive hypothesis is that obligatory source marking trains source monitoring. Evidence is suggestive but not decisive. Research with Turkish-speaking children has linked mastery of evidential forms to the ability to track how knowledge was acquired, and some

cross-linguistic comparisons report advantages in source-memory tasks (Aksu-Koç et al. 2009; Kandemirci et al. 2023). Yet language, schooling, conversational practice, and metacognitive development are difficult to separate. Experimental tasks may inadvertently favour the distinctions prominent in one language. Effects can depend on whether participants expect disagreement or later accountability.

The correct conclusion is therefore conditional. Evidential grammar is a plausible training regime for source attention; it is not a universal vaccine against credulity. Speakers can choose misleading evidentials, exploit conventional exceptions, or repeat false reports with impeccable source marking. What the grammar changes is the structure of omission. A bare claim becomes less normal.

This difference matters institutionally. Contemporary misinformation is not only a problem of false sentences. It is a problem of provenance collapse. A speculative preprint becomes a news article; the article becomes a social post; the post becomes a model-generated summary; the summary becomes “research shows.” At every transition, confidence may rise while contact with the original evidence falls. Citation alone does not prevent the collapse because citations are often copied without being opened, and secondary sources can form circular chains.

Four coordinates of an epistemic checksum. A useful designed system should mark more than sensory source. For scientific and computational communication, a minimal epistemic checksum would have four coordinates:

Term	Explanation
Source mode	observed, measured, inferred, simulated, reported, remembered, or generated
Provenance depth	direct, one transformation removed, two transformations removed, or chain unknown
Confidence state	calibrated probability or a small ordinal band
Revision status	exploratory, provisional, accepted-for-action, contested, superseded, or retracted

In compact notation, a claim might appear as:

The sentence “The bridge is unsafe” should not stand alone. Its representation should also record that it rests on measurement, is one evidential hop from the observation, carries confidence 0.78, and remains provisional.

This is not proposed as literary English. It is a domain-specific layer for reports, dashboards, research environments, AI outputs, and public claims where epistemic detachment is costly. The notation makes visible what ordinary prose hides. Its fields also separate questions that hedging words blur. Apparently may mean reported, inferred, weakly believed, or politely distanced. The checksum does not force one ambiguous adverb to perform four jobs.

Source depth and the illusion of independent confirmation. Natural evidentials usually classify source type, not the full graph of transmission. Digital culture requires source depth and source independence. Suppose five articles report the same finding, but all derive from one press release. The number of visible sources is five; the number of independent evidential roots is one. Ordinary language encourages a counting error because it individuates documents rather than lineages.

A provenance grammar could mark whether a claim is firsthand, transformed from a primary record, aggregated from independent records, or copied through an unknown chain. It could also distinguish quotation from endorsement. Logophoric and reportative systems show that perspective and transmission can be grammatically tracked. Digital infrastructure can extend the principle with cryptographic provenance, persistent identifiers, and machine-readable evidence graphs.

The cognitive gain would be substantial even without perfect verification. A reader who sees roots:1 beside “reported by five outlets” is less likely to confuse repetition with corroboration. A model trained to preserve source depth would be less likely to launder speculation into fact through iterative summarisation.

Confidence is not a decorative hedge. English confidence terms—possibly, probably, almost certainly, I think—are context-sensitive and socially strategic. People use them to protect face, signal politeness, perform authority, or avoid commitment. Numeric probabilities can create false precision, but uncalibrated adjectives make learning from error difficult. A designed grammar should allow both: a coarse mandatory band for routine use and optional numerical calibration where data support it.

The critical innovation is not the number. It is the coupling between confidence and revision. A confidence marker should belong to a forecast or claim whose later outcome can be checked. Otherwise “80 percent” becomes costume. In scientific work, confidence should also distinguish statistical uncertainty, model uncertainty, measurement uncertainty, and uncertainty about whether the chosen question is adequate. These are not interchangeable.

The category unknown is equally overloaded. It can mean that nobody has checked, that available evidence is inconclusive, that the answer is practically inaccessible, that the concept is ill-posed, or that the answer may be unknowable in principle. Treating these states as one encourages both complacency and false certainty. The missing-variables analysis later in this book separates unknown now, unknown to a given community, discoverable with known methods, discoverable only after methodological innovation, and principled indeterminacy.

Mirativity: when information changes the model. Mirativity concerns unexpected information, the “unprepared mind,” and the speaker’s surprise (Aikhenvald 2004). Surprise is not merely an emotion added to a fact. It contains information about a prior model. If the result was unexpected, then either the world is unusual, the evidence is wrong, or the model was inadequate.

Science routinely wastes this information. Papers report a finding but rarely encode what the researchers predicted with enough precision to measure surprise. After results are known, narratives are reconstructed and anomalies become “consistent with” a broad theory. Preregistration partly addresses the problem by preserving prior commitments. A mirative marker could generalise the principle: model-fit, model-strain, or model-break would state whether the evidence fell within expected variation, required parameter revision, or contradicted the operative structure.

The marker would be valuable for AI systems as well. A system that merely emits confidence can remain confidently wrong within a misspecified model. A system that tracks model surprise can notice when incoming evidence lies outside the distribution its explanations assume. The distinction between “I am unsure which familiar answer is correct” and “my current answer space may be wrong” deserves a stable linguistic form.

Memory, testimony, and generated claims. Human memory is reconstructive. Yet English commonly presents remembered events with the same declarative form as current observations. “He said that” may be a verbatim record, a gist, an inference, or a later reconstruction. Legal systems ask witnesses to separate these states, but ordinary grammar does not help.

A future evidential inventory should therefore include memory mode: recorded, contemporaneously noted, repeatedly rehearsed, unaided recall, or reconstructed from later information. It should also distinguish a system’s own inference from generated text that merely resembles one. When an AI states “I found that...,” the first person invites a false analogy with human discovery. Machine outputs need markers for retrieval, transformation, interpolation, calculation, simulation, and ungrounded generation.

This is not a call to burden every conversation with metadata. Natural languages already vary register by context. Scientific claims, legal testimony, medical decisions, intelligence analysis, and high-impact automated outputs can sustain more explicit grammar than casual speech. The central design principle is selective obligatoriness: make a distinction mandatory where its omission creates systematic harm.

The politics of epistemic marking. Any grammar of knowing can become a grammar of authority. Institutions may treat direct observation as superior even when instruments or distributed testimony are more reliable. Marginalised speakers may be forced to over-document claims that elites can assert without markers. A platform may attach “uncertain” to dissent while presenting official statements as unmarked truth. Evidentiality therefore has a conceptual ethics.

The system must mark claims symmetrically, preserve disagreement, and distinguish source from credibility. “Reported by the ministry” is a provenance statement, not a guarantee. “Generated by a model” is a production statement, not a refutation. The aim is not to automate trust but to prevent invisible substitutions among observation, report, inference, and authority.

A well-designed epistemic grammar would change the default question from “Is this sentence fluent?” to “What kind of epistemic object is this sentence?” The answer would not settle whether it is true. It would make truth-seeking less blind.

The typological argument changes the SLEngish design directly. Evidentiality, clusivity, number, classifier choice, possession, perspective tracking, and social deixis are not relegated to anthropological examples. They

become candidate typed operators or profile requirements. The parser remains unchanged because the distinctions live in versioned libraries. Appendix D maps each construction to the proposed operator family and records where the mapping is only an engineering analogy.

Space, Time, Experience, and Conceptual Engineering

A child is shown a row of cards depicting a person at different ages and is asked to arrange them in time. An English speaker is likely to place earlier events to the left and later events to the right, especially in a literate setting where writing runs left to right. Rotate the child, move the experiment outdoors, or ask a speaker whose community uses absolute directions, and the answer may change. Time is not visible as a line. Languages and cultural practices recruit space to make it tractable.

This makes time one of the clearest domains in which linguistic diversity reveals cognitive alternatives. It also exposes the danger of overclaiming. Spatial metaphors can guide reasoning, but task meaning, writing direction, gesture, environment, and experimental framing interact. The aim is not to announce that communities “see time differently” in some total sense. It is to identify the representational tools they habitually recruit.

Relative bodies and absolute worlds. English spatial description is often egocentric or object-centred: left, right, in front of, behind. Many languages make extensive use of absolute frames based on cardinal directions, prevailing winds, coast–inland axes, uphill–downhill, or stable landscape features. To say where an object is, a speaker may need to know orientation beyond the local body.

Stephen Levinson’s comparative programme argued that linguistic frames correlate with memory, gesture, navigation, and non-verbal spatial reasoning (Levinson 2003). Pederson and colleagues found cross-community alignment between preferred linguistic frames and spatial conceptualisation (1998). Haun and colleagues later demonstrated both developmental effects and cognitive plasticity: participants could use alternative frames, but switching imposed a cost (2011).

The important point is not that absolute speakers possess a mysterious faculty. Conversational practice rewards continuous orientation. If ordinary speech requires something like “the cup is north of the plate,” losing north repeatedly creates communicative failure. Grammar and habit turn environmental information into maintained cognitive state.

A comparable deficit exists in modern organisational language. Teams discuss a system component “upstream,” a risk “down the road,” and a decision “behind us,” but often lack a shared map of causal or temporal position. We use spatial metaphors without the discipline of an actual coordinate system. A designed conceptual language could require a scale address and system address for claims whose meaning changes with position.

When the past is in front. English conventionally places the future ahead and the past behind. Aymara has attracted attention because lexical and gestural patterns associate the known past with the front and the unseen future with the back. Núñez and Sweetser interpret this through an experiential logic: what has been witnessed lies before the knower; what has not yet occurred cannot be seen and is therefore behind (2006). Later work in other languages has found additional spatialisations, reminding us that “time moves forward” is not an inevitable cognitive primitive.

The philosophical gain is not a new correct orientation. It is the realisation that temporal metaphors encode epistemology. English ahead suggests goal-directed movement into a visible path. Aymara-style imagery foregrounds the asymmetry between known history and unknowable future. Both reveal one relation while hiding another. Policy discourse needs both: the future is where action is directed, but it is also what cannot yet be inspected.

A richer temporal vocabulary might distinguish action-front from knowledge-front. The future is in front as the target of intervention and behind as the region outside observation. The past is behind as no longer alterable and in front as available evidence. Instead of choosing one metaphor, we can mark which relation is active.

Grammatical distance. English past tense does not normally encode how remote an event is. Time adverbs do the work. Many languages have remoteness distinctions: today’s past, yesterday’s past, recent past, remote past, or legendary past. WALS reports that a substantial minority of sampled languages with past marking make two or more grammatical remoteness distinctions. Yagua is described with five past ranges, from events a few hours earlier to distant or legendary time (Dahl and Velupillai 2013).

These categories demonstrate that temporal distance can be integrated into routine grammar. Yet copying clock intervals would miss the opportunity. Human decisions are organised not only by elapsed time but by institutional horizon. The same delay can be near for climate policy and remote for emergency medicine. A quarterly profit is immediate to a company, while species loss is treated as distant even when irreversible processes begin now.

SLEnglish should therefore use functional horizons:

Term	Explanation
operational	within the cycle of the current action system
strategic	beyond current operations but within planned institutional control
generational	extending beyond present decision-makers
civilisational	affecting durable social or planetary conditions
deep-time	exceeding ordinary institutional continuity

A sentence may state that a technology is beneficial over the operational horizon but hazardous over a generational horizon. Naming both horizons prevents one timescale from impersonating all others.

Aspect and the internal shape of events. Tense locates an event relative to speech; aspect describes its temporal structure—ongoing, completed, repeated, habitual, begun, terminated, or viewed as a whole. Slavic languages often force perfective–imperfective choices more systematically than English, which distributes aspect through auxiliaries, verb forms, lexical choice, and context. Other languages grammatically encode repetition, distribution, or phases of action in different ways.

Aspect is philosophically important because events are not points. “The company changed its policy” can present a completed transition, while “the company was changing its policy” opens an unfinished process. “The model failed” may refer to one event, repeated failures, a persistent disposition, or degradation over time. Public arguments often shift among these aspectual readings.

Systems thinking needs a further distinction between state, event, process, trend, and regime change. English has these nouns, but they are not obligatory frames. A rising temperature can be a measurement event, a local trend, a feedback-driven process, or evidence of a regime transition. A grammatical event-type marker would reduce category errors, particularly in dashboards and risk reports.

Temporal granularity and institutional amnesia. Every temporal representation chooses a grain. A company dashboard aggregates by month; a news feed by minute; a historian by decade. Events visible at one grain vanish at another. A short-term intervention can look successful because delayed costs fall outside the window. A noisy series can look stable after aggregation. A continuous process can be falsely divided into annual units because budgeting demands it.

The missing distinction is temporal granularity. Claims should indicate the interval over which variation is measured and the horizon over which consequences are judged. This is not merely statistical metadata. It changes ontology. A “crisis” can be a spike in a long decline, while apparent stability can be a plateau inside accelerating structural change.

Institutions also have memory horizons. Staff turnover, document loss, changing metrics, and political incentives create institutional amnesia. A policy repeatedly rediscovered as new may belong to the same causal lineage even when its name changes. A temporal grammar linked to provenance could mark recurrence: new-form/old-lineage, first-observed-here, or institution-forgotten. Such markers would be valuable in science, where abandoned ideas return, and in governance, where recurring failures are treated as surprises.

Future people and grammatical absence. Future generations are difficult to represent because they lack names, votes, contracts, and present speech acts. English refers to them as an abstract plural. The grammar of agency privileges existing persons: we decide, they will inherit. The pronouns encode a moral asymmetry without making it explicit.

A posterity-scope marker could attach to decisions and benefits:

Term	Explanation
Present benefit	The benefit is received by participants who can influence the present decision.
Future cost	The cost is transferred to future people who cannot participate in the present decision.

The marker does not solve population ethics or determine how future welfare should be weighted. It prevents temporal externalities from disappearing into an unmarked future. It also distinguishes future people whose existence is highly probable from specific persons whose identity depends on present choices—the non-identity problem in moral philosophy.

A grammar for multiple clocks. Modern systems run on many clocks at once. Biological repair, software release, financial settlement, legal appeal, political mandate, educational development, infrastructure decay, and climate feedback have different characteristic times. A decision can be reversible on one clock and irreversible on another. A pause that is trivial for bureaucracy can be fatal in medicine; a fast optimisation can lock a society into decades of infrastructure.

The proposed temporal address therefore has three fields:

The proposed temporal description therefore records the grain of observation, the horizon of evaluation, and the window within which reversal remains possible.

For example:

The intervention appears efficient when evaluated weekly over one quarter, and it can be reversed within approximately thirty days.

The sentence no longer masquerades as a scale-free truth. Another evaluation may show:

The same intervention appears harmful when evaluated annually over a generational horizon, and no known reversal procedure is available.

Both can be correct. The contradiction existed only because English suppressed coordinates.

Would this language improve foresight? An experiment could give teams complex policy scenarios with delayed effects. One group would reason in ordinary prose; another would be required to tag grain, horizon, and reversibility. Outcomes could measure recognition of delayed costs, consistency across time scales, and resistance to present bias. The markers might add burden without benefit, but that is an empirical result, not a philosophical defeat.

We cannot decide where tomorrow “really” is. We can decide not to discuss it as though all clocks were one.

The argument now changes register. The first ten chapters identified distinctions that natural English often treats cheaply: source of knowledge, identity through change, collective agency, causal depth, temporal scale, reversibility, provenance, and several forms of uncertainty. One response would be lexical reform: invent better words and hope that communities adopt them. Another would be interface design: ask better questions at high-stakes moments. Both remain useful. Neither is sufficient for machines that must store, compare, reason over, and execute knowledge.

A machine cannot execute a philosophical recommendation to “be clearer about sameness.” It requires a typed representation of the entities, the two states being compared, the mode of continuity, the criterion, the temporal interval, the degree, and the evidence. Nor is a conventional controlled sentence enough if it merely removes syntactic ambiguity while preserving conceptual underspecification. The research programme therefore moves from cognitive prostheses to an executable intermediate language.

SLEnglish is not meant to replace ordinary English. It is a compiled semantic layer. An unrestricted sentence remains the human source; an LLM proposes SLEnglish; deterministic tools parse, type, validate, reason, and execute it; and a renderer can produce prose again. The original text, the compiled representation, and the proof or execution trace remain linked. This architecture changes what the target language should optimise. Because humans do not need to author every line unaided, naturalness can be traded for regularity and semantic visibility.

The history of artificial language contains a recurring seduction. If ordinary languages are irregular, ambiguous, culturally partial, and metaphysically confused, perhaps a rational designer can replace them with a transparent system. John Wilkins classified the universe into a hierarchy intended to support a “real character.” Leibniz imagined a *characteristica universalis* in which disputes might be converted into calculation. Later projects pursued logical unambiguity, international communication, feminist expression, minimalism, artistic intensity, or extreme semantic compression. Loglan and Lojban explicitly connected engineered language with the possibility of testing linguistic relativity; Lojban developed a machine-readable formal grammar. Ithkuil explored an extraordinary density of grammatical distinctions. Láadan made evidential and experiential categories central to a language designed from feminist concerns.

These projects are intellectually valuable even when they fail as mass communication. They expose a fundamental choice: should language mirror a pre-existing catalogue of reality, or should it help users act within an open and contested world?

The failure of the perfect taxonomy. Seventeenth-century philosophical languages often assumed that concepts could be placed in one rational tree. The taxonomic ambition produced ingenuity and brittleness. A position in the vocabulary encoded a theory of what kind of thing an entity was. When the theory changed, the name became misleading. Borges’s fictional Chinese encyclopedia remains funny because every classification can be made to look absurd from outside its purpose.

Modern knowledge makes the problem worse. The same object participates in many ontologies. A virus is a molecular structure, evolutionary lineage, pathogen, information carrier, legal object, security concern, and ecological actor. No single hierarchical address captures all legitimate relations. A universal language that fixes one place for the virus would confuse consistency with truth.

Jaap Maat’s historical study of Dalgarno, Wilkins, and Leibniz shows that their projects were more diverse and context-sensitive than caricature suggests (2004). Yet the central lesson remains: a language for better thought should not pretend to finish ontology. It should make perspectives, criteria, and modelling choices explicit.

This is why the present project uses operators rather than a complete world dictionary. A legal boundary can coexist with a statistical boundary, and process continuity can coexist with continuity of role. The system makes the relevant axis visible without declaring one axis universally fundamental.

Conceptual engineering, not word worship. Contemporary philosophy calls the deliberate assessment and improvement of concepts conceptual engineering. Herman Cappelen’s *Fixing Language* examines how words and representational devices can be defective and how difficult intentional meaning change can be (2018). The volume edited by Burgess, Cappelen, and Plunkett maps disputes about targets, methods, ethics, and implementation (2020). David Chalmers proposes a broad definition: conceptual engineering includes the design, implementation, and evaluation of concepts, both by repairing existing ones and creating new ones (2020).

This formulation improves the project in two ways. First, design without implementation is only private notation. A concept succeeds when a community can learn it, use it, negotiate its boundaries, and integrate it into institutions. Second, implementation without evaluation can make language worse. A new category may stigmatise, oversimplify, centralise power, or create false certainty.

Conceptual engineering is therefore closer to protocol design than dictionary writing. A protocol needs semantics, syntax, error handling, governance, interoperability, adoption incentives, and tests. Concepts also need deprecation. A term that once improved analysis may become misleading as reality changes.

From analysis to a compiled language. Vocabulary, annotations, structured schemas, and context-sensitive interpretation can be represented inside one stronger architecture. Ordinary prose remains the source layer. SLEngish is the compiled semantic layer. The practical task of the linguistic chapters is to decide which distinctions deserve first-class verbs, which belong as named parameters, which are type refinements, and which should remain optional contextual metadata. The appendices record these decisions as a revisable operator registry.

This unification matters. Without a canonical intermediate representation, every interface invents its own tags, every ontology binds them differently, and every reasoner requires an adapter. The project becomes a collection of demonstrations rather than an accumulating research programme. A stable target language lets us compare translators, circuits, reasoners, and user interfaces while holding the representation constant.

The language must nevertheless resist the ambition of a final ontology. A virus can be represented simultaneously as a molecular assembly, evolutionary lineage, pathogen, information carrier, legal object, and ecological actor. SLEngish does not decide which view is metaphysically privileged. It lets modules define perspectives and requires claims to identify the context, basis, or profile under which a relation is asserted. Formalisation should expose modelling choices, not conceal them behind a universal taxonomy.

Selective obligatoriness. Natural grammar teaches that obligation creates attention. Interface design teaches that too many mandatory fields create evasion, noise, and box-ticking. SLEngish therefore applies obligation by domain.

Scientific synthesis should require source mode, provenance depth, revision state, and uncertainty type. High-impact automated decisions should require identity of the evaluated system, affected parties, responsibility architecture, and appeal reversibility. Infrastructure proposals should require temporal horizon, threshold distance, externality vector, and option loss. Public polling claims should require population, sample, consensus topology, and scale.

Poetry should require none of these. Intimate conversation should not become an audit log. Even scientific discovery needs unstructured phases in which concepts are unstable. A marker can be mandatory at the transition from exploration to public claim without being mandatory during imagination.

The principle is:

Make distinctions obligatory at the boundary where private ambiguity becomes public consequence.

Design criteria. A candidate distinction should meet six criteria.

Term	Explanation
Error relevance	It addresses a recurring failure with meaningful cost, not merely an interesting philosophical nuance
Compression	It expresses a structure substantially more cheaply than ordinary paraphrase
Compositionality	It combines predictably with other terms and can be nested, compared, or negated
Auditability	Users can ask why a marker was chosen and what evidence would change it
Pluralism	It can state model dependence and contested criteria rather than freezing one worldview
Learnability	People can use it with limited training and without constant reference to a manual

A seventh criterion should be added for AI-mediated environments: machine tractability without machine sovereignty. The distinction should be representable in software, but the software should not silently settle normative choices. A model can prompt for boundary mode; it should not decide which people truly belong to a community.

AI as translator, critic, and danger. Artificial intelligence makes the project more feasible and more urgent. A language model can translate ordinary prose into a structured claim, ask for missing coordinates, preserve tags through summarisation, and render the result back into readable English. It can detect that “same system” shifts from legal to technical identity or that “we decided” hides an exclusive decision-maker.

The same model can also produce the illusion of precision. It may invent provenance, assign numerical confidence without calibration, or turn ambiguous social judgments into categorical fields. Structured language can launder bias more effectively than vague prose because schemas appear objective.

AI should therefore act in three distinct roles. As translator, it proposes structure while preserving the source text. As critic, it identifies omitted distinctions and alternative interpretations. As executor, it may apply validated rules only when governance authorises them. The trace must distinguish the model’s extraction from the human’s endorsement.

This architecture directly uses the perspective distinctions developed above:

An AI-assisted text may have a human author, a structure proposed by a model, a verification supplied by a reviewer, and policy effects produced by the surrounding system. Those roles should remain distinct.

Without such separation, AI-generated conceptual engineering becomes conceptual colonisation by whichever model and dataset dominate the interface.

Why words alone fail. Many reformers assume that once a term exists, cognition improves. History says otherwise. Technical concepts become slogans; precise distinctions blur in popular use; institutions appropriate critical vocabulary without changing practice. Sustainability, resilience, AI, innovation, and evidence-based demonstrate semantic inflation.

A concept survives only if practices reinforce it. Probability becomes useful through scoring rules, forecasting records, and feedback. Provenance becomes useful through citation systems, identifiers, and audits. Reversibility becomes useful through staged deployment and actual rollback mechanisms. A new word without an operational practice can become another layer of rhetoric.

SLEngish therefore links each term to an action. A source marker links to evidence. A revision marker links to version history. A Goodhart marker links to metric audit. A responsibility vector links to control and remediation. The language is a front end for institutions of correction.

Cultural humility and the danger of extraction. A book that harvests grammatical features from minority and Indigenous languages risks reproducing the very Anglocentrism it criticises. It can treat living languages as warehouses of clever devices, detach categories from communities, or imply that an English-speaking designer has finally recognised their value.

The response must be more than a disclaimer. Examples should be cited to grammars and scholars, checked with speakers where possible, and presented with uncertainty. Communities should share authority and benefit when their linguistic knowledge becomes part of a product or research programme. Natural categories should not be copied under the assumption that a brief English label captures their meaning. Often the correct lesson is structural—“source can be obligatory”—rather than lexical—“borrow this morpheme.”

Nor should languages be ranked by depth. Every language enables subtle thought and every language has habitual omissions. English is criticised here because it has unusual global power, not because it is intrinsically primitive.

A language that can admit its own defects. The final design criterion is reflexivity. SLEngish must be able to say that its own categories are provisional, contested, scale-dependent, and shaped by purpose. Each schema needs a revision state and boundary grade. Each formalisation should preserve a link to unstructured language and dissenting interpretations.

The ideal cognitive prosthesis is not a cage made of better concepts. It is an interface that makes its handles visible.

Three grades of executability. The phrase needs separate meanings because parsing, reasoning, and authorised external action impose different guarantees.

Grade	Meaning
Structural executability	A statement can be parsed, typed, canonicalised, checked against a schema, compared across versions, and linked to its source.
Epistemic executability	A system can ask questions, derive consequences, detect conflicts, trace provenance, and explain answers under a declared reasoning profile.
Operational executability	Validated semantic nodes may invoke tools, workflows, simulations, agents, or laboratory procedures only through explicit authority and policy.

The proposed language primarily targets structural and epistemic executability. Operational effects should remain outside the language kernel and be mediated by declared interpreters and governance policies. This separation prevents a claim such as “the sample should be heated” from silently becoming a command to heat it. A semantic proposition may describe, prescribe, hypothesise, or query an action; only an authorised runtime may turn the proposition into an external effect.

The distinction also clarifies safety. Deterministic syntax does not make an action safe. Typed roles do not establish that a source is trustworthy. A proof establishes derivability under premises, not truth of the premises. Executability increases power and auditability simultaneously; governance determines whether the power should be exercised.

Term	Explanation
Linguistic relativity	The defensible thesis concerns salience, habitual attention, compression, and obligatory marking rather than cognitive imprisonment (Lucy 1992a, 1997; Wolff and Holmes 2011; Everett 2013).
Typological diversity	WALS and Grambank document broad structural diversity while warning against treating English categories as universal defaults (Dryer and Haspelmath 2013; Skirgård et al. 2023).
Space and time	Cross-linguistic work on spatial frames and temporal metaphors shows measurable diversity with careful limits on causal interpretation (Levinson 2003; Núñez and Sweetser 2006; Haun et al. 2011).

Term	Explanation
Evidentiality	Grammatical source marking is widespread, but experimental evidence for broad cognitive improvement remains mixed and domain-dependent (Aikhenvald 2004; de Haan 2013a, 2013b; Kandemirci et al. 2023).

The survey is preliminary to the operator inventories rather than an independent catalogue of rare forms. Its engineering use is to show that recurrent semantic distinctions can be conventionalised, compressed, and made difficult to omit. The appendices translate that observation into candidate SLEngish contracts for evidence, perspective, participant structure, time, space, possession, identity, discourse, and social relation.

The mapping is deliberately non-literal. A natural grammatical category may combine several functions, vary across languages, or depend on discourse practice. SLEngish operators are analytical constructions with declared parameters and tests; they do not claim to reproduce a language-specific morpheme exhaustively. Specialist review remains necessary, and a category should be retained only when it improves translation, reasoning, or audit in the intended domain.

CHAPTER 4

Identity, Collective Agency, and Systemic Concepts

Qualified Identity, Process, Boundaries, and Infinity

At noon, a cloud hangs above the lake. At 12:05 it has moved, expanded, lost water, gained water, split at one edge, and merged with vapour at another. A child points and says, “That cloud is getting darker.” The sentence presupposes that the subject persists through change. Without that presupposition, there is no it to get darker. Yet no sharp boundary tells us which droplets compose the cloud, and no material core remains fixed.

Philosophers use clouds to stage the Problem of the Many. Around the vague object are many overlapping aggregates of droplets, each nearly as good a candidate as the others for being the cloud. If one aggregate is the cloud, what disqualifies its almost-identical neighbours? If many aggregates are clouds, why does perception present one? If the boundary is indeterminate, what happens to ordinary identity claims? The problem is not a linguistic trick, but language helps us glide over it. English supplies a count noun, a singular demonstrative, and the relation same. The grammatical package gives a crisp handle to a diffuse process.

The handle is useful. Human action would stall if every reference required a metaphysical inquiry. But when the same linguistic machinery is carried from clouds to organisms, software, firms, states, and artificial agents, useful compression becomes conceptual debt. The word same answers too many questions at once.

Seven continuities hiding inside one word. Consider seven relations that English routinely calls identity.

Continuity criterion	Explanation
Material continuity	asks whether roughly the same matter persists. A repaired stone wall may remain materially continuous; a flame does not. Even material continuity comes in degrees because matter is lost and replaced
Structural continuity	asks whether the organisation or pattern persists. A whirlpool can maintain a form while water passes through it. A company may preserve its reporting structure while every employee changes
Process continuity	asks whether later stages belong to one uninterrupted activity. A flame persists as combustion, a conversation as reciprocal interaction, and an organism as a network of self-maintaining processes
Causal continuity	asks whether each stage arises through the appropriate causal path from its predecessor. A perfect replica assembled elsewhere may match structurally while lacking the same causal history
Genealogical continuity	asks whether one entity descends from another through copying, reproduction, transformation, or branching. This is central to organisms, texts, software repositories, machine-learning models, and institutions
Role continuity	asks whether a socially defined position persists. “The president,” “the court,” and “the company” may remain the same role-bearing entity despite replacement of persons, rules, and material infrastructure
Informational continuity	asks whether an abstract content is preserved. Two files can instantiate the same bit sequence on different media; two models can implement the same function with different numerical representations; two editions can preserve a text while changing typography

A final variable cuts across all seven: scale. At molecular scale, a body is never the same from moment to moment. At the scale of legal responsibility, it may be continuous for a lifetime. Identity is often not a property simpliciter but a relation indexed to purpose, granularity, and tolerated variation.

English can express every item in this list. The problem is that it normally does not. “Is this the same model?” invites a binary answer where the real question may be:

same(model₁, model₂ | weights, architecture, behaviour, lineage, governance, scale)

Without an identity parameter, disagreements can remain invisible. One engineer means identical weights; another means equivalent benchmark behaviour; a regulator means the same accountable system; a user means the service with the same name. All can sincerely answer different questions with the word same.

Things may be stabilised processes. Western metaphysics is often narrated as a conflict between substance and process, with Heraclitus and Whitehead representing processual alternatives to a world of enduring things. The historical picture is more complex, and no natural language maps neatly onto a philosophical doctrine. Still, contemporary philosophy of biology gives a rigorous reason to revisit the noun-centred picture.

In *Everything Flows*, Nicholson and Dupré argue that living systems are better understood as interdependent processes stabilised at different timescales than as substances that merely undergo change (2018). Metabolism, development, immune regulation, symbiosis, and ecological interaction are not optional activities performed by a pre-existing organism. They partly constitute the organism’s persistence. DiFrisco examines criteria by which biological processes can be identified through variation, and Pradeu develops the idea of genidentity—continuity across a series of states—particularly for cases involving metamorphosis, fusion, or fission.

The process view does not abolish objects. It asks what work object language is doing. A rock may be treated as a relatively slow process; a storm as a fast, open one; an organism as a self-producing network; an institution as a norm-governed pattern maintained through documents, roles, memory, and enforcement. The difference lies partly in characteristic timescale and maintenance regime.

This suggests a replacement for the question “What is X?” when X is dynamic:

What must continue happening, at what scale, and within what tolerance, for later stages to count as the continuation of X?

For a cloud, the answer may involve atmospheric dynamics and spatiotemporal continuity. For a river, channel, flow regime, watershed, name, and legal designation compete. For a person, bodily, psychological, narrative, relational, and legal continuities overlap without perfectly coinciding. For a company, control, contracts, brand, assets, organisational memory, and decision procedures matter differently to different observers.

Identity in software and AI. Digital systems reveal the inadequacy of ordinary identity with unusual clarity because copies can be exact while histories diverge.

Suppose model A is copied byte-for-byte to create model B. At the moment of copying, A and B are informationally identical but numerically distinct deployments. If B is fine-tuned, it remains genealogically descended from A but is no longer weight-identical. If A is quantised, architecture and intended function may remain while numerical representation changes. If a serving layer adds retrieval, tools, memory, and policy, users may encounter a system whose behaviour differs substantially despite the same base model. If two companies independently train models that converge on similar behaviour, they may be functionally similar without shared lineage.

Blockchain systems add governance to the puzzle. After a fork, which branch is the original? The answer may depend on state continuity, community recognition, economic weight, protocol rules, name, or normative interpretation. “Original” is not a purely technical property once lineage branches.

The emerging legal and scientific vocabulary of AI will fail if it treats “a model” as a simple object. Audits, licences, safety evaluations, and responsibility need identity criteria. A benchmark result attached to one checkpoint does not automatically transfer to a fine-tuned or tool-augmented descendant. A risk assessment of a service cannot be inherited merely because the marketing name remains stable. The grammar of identity is infrastructure for governance.

The proposed identity operators. Rather than invent unrelated words, SLEngish can begin with compositional operators:

Term	Explanation
matter-same	materially continuous within a stated tolerance
form-same	structurally or organisationally continuous
flow-same	one continuing process
line-same	causally or genealogically continuous
role-same	continuous under a social, institutional, or legal role
info-same	informationally equivalent under a declared representation
effect-same	functionally or behaviourally equivalent for a specified test domain

The terms are deliberately plain. Their value lies not in elegance but in preventing a binary predicate from swallowing its parameter. A claim such as “The upgraded system is line-same and role-same, but not info-same or effect-same” is immediately more informative than “It is the same system.”

These operators require two additional annotations. Tolerance states how much change is permitted. Scale states the level at which comparison is made. The full form becomes:

At the scale of the storm system, two observations may count as the same continuing flow when the interruption is shorter than ten minutes.

or

Two models may count as effect-equivalent for diagnostic triage among European adults when their relevant outcome difference remains within a two-percent tolerance.

This may look technical, but ordinary language already embeds hidden parameters. “The same temperature” means within measurement precision; “the same route” tolerates small deviations; “the same story” tolerates wording changes. Explicit notation is needed only when hidden tolerances matter.

Gradual identity and the tyranny of binary classification. Some continuities are not yes-or-no. A language, city, ecosystem, or institution can change gradually until the point at which observers cease to treat it as the same. There may be no unique threshold. Binary identity is often a decision imposed for law, record keeping, or action, not a discovered boundary.

A boundary-grade marker would distinguish:

Term	Explanation
sharp boundary	sharp boundary fixed by rule
operational boundary fixed	operational boundary fixed for a purpose
fuzzy boundary	fuzzy boundary with central and peripheral cases
contested boundary	contested boundary with rival criteria
indeterminate boundary where	indeterminate boundary where available facts do not settle membership

The distinction matters for clouds, species, diseases, political movements, markets, and AI capabilities. Calling a model “autonomous,” a conflict “war,” or an organisation “terrorist” can activate legal and moral regimes. The world may not provide a sharp natural seam; institutions nevertheless need a rule. A boundary-grade marker would make the imposed character of the rule visible without making action impossible.

Identity as a vector, not a verdict. The strongest proposal of this chapter is that identity should often be represented as a vector of continuities rather than a single verdict. For entities A and B:

A comparison between two entities should distinguish material, structural, processual, causal, genealogical, role-based, informational, and functional continuity rather than compressing them into one relation.

Each component can be binary, graded, or unresolved. A legal system may then define which components are decisive for a particular purpose. A biologist, engineer, historian, and regulator may use different projections of the same vector without pretending that one possesses the metaphysically complete answer.

Would such a vocabulary change thought? The hypothesis is testable. Participants could evaluate Ship-of-Theseus cases, corporate transformations, software forks, and biological metamorphoses. One group would use ordinary same; another would learn the identity operators. Researchers could measure consistency, transfer across domains, detection of hidden disagreement, and calibration to stated criteria. The expected benefit is not a single correct metaphysics. It is a lower rate of unnoticed equivocation.

A cloud does not need a passport. But the institutions, models, and collective systems we are building increasingly do. Before issuing one, we should decide what kind of sameness it certifies.

Numbers feel less like words than like features of reality. Three stones remain three stones whether one says three, trois, tres, or nothing. Yet the perceptual world does not deliver exact symbolic number in the form mathematics requires. It offers small quantities that can be recognised rapidly, larger magnitudes that can be estimated, and collections that can be matched one-to-one. A conventional counting system turns these capacities into an indefinitely extendable technology.

This distinction is visible in research with communities whose languages lack extensive exact number vocabularies. Pirahã speakers can compare quantities and perform direct matching, but exact performance

declines when objects must be remembered or transformed rather than aligned one-to-one (Gordon 2004; Frank et al. 2008). Mundurukú speakers show sophisticated approximate arithmetic despite a restricted exact number lexicon (Pica et al. 2004). Later work argues that exact concepts are constrained by the range of available count words and practices (Pitt, Gibson, and Piantadosi 2022).

These findings have provoked methodological and interpretive debate, as they should. Language, schooling, notation, material practices, and task familiarity are entangled. The safe conclusion is not that people without a long count list “cannot think number.” It is that exact cardinal symbols help preserve and transform quantity across time, memory, and operation. Number words are not mere labels attached to a complete prelinguistic arithmetic. They are parts of a cognitive machine.

When counting conceals the unit. Modern societies count compulsively: users, jobs, papers, deaths, incidents, companies, species, models, and votes. The arithmetic can be exact while the ontology is unstable. What is one active user? One disease case? One AI model? One conflict? One scientific contribution? An exact number attached to an underdefined unit creates the appearance of precision.

Numerical-classifier systems remind us that counting can require a category mediating between numeral and noun. Their conventional classes are not solutions to contemporary measurement, but the grammatical principle is powerful: one must state what kind of one is being counted.

A designed count classifier could mark the unit principle:

Term	Explanation
Bounded physical object	a unit individuated primarily by a physical boundary
Individuated event	a unit defined as one occurrence under an event-counting rule
Continuing process	a unit defined by the persistence of an organised dynamic rather than fixed matter
Legal unit	an entity whose countable identity is fixed by law, contract, or registry
Lineage unit	a genealogical continuation that may survive copies, branches, or versions
Deployment unit	one operationally instantiated system in a particular environment
Statistical cluster	a grouping whose unity depends on a stated method and parameterisation

Thus “twenty models” becomes “twenty deployments of four lineages” or “twenty weight files from one architecture.” The number becomes more complex and more honest.

Zero and the invention of absence. Zero is often described as a discovery, an invention, or both. The physical world contains empty containers and missing expected objects, but the mathematical zero is not identical to any particular absence. It functions as a number, a placeholder, an identity element, a coordinate, and a boundary between signs. Stable notation allowed absence to participate in calculation.

Ordinary English still handles absence badly. “There is no evidence” can mean that nobody looked, that a search found nothing, that available methods cannot detect it, that evidence once existed but was lost, or that the hypothesis predicts no observable trace. “The feature is absent” may mean nonexistence, non-detection, deactivation, irrelevance, or failure of a classification rule.

We need at least four negative states:

Term	Explanation
empty	a defined container has zero members
missing	an expected item is not present
undetected	a search did not find an item within known sensitivity
nonexistent-under-model	the operative ontology contains no such item

A fifth, uninstantiated, applies to a possible type with no current token. A sixth, inapplicable, says that the question itself does not fit the object. Databases often collapse all six into null. Natural language often does the same.

The several infinities hidden in infinity. The word infinite performs even more work than same. It can mean that a process has no last step, that a set is treated as a completed totality, that a magnitude is unbounded, that

a series diverges, that a mathematical object has transfinite cardinality, that resolution continues without practical limit, or merely that something is unimaginably large.

The historical distinction between potential infinity and actual infinity remains useful. Potential infinity characterises a process that can always be continued: for every number, another can be produced. Actual infinity treats an infinite collection or structure as complete enough to quantify over mathematically. Modern set theory formalises actual infinite sets; Cantor's work distinguishes different infinite cardinalities. Analysis uses limits to reason rigorously about unbounded processes and arbitrarily fine variation.

The claim that infinity has “no counterpart anywhere in the real world” is too strong. Whether physical space, time, divisibility, field values, or the cosmos instantiate any form of actual infinity is a live question in physics and metaphysics. Some theories contain infinities; some physicists treat singularities as signs of theoretical breakdown; cosmological observations do not settle whether the universe is spatially finite. The responsible claim is narrower: human beings do not encounter a completed infinite collection through ordinary perception, yet formal infinity has transformed mathematics, theology, cosmology, economics, and culture. The Stanford Encyclopedia's recent survey rightly treats mathematical, physical, cosmological, and theological infinity as related but distinct problems (Easwaran et al. 2025).

A better ordinary vocabulary should separate:

Term	Explanation
endless	no terminal stage is specified
unbounded	no finite upper bound exists in the model
divergent	a sequence or process fails to converge in a specified way
actual-infinite	a completed mathematical totality governed by an axiom system
transfinite	belonging to formal orders of infinite size
practically inexhaustible	finite or unknown in extent but impossible to enumerate with available resources
outfinit	open-ended beyond every current operational boundary while every realised stage remains finite

The last term is a deliberate conceptual invention. Outfinit is not a substitute for mathematical infinity. It names systems whose continuation has no relevant designed endpoint, although every physical or computational realisation is bounded. An expanding research programme, evolutionary process, generative search, or indefinitely extensible proof practice can be outfinit without containing a completed infinite object. The term protects us from two errors: treating the unfinishable as literally infinite, and treating finite implementation as evidence of a final conceptual limit.

Infinity as cultural operator. Infinity does not remain inside mathematics. Religions use it for divine attributes and eternity. Economics invokes indefinite growth. Technology markets “unlimited” storage or scale while relying on finite infrastructures. Political narratives imagine permanent institutions and inexhaustible progress. Personal identity is sometimes invested in digital traces expected to persist without horizon.

These transfers are rarely innocent. A model that assumes infinite substitutability of resources can erase ecological thresholds. A cultural image of endless growth makes saturation or sufficiency appear as failure. Conversely, an overreaction against infinity can cripple abstraction: calculus, probability, topology, and modern physics rely on disciplined idealisation.

The relevant distinction is between formal permission and ontological commitment. One may quantify over infinite structures inside a mathematical theory without claiming that the universe contains a corresponding completed object. One may use an infinite-horizon optimisation while recognising that institutions and agents are finite. A grammar for infinity should mark the mode:

The vocabulary should distinguish formal infinity, infinity as a physical hypothesis, procedural endlessness, and operational outfinit.

Much philosophical confusion occurs when a property moves silently from one mode to another.

Negative, imaginary, and probabilistic objects. Negative numbers once appeared less natural than positive counts; imaginary numbers carried their scepticism in their name. Their power came from rules of operation and integration into systems, not from resemblance to visible objects. Probability similarly converts uncertainty into quantities that can be combined, updated, scored, and compared—imperfectly, but far beyond vague confidence words.

These cases support the book's constructive thesis. A useful concept need not correspond to a simple object. It must provide stable distinctions, compositional operations, inferential discipline, and practical gain. The goal for collective agency or identity is not to find a hidden "thing" matching the term. It is to create representations that support better reasoning.

The analogy has limits. Mathematical symbols work in highly controlled systems; social and metaphysical concepts are open-textured and norm-laden. A responsibility vector cannot attain the universality of arithmetic. But it can still outperform a single vague noun, just as a multidimensional medical profile can improve on the binary "healthy/sick" without becoming pure mathematics.

What else is hidden by exactness. Modern metrics often combine exact numbers with uncertain constructs. A university ranks seventh; a model scores 91.3; a country has a freedom score of 78; a worker has a productivity index. The precision belongs to the calculation, not necessarily to the concept. Goodhart's law captures one failure mode: when a measure becomes a target, optimisation can detach it from the underlying goal.

A metric-mode marker should therefore distinguish measurement, proxy, target, and constitutive rule. Temperature is measured; citation count is a proxy for influence; a benchmark becomes a target when developers optimise against it; a legal threshold can constitute eligibility rather than merely detect it. The same number behaves differently in each mode.

Counting is among humanity's greatest cognitive achievements. Its next refinement may be learning when the unit, boundary, and mode of infinity are doing more philosophical work than the numeral.

Collective Agency, Causation, and Distributed Responsibility

Ten thousand people stand in a railway station. Ten thousand people march toward parliament. Ten thousand birds change direction in a wave. Ten thousand employees contribute to a corporation. Ten thousand citizens are sampled in an opinion poll. English can call each collection a group. The noun performs useful compression, but it erases the very property that makes collective phenomena difficult: the relation between members and the whole.

A crowd can act without deciding. An organisation can decide without every member agreeing. A market can produce a price nobody intended. A state can persist while governments and citizens change. A swarm can display coordinated motion without central control. A committee can issue a judgment that no member would have written alone. When we say "the group did X," we may attribute statistical regularity, coordinated behaviour, authorised decision, emergent effect, or legal fiction. These are different ontologies of the plural.

From aggregate to agent. A useful scale begins with the aggregate: entities grouped by an observer but not coordinated with one another. "People wearing red" may be a category with no shared awareness. A crowd shares location and may respond to common signals. A network has patterned relations through which information or influence travels. A swarm achieves distributed coordination through local interactions. An organisation has roles, membership rules, decision procedures, and persistent records. An institution adds normative authority, public recognition, and mechanisms that outlast particular occupants.

These forms can overlap. A corporation is an organisation, a legal person, a network of contracts, a set of buildings and systems, a brand, and a coalition of human interests. A nation is population, territory, historical narrative, legal order, administrative apparatus, imagined community, and external recognition. A scientific field is a network of researchers, journals, methods, prestige relations, and shared objects of inquiry. Treating any of these as a simple individual creates both analytical power and moral hazard.

The philosophy of collective agency asks when a group can literally bear beliefs, intentions, and responsibility. Christian List and Philip Pettit argue that suitably organised groups can qualify as rational agents when they possess decision procedures that integrate member inputs into coherent group-level attitudes (2011). Group judgments can be systematically related to yet not reducible to a simple majority of individual judgments. Raimo Tuomela distinguishes acting in an individual "I-mode" from a "we-mode" in which participants understand themselves as members committed to a group reason or ethos (2013). More recent work emphasises roles, membership rules, and procedures that explain how collectives persist and exercise causal powers despite turnover.

One need not accept every philosophical criterion to see the linguistic problem. English moves too easily from “members of the organisation produced an outcome” to “the organisation wanted the outcome.” The singular verb invites a mind where there may be a procedure, an incentive system, or an emergent pattern.

The many meanings of “France wants”. Consider the sentence “France wants stricter regulation.” It could report:

Term	Explanation
a policy adopted	a policy adopted through authorised government procedure
the preference of	the preference of the current executive
a parliamentary	a parliamentary majority
a polling	a polling majority among residents
a dominant	a dominant media narrative
a strategic	a strategic position attributed by foreign analysts
an outcome	an outcome produced by institutions without majority support

The sentence may be acceptable shorthand, but it is not ontologically innocent. It creates a unitary subject, suppresses internal dissent, and transfers the phenomenology of individual wanting to a complex polity. Similar problems occur in “the market believes,” “science says,” “the company decided,” and “society rejected.”

A collective-mode marker would specify what licenses the singular attribution:

Term	Explanation
Authorised state decision	A policy was adopted through a recognised governmental procedure.
Survey majority	A measured majority of respondents expressed a preference; this is not automatically a state decision.
Institutional output	An output is attributed to the state through a named institutional procedure.
Analytical attribution	An external analyst attributes a strategic objective under an explicit interpretation.

The marker does not forbid metonymy. It audits it.

Clusivity as a primitive of political honesty. Inclusive and exclusive “we” is the natural linguistic starting point. English speakers routinely use we to manufacture solidarity or diffuse responsibility. A leader may say “we made mistakes” when decisions were concentrated among a few, or “we must adapt” when costs fall on people outside the decision process. A researcher may write “we know” to mean “a specialised literature currently supports.” A platform may say “we value privacy” while no coherent human subject corresponds to the pronoun.

Natural clusivity distinguishes whether the addressee is included. A modern extension should also distinguish participation, authorisation, and burden:

Term	Explanation
we-including-you	we-including-you as participant
we-speaking-on-your-authority	The speaker claims authority to speak for the addressee or included group; the mandate must be declared.
we-sharing-the-consequences	The participant set is defined by shared exposure to consequences rather than by shared decision authority.
we-excluding-you but acting upon you	The speaker excludes the addressee from the deciding group while including the addressee among those affected by its action.

These distinctions would not make rhetoric honest by force. They would make some evasions syntactically visible. A political sentence that says “we-exclusive-decider require sacrifices from you-inclusive-affected” cannot borrow the warmth of collective inclusion quite so cheaply.

Agency architecture. The next distinction concerns how collective action is produced. Four architectures recur:

Term	Explanation
Centralised agency	an authorised centre selects and executes a decision. A chief executive signs, a court rules, a commander orders

Term	Explanation
Distributed coordinated agency	participants follow shared protocols or reciprocal expectations without one complete controller. Open-source projects and federated networks often approximate this form
Emergent outcome	local actions produce a macro-pattern no participant selected. Congestion, bank runs, segregation, and many market dynamics belong here
Statistical attribution	a claim describes a distribution of individual states rather than a collective decision—"consumers prefer," "voters fear," "scientists believe."

A fifth category, legal-person agency, records that law attributes acts, rights, and duties to an entity. Legal agency may track a real decision architecture or may function as a practical fiction that bundles responsibility.

The grammatical problem appears when the same active verb applies to all five. "The platform discriminated" may identify a centrally chosen policy, an emergent effect of optimisation, a data pattern, or a legal attribution. Responsibility depends partly on which architecture generated the harm.

Distributed responsibility without diluted responsibility. Complex systems create a familiar paradox: many people contribute to an outcome, yet no individual contribution seems sufficient. Ordinary language oscillates between scapegoating one visible actor and declaring that "the system" is responsible, which can leave nobody accountable.

A better grammar separates at least four relations:

Term	Explanation
causal contribution	the action changed the probability or form of the outcome
control	the actor could alter the process or stop it
knowledge	the actor knew or should have known the relevant risk
role obligation	the actor had a duty to monitor, disclose, or intervene

Responsibility is not identical to any single coordinate. A low-level engineer may make a causal contribution with little control over deployment. An executive may never touch the system yet possess control and role obligation. A regulator may lack direct causation but have failed a duty to act. A user may participate in an emergent pattern without meaningful alternatives.

SEnglish therefore proposes a responsibility vector, not a single blame marker:

Responsibility for an outcome should be represented through separate dimensions of contribution, control, knowledge, duty, benefit, and vulnerability.

Again, this is not intended for every sentence. It is a schema for incident reports, audits, legal reasoning, and institutional design. Its purpose is to prevent distributed causation from becoming distributed invisibility.

The persistence of institutions. Institutions survive member replacement through records, rules, routines, assets, symbols, recognition, and enforcement. Their identity is neither purely material nor merely fictional. A university can persist while its faculty, students, buildings, curriculum, and charter change, but at some point continuity becomes disputed. A company can be acquired, renamed, emptied, and repurposed while remaining legally continuous. A state can undergo revolution while claiming inherited debts and treaties.

Institutional persistence therefore combines the identity distinctions developed earlier in this chapter with an explicit account of collective mode. A robust description should answer:

Term	Explanation
Which decision	Which decision procedure persists?
Which obligations and	Which obligations and claims are inherited?
Which memories	Which memories are preserved, and by whom?
Which members	Which members recognise the continuity?
Which external	Which external institutions recognise it?
Who benefits from	Who benefits from claiming sameness, and who bears inherited liability?

The phrase “the same institution” is often a strategic achievement. Continuity may be invoked to preserve property and authority, while discontinuity is invoked to evade responsibility. A grammar that marks role-same and line-same separately would expose such selective inheritance.

Swarms, AI teams, and synthetic collectives. Collective ontology becomes more urgent as artificial agents participate in organisations. A multi-agent system can divide tasks, critique outputs, maintain shared state, and select actions. Is it one agent, a team, or an orchestration layer? If an AI system acts through dozens of transient model calls, retrieved tools, and human approvals, where does agency reside?

The wrong answer is to copy human singulars. Calling the system “an AI” may hide the architecture; calling every component an independent agent may exaggerate autonomy. The relevant unit depends on memory, coordination, boundary, goal structure, and control. A collective-mode grammar could mark:

Term	Explanation
Central orchestration	One control centre coordinates the system and can be treated as the principal unit of agency.
Distributed consensus	Agency is distributed across components governed by a consensus mechanism.
Human-authorised chain	Several AI components act through a chain whose authority remains explicitly human.
Emergent coordination	System-level behaviour emerges without one controlling component.

Such markers would improve not only philosophy but engineering. Failures could be assigned to coordination protocols, shared knowledge, role definitions, or control points rather than attributed vaguely to “the model.”

Collective identity as a measurable profile. The earlier brainstorming proposed a scale based on coordination, memory, boundary, goals, decision, and persistence. It is worth retaining in a refined form. For a collective G:

The relevant profile of a collective includes its boundary, coordination, memory, integration of goals, decision procedure, persistence, and accountability.

The profile does not generate a metaphysical threshold above which a group becomes “really” an agent. It supports comparison. A flash mob has coordination but little memory or persistence. A market has persistence and distributed information but weak integrated intention. A corporation has boundaries, records, decisions, and legal accountability, though its goals may be internally contested. A nation has extraordinary persistence and symbolic memory but highly plural agency.

This profile could be used experimentally. Participants could receive descriptions of groups and assign mental-state verbs. One group would use ordinary English; another would use collective-mode tags. Researchers could test whether the tags reduce over-attribution of unified beliefs, improve responsibility allocation, or make internal dissent more salient.

The ultimate purpose is not to ban sentences such as “the country decided.” It is to make singular collective language answerable to the architecture underneath. Many can become one—but not always in the same way.

A child knocks over a glass. A gust knocks it over. A poorly designed table makes it unstable. A manager ignored repeated warnings. A procurement rule favoured the cheapest surface. An optimisation system rewarded speed over safety. Which of these caused the spill?

Natural language answers by selecting a subject and a verb. “The child broke the glass.” “The wind broke the glass.” “The design caused the accident.” Grammar packages a causal field into an event with an agent, an action, and a patient. This compression is indispensable for action and memory. It is also one of the main ways language assigns blame.

Cross-linguistic work suggests that preferred descriptions of accidental events can influence memory for agents. English speakers often use agentive constructions even for accidents—“She broke the vase”—where Spanish or Japanese speakers may prefer less agentive forms roughly corresponding to “The vase broke.” In experiments, linguistic framing affected memory for who was involved, and habitual language differences correlated with agent memory for accidental but not intentional events (Fausey et al. 2010; Fausey and Boroditsky 2011). These studies concern specific tasks and languages, not a universal moral psychology. They nevertheless show that syntax can guide attention toward or away from an agent.

Cause is a role in an explanation. Philosophy offers many accounts of causation: regularity, counterfactual dependence, intervention, mechanism, probability raising, process transmission, and more. Ordinary speech uses cause pragmatically. We select the factor that is abnormal, controllable, morally relevant, or useful for intervention. Oxygen is causally necessary for a fire but is rarely named the cause; a spark is selected because it differentiates this fire from normal conditions.

Complex systems defeat this habit because no single factor plays every causal role. A harmful algorithmic decision may involve:

Term	Explanation
Trigger	the immediate input or threshold crossing selected as the proximate cause
Enabling condition	infrastructure or background conditions that made the event possible
Mechanism	the process that translates inputs or conditions into the observed outcome
Structural cause	an incentive, policy, or distribution that repeatedly produces the relevant risk
Maintenance cause	the factor that keeps a harmful regime stable after its initial emergence
Selection cause	the reason this design, institution, or behaviour persisted among alternatives

Term	Explanation
an absence cause	a safeguard that was not implemented

English can list these, but the unmarked sentence “the algorithm caused it” collapses the chain around the most visible technical component. The result is often a form of causal laundering: organisational choices become properties of software, while software’s real causal contribution is either mystified or denied.

A grammar of causal depth. SLEngish can introduce a causal-layer marker:

Term	Explanation
Trigger cause	the immediate event or threshold crossing that initiates the sequence
Enabling cause	the condition or infrastructure without which the outcome would not occur
Mechanistic cause	the process that carries the influence from source to outcome
Structural cause	the recurrent arrangement of incentives, rules, or resources that generates the risk
Maintenance cause	the factor that keeps a regime or process stable after it begins
Omission cause	the absent action whose performance was expected under a stated duty or counterfactual

A complete incident account need not identify only one. “The sensor error triggered the denial; the scoring mechanism propagated it; the procurement policy structurally favoured a system without review; the absent appeal path maintained the harm.” This language does not eliminate disagreement about causation. It prevents disagreement about different causal roles from masquerading as disagreement about one binary relation.

The marker also improves scientific explanation. A gene may be part of a mechanism, a developmental condition, or a population-level difference maker. A social variable may predict an outcome without identifying an intervention target. A model feature may be statistically important without carrying the causal pathway. Scientific prose often moves silently among association, prediction, explanation, and intervention.

A minimal claim should therefore distinguish:

Term	Explanation
associated-with	Records an association without asserting prediction, mechanism, or intervention effect.
predicts-under-model	Records predictive performance under a named model and evaluation context.
counterfactually-relevant	States that changing the factor changes the outcome under a declared counterfactual model.

Term	Explanation
mechanistically-linked	Records a specified mechanism connecting the factor to the outcome.
and intervention-supported	Explicitly records the state and intervention-supported.

These are not synonyms. A language that treats them as separate verbs would remove a large amount of manufactured certainty from research communication.

Intention is not one switch. English often contrasts intentional and accidental action. Real responsibility occupies a richer space. An outcome may be:

Term	Explanation
directly intended	The outcome was adopted as an explicit goal of the action.
used	used as a means to another goal
foreseen and	foreseen and accepted
recognised	recognised as a risk but discounted
foreseeable	foreseeable under a role obligation but not actually considered
genuinely unforeseeable	The outcome was not reasonably predictable under the actor's information, role, and available methods.
produced while	produced while the actor lacked control or awareness
contrary to intention	contrary to intention but recurrently generated by the actor's system

Legal traditions distinguish forms such as purpose, knowledge, recklessness, and negligence, but ordinary public language repeatedly collapses them. “They did not intend the harm” is used as though it answered whether they foresaw it, benefited from it, ignored warnings, or designed conditions that made it likely.

An intent–effect marker could encode the relation between an actor's plan and an outcome:

Term	Explanation
Intended effect	The outcome belongs to the agent's stated goal.
Instrumental effect	The outcome is accepted as a means to another goal.
Foreseen and accepted	The outcome is anticipated and accepted without being selected as the principal goal.
Known risk	The outcome is recognised as possible but is not selected as a goal.
Role-foreseeable effect	A competent holder of the relevant role should have anticipated the outcome.
Unforeseeable effect	The outcome falls outside the actor's declared reasonable predictive capacity.
Systemic effect	The system repeatedly produces the outcome despite the stated intentions of local participants.

The final category matters because institutions can repeatedly produce outcomes they officially oppose. Intention at the level of mission statements may diverge from incentive-compatible behaviour. A company can sincerely contain many people who do not intend discrimination while maintaining a system that predictably generates it. Grammar should not allow individual innocence to settle institutional responsibility.

Responsibility as control over alternatives. Responsibility becomes clearer when linked to available alternatives. Did the actor have the ability, authority, information, time, and resources to choose differently? Was refusal realistically possible? Could the harm be mitigated later? Did the actor create the constraint under which later actors operated?

This leads to counterfactual reach: the set of relevant outcomes an actor could have changed. A senior manager may have wide reach but little direct contact with the event. A frontline worker may have immediate contact but narrow reach. A software developer may change code but not deployment policy. A state may regulate a firm but not foresee a particular failure.

A responsibility language should record both actual contribution and counterfactual reach. Otherwise those near the event are blamed because they are visible, while those who designed the choice architecture remain grammatically distant.

Reversibility changes the moral tense. Actions differ not only in expected benefit and risk but in whether their consequences can be reversed. A reversible experiment permits learning. An irreversible intervention converts uncertainty into permanent exposure. English commonly says “we can change course later” without marking the time window, recovery cost, or whether some losses cannot be restored.

A reversibility grade should distinguish:

Term	Explanation
immediately reversible	immediately reversible at low cost
reversible	reversible within a finite window
technically reversible	technically reversible but socially or economically locked in
partially compensable	partially compensable but not restorable
irreversible	irreversible under known methods
reversibility unknown	Available evidence does not establish whether restoration, compensation, or rollback is possible.

The distinction interacts with uncertainty. A low-confidence decision may be rational when reversible and reckless when irreversible. A high expected value can be unacceptable if downside crosses a catastrophic threshold. Risk language that omits reversibility is incomplete.

Path dependence and lock-in. A system is path-dependent when its present state depends on the sequence through which it arrived, not only on current conditions. Lock-in is a stronger practical state in which alternatives may be better but switching costs, network effects, standards, contracts, skills, or political coalitions stabilise the incumbent arrangement.

English has the terms, but they remain specialist vocabulary. Everyday causal narratives favour current reasons: “we use the system because it is efficient.” Path dependence asks whether it appears efficient because infrastructure, training, compatibility, and expectations accumulated around it. The present can become evidence for a choice that the past manufactured.

A path marker would distinguish:

Term	Explanation
selected-now	The current configuration was selected through a present choice.
inherited-path	The configuration is inherited from earlier decisions and accumulated dependencies.
locked-by-cost	Transition is constrained by economic or operational switching cost.
locked-by-network	Transition is constrained by compatibility, coordination, or network effects.
locked-by-law	Transition is constrained by legal or regulatory commitments.
and locked-by-identity	Explicitly records the state and locked-by-identity.

The final form captures cases where communities resist change because the incumbent practice has become part of who they believe themselves to be. Such lock-in cannot be solved by proving technical inferiority.

Externalities and displaced grammar. An externality occurs when costs or benefits fall outside the decision-maker’s accounting boundary. Ordinary syntax mirrors the boundary. “The factory produced affordable goods” is grammatically complete even when pollution, health costs, and ecosystem damage appear in other sentences with other subjects. The language separates what the system couples.

An externality vector should mark direction:

Term	Explanation
Benefit recipient	The benefit accrues to the decision-maker.
Non-consenting cost	The cost is borne by neighbouring parties who did not consent.
Future-person cost	The cost is transferred to people who are not yet able to participate.
Publicly borne risk	A public institution bears or guarantees the downside risk.
Knowledge asymmetry	Relevant knowledge is distributed asymmetrically among affected parties.

This makes the distribution of agency and burden part of the proposition rather than an optional moral appendix. It also prevents aggregate welfare from hiding concentrated harm.

From blame to intervention. A richer causal grammar could become oppressive if used primarily to assign blame. Its better purpose is intervention. Different causal layers imply different remedies. Remove the trigger, redesign the mechanism, change the incentive, add a safeguard, alter the institutional memory, or redistribute control. A single causal noun encourages a single culprit; a causal map supports layered prevention.

The empirical question is whether such markers improve reasoning. Participants could analyse accidents, algorithmic harms, public-health failures, and environmental events. Researchers could measure whether causal-layer language increases identification of structural interventions without erasing individual responsibility. Legal and organisational teams could compare ordinary incident reports with tagged reports for actionability and disagreement detection.

Language should not decide guilt. It should make the architecture of action harder to falsify by omission.

Relations, Systems, and Candidate Conceptual Operators

A noun gives the impression that the world has already decided where one thing ends and another begins. Tree, company, disease, market, intelligence, community: each word packages a region of reality as a unit available for counting, predication, and possession. This capacity is a triumph of cognition. It is also a source of systematic error when relations and processes are treated as intrinsic properties of isolated things.

A person “has” a disease, a company “owns” data, a country “possesses” resources, a model “contains” bias, a community “shares” values. The grammar offers owner and owned, subject and property. Yet many of these phenomena are relational. Disease can depend on organism, environment, classification, and threshold. Data rights can involve subjects, collectors, custodians, processors, beneficiaries, and affected non-subjects. Bias can arise through interaction among objective, dataset, model, interface, deployment, and social feedback. Values can be contested practices rather than items stored in a collective mind.

Possession, constitution, and stewardship. Languages with possessive classes show that possession is not universally treated as one relation. Body parts and kin may receive different marking from transferable objects; food and drink may be classified by intended use or control; spatial and social relations may have dedicated constructions. The common label “alienable versus inalienable” captures only part of this diversity (Nichols and Bickel 2013).

Contemporary English needs at least these relations:

Term	Explanation
title	legally recognised ownership
control	practical ability to use, exclude, or alter
custody	temporary responsibility for keeping
stewardship	authority constrained by duties to others or the future
authorship	origination of an expression or design
provenance	causal contribution to an artefact
membership	belonging within a collective
constitution	being a part without which the whole is not the same kind of thing
dependency	requiring a relation for continued existence or function
exposure	bearing effects without ownership or control

“The company’s data” can then be unpacked. The company may hold custody and processing control; individuals may be the data subjects; a public institution may impose stewardship duties; several actors may contribute provenance; communities may bear exposure. Possessive apostrophe should not settle the politics.

Part, member, ingredient, and participant. English part of similarly combines different mereological relations. A wheel is a component of a bicycle. A person is a member of a committee. Water is an ingredient in soup. A scene is a segment of a film. A cell participates in a tissue. A subcontractor participates in a supply chain. A rule belongs to a legal system. These relations support different inferences.

If a component is removed, the machine may fail. If a member leaves, the organisation may persist. An ingredient loses some independent identity in a mixture. A participant may influence a process without being contained by it. A boundary dispute becomes confused when one form of parthood is substituted for another.

Collective identity requires especially careful distinctions between member, role-holder, stakeholder, affected party, beneficiary, and represented party. Someone can be affected without membership, represented without authorisation, and formally included without power. An institutional grammar that records only members will systematically miss its moral perimeter.

Boundaries are operations. The natural world contains gradients, overlaps, nested systems, and phase transitions. Human institutions need discrete decisions. We draw a line around a wetland, a disease category, a labour market, a species, or a conflict. The line can be scientifically useful and legally necessary even when no equally sharp boundary exists in the phenomenon.

The mistake is not drawing boundaries. It is forgetting the operation that drew them. A category should therefore carry a boundary mode:

natural-discontinuity, measurement-threshold, legal-rule, operational-convenience, statistical-cluster, self-identification, external-attribution, or contested.

This marker changes debate. Two researchers may agree on all observations but use different thresholds. Two communities may contest whether identity is self-ascribed or institutionally assigned. A regulator may intentionally create a legal boundary that does not claim to mirror a natural kind. Without a boundary mode, disagreements are misdiagnosed as factual when they are methodological or normative.

Emergence as a system-level relation. A property is emergent when it belongs to an organised system in a way not straightforwardly attributable to isolated parts. Temperature, flocking, traffic waves, institutional reputation, and market price are standard examples at different levels. The term is often abused as a synonym for surprising or unexplained.

A useful emergence marker should specify the relation:

Term	Explanation
macro-pattern generated	macro-pattern generated by local interaction
property predictable	property predictable only at a higher-level model
property robust	property robust under replacement of parts
property exerting constraints on parts	property exerting constraints on parts through organised structure
property merely redescribed	property merely redescribed at aggregate level

This prevents “emergence” from functioning as a mystical stop sign. It also prevents reductionism from assuming that because a macro-property depends on parts, talk of the whole is dispensable. A traffic jam is made of cars but is not located inside any one car. A company policy is enacted by people and documents but cannot be assigned to one molecule or employee.

Scale-indexed truth. Many apparently contradictory claims are true at different scales. Individual ants follow local rules; a colony allocates labour. A neuron fires; a person decides. A firm cuts costs; an industry becomes fragile. A local intervention lowers risk; a network adapts and restores the original pattern. Without a scale index, explanations slide between levels.

A distinction among micro, meso, and macro scale is only a beginning. Scale can be spatial, temporal, organisational, informational, or causal. A claim should identify the unit whose behaviour is described and the aggregation rule connecting levels. “Competition improves efficiency” may hold for a transaction while failing for infrastructure resilience. “The model is accurate” may hold on average while failing for a vulnerable subgroup. “The country is richer” may coexist with most citizens losing.

Scale-indexed language resists the fallacy of composition and the fallacy of division. Properties of parts need not belong to wholes; properties of wholes need not belong to every part.

Feedback and the disappearance of the first cause. Linear grammar prefers A caused B. Systems contain loops: A changes B, B changes C, and C changes A. Positive feedback amplifies deviations; negative feedback counteracts them; delayed feedback can create oscillation; adaptive agents can alter the rules in response to measurement.

Once a loop is established, asking for “the cause” may be less useful than asking what maintains the cycle and where intervention has leverage. A feedback marker would identify reinforcing, balancing, delayed, and adaptive loops. It could also state whether an observed relation is one pass through a loop or a stable regime.

This matters for polarisation, poverty traps, ecological collapse, addiction, platform engagement, and arms races. A sentence such as “people consume sensational content because they prefer it” ignores the loop through recommendation, production incentives, habituation, and social visibility. Preference is both input and product.

Thresholds and phase changes. Linear language also encourages proportional expectations: a little more cause yields a little more effect. Many systems contain thresholds beyond which behaviour changes qualitatively. An ecosystem collapses, a material changes phase, a network cascades, an institution loses legitimacy, or a bank run becomes self-fulfilling.

A threshold-distance marker would state whether a system is far from a known threshold, approaching one, within an uncertainty band, or already beyond it. Importantly, the threshold itself may be unknown or moving. “Risk is currently low” can be dangerously misleading if the system is close to a tipping point with uncertain location.

The marker should distinguish probability of crossing from severity after crossing. Low-probability irreversible thresholds require different reasoning from frequent reversible fluctuations.

Commons and the grammar of no one. Resources such as atmosphere, biodiversity, shared knowledge, public trust, and open infrastructure do not fit ordinary possession. If nobody owns them, grammar can imply that nobody is responsible. If humanity “owns” them, the possessive remains too vague to allocate duties.

The concept of commons is already available but undergrammaticalised. A commons relation should mark access rights, contribution duties, depletion effects, governance, exclusion, and future claims. “We used public data” should not be complete without asking who maintained it, who consented, who can withdraw, and who receives benefit.

A commons-burden marker could attach costs to users who would otherwise appear as free beneficiaries. Open-source software offers a simple example: millions may depend on a library maintained by a handful of people. The code is accessible, but maintenance capacity is rival and exhaustible. “Free” describes price, not burden.

Unknown, unknowable, and not yet conceptualised. The final relation hidden by nouns is the relation between a question and its possible answer. We speak of “the unknown” as though it were a region containing facts waiting to be collected. Some unknowns are unanswered questions with known methods. Some require new instruments. Some are underdetermined. Some are inaccessible for physical or computational reasons. Some are not yet questions because we lack the concepts to formulate them.

The phrase unknown unknowns gestures toward the last category but remains paradoxical. Once named, a specific unknown unknown becomes at least a known gap. What matters is the epistemic perimeter: evidence that our present ontology, variable set, or question space is incomplete. An anomaly, repeated model failure, category instability, or perspective excluded from the inquiry can signal such a perimeter.

A language for deeper thinking should therefore contain not only unknown but method-known, instrument-missing, underdetermined, computationally-intractable, concept-missing, and question-contested. The distinctions prevent ignorance from being mistaken for mystery—and mystery from being prematurely converted into a routine data gap.

Nouns let us carry the world. Relations remind us that what we carry was cut from a larger fabric. The final section of this chapter turns these insights into a compact inventory of missing variables.

The following proposals are not claims that English speakers lack the corresponding ideas. Most can be expressed through technical prose. The problem is that expression is expensive, inconsistent, or confined to specialist disciplines. Each proposal asks whether a compact lexical item, operator, or grammatical prompt could make a distinction easier to retain across reasoning.

The inventory also avoids a common trap. A new word is not automatically a new thought. Coinages become useful only when they organise contrasts, support inference, and survive use. The strongest candidates therefore have four properties: a recurring cognitive failure, partial analogues in natural languages or formal systems, a compact compositional design, and an experimentally testable benefit.

Source mode. A claim should be able to carry a compact answer to “How was this known?” Natural evidentiality supplies the model: direct perception, non-visual sensation, inference, report, and quotation can be grammatical categories. Scientific and computational settings require a wider inventory: observed,

instrument-measured, experimentally manipulated, statistically inferred, simulated, retrieved, reported, remembered, and generated.

A future interface could make source mode visible as a compact annotation or a spoken marker in specialised settings. A mortality claim reported from a trial is epistemically different from one produced by causal inference. Making the distinction visible does not validate either claim; it preserves its epistemic species.

The cognitive hypothesis is that repeated source marking improves source memory and reduces the tendency to treat fluent repetition as independent evidence. Experiments should test delayed recall, transmission across summarisation chains, and resistance to authority effects. Existing work on Turkish evidentiality gives a plausible starting point but not a settled answer.

Provenance depth and independence. Source mode identifies a channel; provenance depth identifies a lineage. A newspaper may be one step from a press release, three steps from primary data, and zero steps from an anonymous social post. Five visible documents may have one evidential root. Ordinary citation formats represent links but rarely express independence at a glance.

A claim could record that it is two transmission steps from one primary root and has undergone both summarisation and translation. The most important distinction is between multiplicity and independence. “Confirmed by five sources” should not be licensed when all five copy one origin.

The marker would change research review, intelligence analysis, and AI retrieval. Its hypothesis is that visible source depth reduces false corroboration and better preserves uncertainty through chains of abstraction. A cryptographic or database layer can verify parts of the metadata, but the linguistic category remains useful even when the record is incomplete: depth:unknown is more honest than silence.

Confidence state and uncertainty kind. A bare probability answers too little. Uncertainty may arise from sampling, measurement, model misspecification, missing variables, disagreement among experts, stochastic processes, or unknown mechanisms. Two claims with confidence 0.7 may require different actions.

The proposed state pairs an ordinal or numeric confidence with an uncertainty type:

Term	Explanation
Sampling uncertainty	The confidence estimate is conditioned mainly by sampling variation.
Model uncertainty	The numeric confidence is similar, but uncertainty concerns the structure of the model.
Deep uncertainty	Relevant alternatives, models, or probabilities are not sufficiently characterised.

Deep uncertainty applies when actors do not agree on the model, probabilities, or relevant outcome set. It should not be disguised as a wide error bar inside one accepted model.

The hypothesis is that separating confidence from uncertainty type improves decision choice. Participants should prefer information gathering under measurement uncertainty, robustness under model uncertainty, and reversible experiments under deep uncertainty. Ordinary “not sure” encourages one generic response to heterogeneous ignorance.

Revision state. Knowledge is not only more or less certain; it occupies a position in a revision process. A claim can be an exploratory lead, a working assumption, a preregistered prediction, a provisional conclusion, an operational decision, a contested result, a superseded model, or a retracted assertion.

Scientific documents encode some revision states through genre and publication history, but sentences detach from their lifecycle. A preliminary estimate quoted years later may look final. Revision status such as exploratory or superseded should therefore travel with the claim.

The expected effect is improved temporal calibration of knowledge. It may also reduce the rhetorical exploitation of science’s self-correction. A superseded claim is not simply “false information scientists once believed”; it is a state transition in an error-correcting practice. Conversely, publication is not an automatic transition to settled truth.

Absence status. “No X” can mean an empty set, a missing expected item, non-detection, nonexistence, deactivation, or inapplicability. Medical tests, security scans, databases, and scientific null results are routinely misinterpreted because these states share negative syntax.

A compact family could be:

Term	Explanation
zero-X	A defined collection contains no members.
missing-X	The item was expected in the relevant context but is absent.
undetected-X	A search found no item within a declared sensitivity or coverage.
uninstantiated-X	The type is admissible, but no current token is known.
Model-relative non-existence	Model M contains no entity of the specified kind; the statement does not establish metaphysical non-existence.
inapplicable-X	The predicate is not applicable to the object.

The hypothesis is straightforward: explicit absence status should reduce false reassurance from negative tests, confusion between missing and zero data, and ontological inflation from treating every null field as an unknown object.

Knowability and modal mode. Unknown says nothing about whether an answer can be obtained. Possible says nothing about whether possibility is logical, physical, technological, institutional, or merely not yet excluded. A combined modal–epistemic operator should distinguish:

Term	Explanation
unknown with	unknown with known method
unknown because an	unknown because an instrument is missing
underdetermined by	underdetermined by all available evidence
computationally intractable	computationally intractable under current bounds
unknowable	unknowable under a stated physical or logical assumption

Concept missing: the current vocabulary does not formulate the relevant variable.

Term	Explanation
counterfactual	counterfactual under model M
impossible	impossible under model M but open under rival models

The crucial distinction is model relativity. “Impossible under model M” is weaker and often more accurate than an unqualified “impossible.” A perpetual-motion machine is impossible under accepted thermodynamics; a policy is often called impossible only under current institutions.

The hypothesis is that model-indexed modality reduces premature closure. It should help teams distinguish “we lack data” from “we may be asking the wrong question,” and distinguish robust physical constraints from social arrangements presented as necessities.

Matter-same. Matter-same records continuity of substrate. It applies naturally to artefacts whose identity depends partly on retained material: a restored painting, a biological sample, a historical building. The marker must include tolerance because no macroscopic object preserves every particle.

Material continuity may be judged from the retained mass and the preservation of critical original components; one example retains seventy-two percent of its mass and all designated critical components.

This operator prevents material continuity from being confused with function or role. A ship rebuilt plank by plank may remain role-same and flow-same while losing matter-sameness. In conservation, heritage, and forensic evidence, that difference is decisive.

The experimental prediction is that an explicit matter operator will reduce inconsistent identity judgments driven by superficial wording. It may also reveal genuine value pluralism: people can agree on the facts while weighting continuities differently.

Form-same. Form-same marks preserved structure, organisation, or relational pattern. A copied key, reconstructed molecule, replicated database schema, and reorganised team can be form-same at one level while materially different.

Any claim of structural sameness needs a declared comparison vocabulary. Two neural networks may share architecture but not weights; two firms may share an organisation chart but not communication patterns. Architectural similarity must not silently become behavioural equivalence.

The value of the term is to make pattern ontology explicit. Its risk is Platonism by grammar: users may treat “the pattern” as uniquely defined when many abstractions are available. The scale address of proposal 28 is therefore mandatory whenever form-same has practical consequences.

Flow-same. Flow-same marks continuity of a process through changing material and state. A flame, storm, river, metabolism, conversation, scientific programme, or institution can persist by continuing to happen.

Process continuity may instead be judged by an unbroken causal path, with interruptions shorter than a declared five-minute tolerance.

The interruption parameter matters. Is a conversation the same after a week? Is a fire the same after smouldering underground? Is an organisation the same after dissolution and restoration? The process operator does not settle thresholds; it exposes them.

The predicted cognitive benefit is a shift from static essentialism toward maintenance questions. Instead of asking what fixed property an institution possesses, users ask which processes reproduce it and what would interrupt them. That change could improve intervention design in biology, organisations, and social policy.

Line-same: copies, forks, and merges. Line-same marks causal or genealogical descent. It is the natural identity relation for software, texts, organisms, models, and traditions. Unlike form-same, it can survive extensive transformation; unlike role-same, it does not depend on public recognition.

The marker needs branch vocabulary: copy-of, fork-of, merge-of, derived-from, independent-convergence, and unknown-lineage. In AI governance, a model card should record whether a system is a fine-tune, distilled descendant, merged checkpoint, retrieval-augmented deployment, or independent model with similar branding.

The hypothesis is practical: line-same language should improve transfer judgments. Evaluators should be less likely to assume that safety evidence for an ancestor applies unchanged to a descendant, while still recognising inherited risk and data provenance.

Boundary grade. Boundary grade identifies how a category separates members from non-members. The proposed values are sharp-by-nature, sharp-by-rule, operational, fuzzy, contested, and indeterminate.

Boundary example	Explanation
Legal boundary	A wetland may be delimited by a legal rule.
Theory-dependent boundary	A species boundary may depend on a contested scientific theory.
Fuzzy dynamic boundary	A cloud has a fuzzy and dynamically changing boundary.

This marker prevents legal precision from being misreported as natural discovery and prevents natural vagueness from being used to block necessary regulation. A boundary can be operationally exact and ontologically conventional.

The hypothesis is that boundary-grade language improves tolerance for principled pluralism. Participants may be less likely to treat rival thresholds as factual incompetence, while becoming more alert to strategic boundary manipulation.

Authenticity profile. Authentic is another overloaded verdict. It may concern origin, material integrity, authorial intention, cultural continuity, unmodified transmission, authorised performance, or honest self-presentation. Digital generation intensifies the ambiguity. A synthetic image may be clearly labelled and creatively authentic; a genuine photograph may be deceptively captioned.

An authenticity profile separates:

Term	Explanation
Origin	Identifies who or what produced the artefact.
Integrity	States whether the artefact was altered after production.
Lineage	Records transformations from the source to the present form.
Authority	States whether relevant parties authorised the representation.
Endorsement	Identifies who currently stands behind the claim.
Fidelity	Declares the relation asserted between the present artefact and an earlier object.

The design hypothesis is that profiles outperform binary “real/fake” labels. They should help users distinguish synthetic origin from deception, and unaltered origin from false context. This is particularly important for AI media, cultural artefacts, scientific images, and digital identity.

Clusivity and mandate. Natural inclusive/exclusive pronouns answer whether the listener belongs inside we. Modern collective speech also needs to answer whether inclusion is participatory, representative, affected, or merely rhetorical.

A compact pronoun system could distinguish:

Term	Explanation
Inclusive participant	The addressee is included as a participant in the referenced group.
Inclusive affected set	The addressee is included among those affected, without implying decision authority.
Exclusive decision-maker	The deciding group excludes the addressee.
Authorised representation	The speaker claims a verified mandate to act or speak for the addressee.
Unverified representation	The speaker claims to act or speak for the addressee without a verified mandate.

In ordinary prose these may appear as tags rather than new pronouns. The hypothesis is that visible clusivity and mandate reduce strategic ambiguity in policy statements and organisational communication. The strongest test would examine whether readers better identify who decided and who bears costs.

Collective form. Collective nouns should be classified by coordination structure: aggregate, category, crowd, network, swarm, team, organisation, institution, polity, or ecosystem. Each supports different inferences about boundary, memory, and action.

“The community believes” becomes questionable until community form is stated. A statistical category does not believe; a deliberative organisation may adopt a position; a network may amplify a view without endorsing it.

The cognitive hypothesis is that collective classifiers reduce anthropomorphism and ecological fallacies. They should improve reasoning about whether a group-level pattern implies individual agreement, and whether member replacement threatens identity.

Agency architecture. Collective form describes what the group is; agency architecture describes how an outcome is generated. The values are central command, delegated hierarchy, distributed protocol, negotiated consensus, emergent local interaction, statistical aggregation, and legal attribution.

The platform amplified the outcome through emergent optimisation rather than one central decision.

The platform prohibited the action through an executive policy decision.

The same noun can carry different architectures across actions. A company may set policy centrally while producing market effects emergently.

The hypothesis is that architecture markers improve intervention choice. Central actions invite governance of decision-makers; emergent effects may require feedback redesign; statistical claims require distributional evidence; legal attributions require procedural verification.

Responsibility vector. Responsibility should be decomposed into causal contribution, control, knowledge, role duty, benefit, and vulnerability. The vector does not assign a final moral score. It records the dimensions from which judgments are made.

A responsibility profile might record moderate causal contribution, high control, substantial knowledge, high role duty and benefit, and low vulnerability. These dimensions should not be compressed into a single blame score.

Numbers are optional and should not create false objectivity. Ordinal labels may be better. The essential gain is that two actors can be responsible in different ways.

The hypothesis is that vector language reduces both scapegoating and responsibility dilution. Incident teams should identify a wider set of actionable responsibilities while avoiding the conclusion that distributed causation means no responsibility.

Consensus topology. “Experts agree” can mean unanimity, a large majority, convergence among independent methods, negotiated compromise, compliance with an authority, or a public appearance produced by selection. Dissent can concern evidence, values, methods, or definitions.

A consensus marker should state topology:

unanimous, supermajority, plurality, overlapping-convergence, compromise, procedural-decision, authority-imposed, surface-consensus, or polarised-clusters.

It should also mark independence of routes. Ten experts trained in one school are not equivalent to convergence across methods and institutions.

The hypothesis is that consensus topology improves calibrated trust. Readers should distinguish genuine convergence from mere headcount, while avoiding the opposite error of treating any dissent as evidence that “science is divided.”

Perspective stack. A sentence can involve speaker, grammatical subject, experiencer, source, perspective-holder, endorser, beneficiary, and affected party. Logophoricity and obviation show that grammar can track discourse perspective more richly than English pronouns usually do.

A perspective stack might render an AI-assisted sentence as:

A claim may be uttered by an analyst, grounded in a dataset, inferred through a model, endorsed by the analyst, and consequential for patients. These relations are not interchangeable.

In fiction, law, journalism, and research synthesis, the stack prevents voice from being confused with commitment. A narrator can report a belief without endorsing it; a model can generate wording without being the epistemic source.

The hypothesis is that perspective marking reduces attribution errors and improves detection of hidden stakeholders. It may also make multiple-perspective reasoning easier by giving each viewpoint a stable slot rather than forcing readers to reconstruct it from pronouns.

Causal layer. Causal layer separates trigger, mechanism, enabling condition, structural cause, maintenance cause, selection history, and omission. It turns a one-word relation into a map of intervention roles.

The outage was triggered by a certificate expiry, enabled by absent redundancy, structurally caused by cost policy, and maintained by fragmented ownership.

The sentence is longer than “the certificate caused the outage” but much more actionable. A compact tag system can preserve the structure in technical reports.

The hypothesis is that layer marking increases systemic interventions without reducing recognition of proximate causes. It should be tested against the common failure in which “root cause analysis” selects one endpoint and stops.

Intent–effect relation. This marker separates intended, instrumental, foreseen-and-accepted, risk-known, role-foreseeable, genuinely unforeseen, and systemic-against-stated-intent outcomes.

The last category is critical for organisations. Repeated effects can contradict every participant’s stated values while remaining stable under incentives. Language should not let declaration of benevolent intent erase predictable structure.

The hypothesis is that the marker improves moral and legal reasoning by preventing a binary intent/accident frame. It should also improve design review by making foreseeable side effects part of the action description before deployment.

Reversibility grade. Reversibility is a relation among action, time, cost, and restoration. The proposed grades are low-cost reversible, reversible within window, technically reversible but practically locked, compensable-not-restorable, irreversible under known methods, and unknown.

A policy can be legally reversible while infrastructurally irreversible. Data disclosure can be impossible to retract even if access rights later change. Ecosystem loss may be compensable in money but not restorable in kind.

The hypothesis is that reversibility marking improves decisions under uncertainty and increases preference for staged experimentation where appropriate. It may also expose deceptive “we can always stop” claims whose window is already closing.

Path mark. Path mark indicates whether the current state is selected for present reasons or inherited through contingent history. Values include current-optimum, legacy-compatible, accidental-standard, crisis-born, institutionally-inherited, and path-unknown.

The marker does not imply that inherited systems are bad. It asks whether current justification is independent of sunk history. A keyboard layout, legal doctrine, scientific taxonomy, or software stack may persist because surrounding systems adapted to it.

The hypothesis is that path marking reduces status-quo naturalisation. Decision-makers should more readily distinguish “best now” from “expensive to replace,” enabling better counterfactual evaluation.

Lock-in pressure. Lock-in is path dependence plus resistance to exit. The marker identifies the source: switching cost, network effects, contractual commitments, regulation, data format, skill base, political coalition, psychological identity, or physical infrastructure.

Saying that a system is locked in through network, data, and skill dependencies is more informative than saying that it is merely too difficult to change. The diagnosis also suggests different interventions: interoperability for network lock-in, conversion tools for format lock-in, retraining for skill lock-in, and compensation for political lock-in.

The hypothesis is that decomposing lock-in increases perceived agency. A monolithic “impossible to change” often dissolves into several constraints, some of which are tractable.

Externality and commons vector. Externality language should state who receives benefit, cost, risk, control, information, and future burden. It should also mark whether the affected resource is private, public, common-pool, club, or ungoverned.

A decision may benefit a firm, impose costs on neighbours, transfer tail risk to the state, consume a shared resource, and proceed without the consent of those affected.

The vector integrates what accounting and grammar separate. It also records maintenance burden for “free” resources such as open-source infrastructure and public data.

The hypothesis is that the vector reduces moral discounting caused by syntactic distance. Costs borne by absent or future parties become part of the action representation, not an optional second sentence.

Effect order. Every intervention has a direct effect and can have effects produced by responses to that effect. A subsidy changes price; changed price alters demand; demand changes production; production changes politics. English uses phrases such as second-order and unintended consequence, but the index is rarely maintained.

The proposed marker is E1, E2, E3, with optional rebound for an effect that offsets the intended gain. The notation does not claim complete foresight. It prompts a search beyond the first causal edge.

The hypothesis is that effect-order prompts improve policy analysis and reduce one-step optimism. A control group could list expected consequences freely; a treatment group would be required to populate E1–E3 and indicate uncertainty. Quality, not mere quantity, should be evaluated.

Feedback regime and antifragility. Feedback markers identify reinforcing, balancing, delayed, oscillatory, adaptive, and reflexive loops. Antifragile systems, in Taleb’s terminology, improve under some range of stress rather than merely resist it. The term is useful but should be indexed to stress type and range; no system benefits from every shock.

A reinforcing feedback loop with an approximately six-month delay behaves differently from an immediate causal chain.

The same system may improve under small demand shocks yet remain fragile to correlated supply loss.

The marker prevents resilience, robustness, and antifragility from being treated as synonyms. A robust system resists change; a resilient one recovers; an adaptive one changes; an antifragile one gains under specified variation.

The hypothesis is that feedback grammar reduces linear-causality errors and improves anticipation of rebound, escalation, and delayed overshoot.

Threshold distance. Threshold distance identifies whether a system is far from a known transition, approaching it, inside an uncertainty band, beyond it, or governed by an unknown threshold. It should pair with direction and reversibility.

A threshold may be approaching even when its exact location is uncertain and crossing it may be irreversible.

This form is more informative than “risk remains low” when the probability distribution has a severe tail or the threshold is poorly located.

The hypothesis is that threshold markers improve precaution under nonlinear risk without automatically producing alarmism. The test is whether users distinguish low current damage from high proximity to regime change.

Scale address. Scale address states the unit, aggregation level, spatial grain, temporal grain, and decision horizon. It is the grammar of “true where?” and “true for whom?”

Accuracy measured per patient, for a rare-disease subgroup, at the diagnostic event is one specific metric definition.

Profit measured per firm, aggregated quarterly, over a one-year horizon is another metric definition.

A claim at one scale need not transfer. Average model accuracy can coexist with subgroup failure; individual prudence can produce collective panic; local efficiency can reduce system resilience.

The hypothesis is that scale marking reduces composition fallacies and improves cross-level explanation. It may be the single most generally useful proposal in the book because so many disputes are hidden scale shifts.

Posterity, opportunity, and option scope. Present decisions alter not only outcomes but future choice sets. Opportunity cost is the value of the best forgone alternative. Option value is the value of preserving the ability to choose later under uncertainty. Future people may inherit benefits, costs, constraints, and lost options without participating in the decision.

A posterity–option marker could state:

Term	Explanation
Present benefit, future cost	The policy creates present benefits, transfers costs to a future generation, and reduces later options.
Short-term cost, preserved options	The decision imposes an immediate cost while preserving options and producing information.

The hypothesis is that this marker reduces present bias and makes irreversible opportunity loss visible. It should be tested in infrastructure, environmental, and technology-governance scenarios.

Goodhart pressure. A measure can be an observation, a proxy, a target, a constitutive rule, or a gamed signal. Goodhart pressure rises when rewards depend on the measure and agents can alter the measure more easily than the underlying goal.

The proposed marker has stages:

Term	Explanation
M0 — observation	The measure is used descriptively as an observation.
M1 — proxy	The measure is treated as a proxy for another target property.
M2 — target	The measure becomes an explicit optimisation target.
M3 — optimised	Actors adapt behaviour to improve the measure.
M4 — gamed	Actors exploit weaknesses in the measure without improving the intended goal.
M5 — goal displaced	Optimisation of the measure displaces the original objective.

A benchmark may move from M1 to M3 as developers train against it; a citation count may reach M4 through strategic publication; a school metric may become M5 when teaching is reorganised around the score rather than learning.

The hypothesis is that explicit Goodhart state improves metric governance. Teams should become more likely to rotate measures, audit gaming, preserve qualitative evaluation, and distinguish progress on the score from progress on the goal.

What should become grammar. Not every entry deserves a word, and even fewer deserve obligatory grammar. A practical hierarchy is needed.

A lexical distinction is appropriate when the concept is useful but context-specific, as with lineage continuity, boundary grade, or Goodhart pressure. A compositional operator is appropriate when the same variable, such as scale, source, or horizon, applies across many domains. A mandatory field is justified only when omission creates recurring high-stakes error, as with provenance in scientific synthesis or reversibility in risk approval. A visual or computational representation is superior when relations exceed the capacity of linear text, as with causal graphs and responsibility profiles.

The purpose of the thirty variables is therefore not lexical maximalism. It is to identify points where ordinary English repeatedly deletes structure. Some can be restored by words. Some require notation. Some should be embedded in software. The right unit of conceptual engineering is not always the dictionary; sometimes it is the interface.

SLENGLISH AND EXECUTABLE NATURAL LANGUAGE

A canonical intermediate language between prose, knowledge, and executable semantic representations

Term	Explanation
Process ontology	Clouds, organisms, institutions, and software motivate processual and qualified identity rather than one generic equality relation (Nicholson and Dupré 2018; Pradeu 2018; Weatherson 2023).
Collective agency	Groups can possess decision procedures and rational agency without reducing every outcome to one person (List and Pettit 2011; Tuomela 2013).
Causation and intervention	Causal language should distinguish mechanisms, interventions, enabling conditions, structures, and explanatory contrast rather than use one undifferentiated predicate.
Systems variables	Feedback, thresholds, externalities, path dependence, lock-in, scale, and Goodhart pressure are candidates for explicit semantic operators and executable profiles.

CHAPTER 5

SLEnglish: Design, Syntax, and Semantic Libraries

Design Objectives and the Uniform Sentence Form

SLEnglish, short for SOP Lang English, is the controlled executable layer proposed here. It receives task-relevant interpretations of natural language and represents them through a uniform statement form, explicit SSA identities, path-addressed verbs, named parameters, typed values, contexts, and references. The design concentrates semantic variation in operator contracts so that parsing, validation, diffing, provenance, and execution can use a stable outer grammar.

The design starts from a practical observation. When language models translate prose into JSON, triples, code, graphs, or solver formulas, each target recreates a different surface convention for entities, relations, claims, evidence, variables, contexts, and questions. This proliferation makes semantic comparison difficult. SLEnglish introduces one regular outer envelope while leaving the operator libraries and interpretation profiles open. The intended gain is not expressive novelty alone; it is stable identity, explicit roles, source alignment, canonical diffing, and portable validation across heterogeneous methods.

The canonical form is deliberately visible before its details are elaborated: ``@ssa-id x/y/name.sle parameter-name value parameter-name value ... .``. The token beginning with ``@`` creates a new Single Static Assignment identity. The following path selects a versioned operator contract. The remaining tokens form named parameter–value pairs. The final period is a separate terminator. An operator path may contain any number of directory levels. ``knowledge/evidence/report.sle``, ``language/english/aspect/progressive.sle``, and ``science/clinical/trial/adverse-event.sle`` are all valid structural shapes when their files and dependencies are declared.

The folder path is not decorative punctuation. It is the namespace. A repository can place related operator files under nested directories, and ordinary file-system tools can browse, version, hash, review, package, and distribute them. A project may organise a local operator at ``projects/climate/region-a/ice-loss.sle``; a shared standard may expose ``knowledge/causation/structural.sle``; a language-specific library may contain ``language/quechuan/evidential/reported.sle``. No global rule limits the depth. The root manifest and import lock determine which paths are visible and which version each path denotes.

A slash-separated operator path follows the structure of the package folders. A verb can therefore be addressed at arbitrary depth, for example ``knowledge/evidence/report.sle``, ``science/measurement/observe.sle``, or ``social/agency/collective/decide.sle``. The path provides an intuitive namespace, supports ordinary version-control tooling, and allows independent libraries to grow without imposing a fixed number of namespace levels.

The uniform sentence form is stricter than most controlled natural languages. It does not preserve subject–verb–object grammar, articles, agreement, conjugation, passive alternations, or free attachment. English remains present in operator names, parameter names, labels, and reversible renderings. The execution syntax is closer to a semantic assembly language. This is acceptable because an LLM or another semantic parser performs most initial translation. The human task is review and correction rather than unaided composition of every line.

Strictness is useful only when it targets the right layer. The parser should remain small: it recognises bindings, paths, atomic values, quoted values, references, parameter names, and the final period. It should not contain a universal grammar of science or encode evidentiality, causation, identity, and social authority as special productions. Those distinctions belong to operator contracts stored in ``.sle`` files. Extension therefore adds semantic libraries without repeatedly changing the parser.

A SLEnglish operator is called a verb because it constructs or transforms a semantic object, not because it must correspond to an English lexical verb. ``knowledge/evidence/report.sle`` links a proposition to a report. ``language/reference/inclusive-we.sle`` constructs a participant relation. ``knowledge/identity/process.sle`` qualifies continuity. ``social/collective/decide.sle`` represents an authorised or otherwise specified collective decision. ``execution/query/why.sle`` asks for a derivation, provenance chain, causal path, or assumption record. The operator owns the legal parameter names and the semantic contract associated with its output.

The language distinguishes a value from the node that produced or states it. ``${name}`` requests the primary value associated with an earlier SSA identity and creates an explicit data dependency. ``${name}`` refers to the

node itself, including its semantic identity, provenance, status, alternatives, and source alignment. A measurement value may be compared through ``$measurement``; the measurement statement may be cited through ``~measurement``. This separation prevents a common error in which a proposition, its numerical result, and its evidence record are treated as interchangeable.

Single assignment preserves history. A node is created once within a source revision and is never mutated in place. A correction creates a successor node or a new execution epoch, linked by supersession or derivation. This makes it possible to explain why an earlier conclusion was accepted, identify which downstream results depend on a revised premise, and compare two interpretations without erasing either one. SSA is therefore not merely a compiler convention. It is part of the provenance and semantic-diff model.

The canonical source is intentionally verbose where explicitness reduces error. Parameter order is not used to infer semantic roles. A transfer states agent, theme, recipient, origin, destination, time, and context through names when the active contract requires them. A profile can require source, valid time, unit, uncertainty, or identity mode for particular classes of statement. Editors may present compact forms, but the canonical representation does not rely on a hidden default that changes across tools.

The decision to expose all semantic constructions as operators is broader than the initial core proposal. Ordinary English constructions and cross-linguistically attested categories can enter the same library system. Progressive aspect, reportative evidence, switch-reference, logophoric perspective, alienable possession, collective agency, structural causation, process identity, and threshold effects differ in content but not in the outer mechanism by which they become inspectable and executable. The appendices provide a candidate inventory rather than a fixed final standard.

Element	Meaning in canonical SLEnglish
<code>`@ssa-id`</code>	Creates one stable node identity in the current source revision. The identity is assigned once and can be referenced by downstream statements.
<code>`x/y/name.sle`</code>	Selects a path-addressed operator contract. The path may contain arbitrary directory depth and is resolved through manifests, imports, versions, and content digests.
<code>`parameter-name value`</code>	Supplies a typed role to the selected operator. The contract declares required, optional, repeated, and mutually constrained parameters.
<code>`\$name`</code>	Dereferences the primary semantic value produced by a previous node and establishes a value dependency.
<code>`~name`</code>	Passes a reference to the semantic node, preserving proposition identity, provenance, status, alternatives, and source links.
Quoted literal	Preserves textual content including spaces. Its datatype, language, or source role is supplied by the operator contract or an explicit parameter.
Atomic symbol	Names a declared entity, type, unit, enum member, context, model, or other symbol without natural-language inflection.
Final period	Terminates the statement. It is separated by whitespace from the final value, so the <code>.sle`</code> extension remains unambiguous.

```
@core slenglish/core/import.sle module "slenglish-core" version "0.3" root "slenglish" .
@alice slenglish/core/entity.sle type person label "Alice" .
@sample slenglish/core/entity.sle type biological-sample label "Sample A" .
@transfer language/event/transfer.sle agent $alice theme $sample origin freezer-1 destination lab-2 time "2026-08-22T09:00Z" .
@reported knowledge/evidence/report.sle proposition ~transfer source lab-log source-depth 1 confidence 0.98 .
@accepted slenglish/core/assert.sle proposition ~transfer context actual-world .
```

The example illustrates the division between proposition construction and commitment. The transfer node states a semantic event. The report node records an evidential relation to that event. The assertion node accepts the event in a named context. Another context may retain only the reported claim without accepting its content. This separation permits testimony, belief, quotation, hypothesis, denial, and assertion to share one proposition without collapsing their epistemic status.

Canonicalisation is required for reproducible tooling. Whitespace, numeric representation, booleans, parameter order, path resolution, aliases, repeated parameters, strings, and missing-value atoms receive deterministic serialisations. Two source views that resolve to the same operator identity and parameter structure should produce the same canonical node when their meaning under the selected contract is the

same. Canonical equality does not imply metaphysical equivalence; it means identity within the declared representation.

The syntax remains a proposal subject to evidence. The final period might prove unnecessary, the `\$` and `~` distinction might require clearer notation, one file per operator may be too granular, or path-addressed names may need a registry alias for portability. These choices should be tested against parser complexity, translation accuracy, correction time, semantic diff quality, repository maintenance, and user comprehension. Their provisional status is part of the specification rather than a weakness to conceal.

Files, Operator Contracts, Parameters, Contexts, Rules, and Queries

The semantic library is the substantive part of SLEngish. A path becomes executable only when its file declares a contract. The contract specifies the operator's output kind, required and optional parameters, accepted value types, cardinalities, determinism, purity, monotonicity, context behaviour, effect class, compatible interpreter profiles, canonical parameter order, inverse rendering guidance, validation cases, and version history. A directory of names without these properties is a vocabulary; it is not an executable language library.

Operator files can be organised at any useful depth. A root manifest declares package identity, dependencies, version constraints, content digests, exported roots, default labels, and conformance expectations. A directory may contain a shared role registry, type declarations, examples, tests, lowering adapters, and migration rules. The visible path remains intuitive because the folder layout corresponds to conceptual scope. The compiled identity remains stronger because it includes the package digest and resolved version.

The path tree should not be mistaken for a single taxonomic truth. Different communities may publish competing causal, social, clinical, or linguistic libraries. A project can import both under distinct roots, declare bridge mappings, and record where the mappings are partial. The standard concerns resolution, typing, provenance, and comparison. It does not require every discipline to agree on one ontology before executable work can begin.

Parameters provide semantic roles shared across operators. A common registry contains highly recurrent names such as agent, experiencer, patient, theme, source, target, proposition, context, time, value, unit, confidence, evidence-mode, identity-mode, coverage, authority, and source-span. Domain libraries may introduce deeper path-qualified roles when a distinction cannot be represented safely through the common set. Promotion into the common registry should require corpus evidence, independent use, migration rules, and a demonstration that the role has stable cross-domain meaning.

Values require more precision than a generic null. `unknown` indicates that the value is conceptually applicable but not known. `not-applicable` states that the role does not apply. `withheld` records that a value is intentionally unavailable. `not-measured` states that no measurement was performed. `conflicted` records incompatible supported values. Omission means that the statement makes no commitment about the parameter. These distinctions determine which questions can be answered and which inferences are licensed.

Types constrain values but should remain plural and versioned. A type may describe an entity, event, proposition, quantity, unit, interval, agent, collective, source, model, or domain-specific object. Subtyping, disjointness, cardinality, and conversion can be delegated to an ontology or type interpreter when appropriate. The SLEngish source records which type system is active rather than treating one backend's class semantics as universal.

Contexts prevent incompatible commitments from being merged. A statement may belong to the actual-world context, a quoted report, a hypothesis, a simulation scenario, a legal jurisdiction, a historical reconstruction, a counterfactual branch, or a reviewer's interpretation. Contexts can inherit from one another under explicit rules, but inheritance is not ambient. A proposition accepted in a simulation does not become a fact about the actual world; a claim in one jurisdiction does not automatically apply in another.

Commitment operators distinguish what is being done with a proposition. Assertion accepts it in a context. Denial accepts its explicit negation. Claim records that a source supports it. Hypothesise marks it as an object of investigation. Assume makes it available under a declared scope. Quote preserves reported language. Retract and supersede change the status of earlier commitments without deleting their history. These operators are among the small bootstrap library because every domain needs to separate content from commitment.

Rules construct consequences under named semantics. A strict rule can resemble Horn or Datalog inference. A defeasible rule can specify exceptions and priority under a non-monotonic profile. A constraint can state illegal combinations, required cardinalities, dimensional relations, temporal obligations, or optimisation criteria. The outer syntax stays uniform because premises, conclusions, exceptions, and profile names are parameters. The selected interpreter determines the formal meaning and must return a profile-specific receipt.

Queries are first-class nodes. A whether query requests status in a context; a select query requests bindings; a count query requires an explicit classifier and coverage boundary; a why query requests proof, provenance, causal path, or assumptions; a what-if query introduces counterfactual changes; a compare query states the dimensions along which artefacts, models, versions, or scenarios should be compared. Treating a query as a node allows its scope, budget, guarantee, source, and answer lineage to remain inspectable.

Ordinary linguistic constructions enter this contract system when they alter semantics. English progressive aspect can be represented by an operator that relates an event description to an internal temporal view. Passive voice can normalise surface order while preserving the fact that the agent was omitted, unknown, backgrounded, or deliberately concealed. A reportative evidential records source mode; switch-reference records participant continuity across clauses; logophoricity records whose perspective anchors a report. The operator inventory is therefore both a grammar-to-semantics bridge and a knowledge-representation library.

Not every lexical distinction should become an operator. A new file is justified when the distinction changes denotation, inference, evidence status, participant structure, temporal organisation, social authority, validation, or execution policy; when a stable contract can be stated; and when hiding the distinction inside an untyped label creates measurable error. Pure stylistic variation can remain a rendering choice. The boundary between operator, parameter, enum value, type, and label is itself a research question.

The directory layout supports governance. Shared roots can use semantic versioning, deprecation periods, conformance suites, migration scripts, and signed releases. Domain communities can govern their own subtrees. Local projects can experiment under private paths without polluting the common registry. A lock file records the exact roots and digests used by a source package. Competing libraries are acceptable; silent divergence under the same path is not.

A SLEngish document is therefore both text and dependency manifest. Operator paths determine semantic dependencies. `\$` references determine value dependencies. `~` references determine proposition and provenance dependencies. Imports determine library dependencies. Rules and queries determine demand. These relations can be lowered into many execution forms, but they are useful even before execution because they support validation, source alignment, semantic diffing, impact analysis, and review.

Contract field	Purpose
Resolved identity	Combines package or repository digest, arbitrary-depth relative path, file version, and exported contract identity.
Output kind	Declares whether the operator produces an entity, proposition, value, context, rule, query, diagnostic, request, or another typed object.
Parameter schema	Defines required, optional, repeated, ordered, mutually exclusive, and conditionally required roles.
Type and cardinality	Constrains admissible values and the number of occurrences for each role.
Semantic profile	Names the interpretation family under which the operator has formal consequence.
Purity and effects	States whether evaluation is declarative, deterministic, cacheable, state-sensitive, or capable of requesting an external effect.
Monotonicity and closure	Records whether adding knowledge can retract conclusions and which open- or closed-world assumptions apply.
Canonical order	Provides a deterministic serialisation independent of authoring order.
Lowering adapters	Describe translations to compatible rule, graph, constraint, temporal, proof, code, simulation, or circuit backends.
Inverse rendering	Provides controlled natural-language views without granting the renderer authority to strengthen the statement.
Conformance evidence	Contains positive cases, negative cases, semantic mutations, migration tests, and expected diagnostics.

The core should remain small even though the library is open. The bootstrap files must support imports, definitions, entities, variables, contexts, commitments, rules, constraints, queries, validation, explanation, and diagnostics. Rich concepts belong in versioned libraries. This separation allows generic tooling to parse and transport unfamiliar operators while specialised interpreters decide whether they can execute them.

The appendices list candidate parameters and operator families derived from English grammar, linguistic typology, philosophy of identity, social ontology, scientific practice, and systems reasoning. The tables are intentionally extensive because the book proposes a research corpus, not a closed standard. Some entries will collapse into shared operators; some will become enum values or parameters; some will prove too ambiguous for stable implementation; some important distinctions are certainly missing. The inventory is a falsifiable design surface.

The strongest reason to retain the source-language notation is correction locality. A reviewer can reject one identity link, replace one evidence mode, or change one scope relation without rewriting a complete formal theory. The stable surrounding nodes preserve their identities. The revised document can then be revalidated and reinterpreted, and the affected results can be identified from dependency records. Whether this advantage outweighs the additional representation layer must be measured rather than assumed.

Interpretation and Execution across Multiple Backends

LLM Translation, Deterministic Validation, and Auditable Meaning

The principal advantage of using a language model at the front end is economic. A rigid target language no longer needs to anticipate every surface form through hand-written parsing rules. The model can propose entities, events, relations, scopes, evidence links, and candidate operator paths from unrestricted prose. The principal danger is equally clear: the model can introduce precision that the source did not contain. SLEngish therefore treats model output as a candidate compilation, not as accepted meaning.

A translation request includes the source revision, active operator libraries, type environment, task contract, desired interpretation profile, and examples of valid canonical source. The model returns SLEngish candidates together with source-span mappings, unresolved references, alternative readings, introduced assumptions, unsupported constructions, and a translation confidence that is explicitly not a truth probability. The output must parse without an automatic repair stage that silently changes semantics. A failed parse is a visible compiler failure.

The deterministic parser performs only lexical and structural work. It recognises SSA identities, operator paths, parameter–value pairs, strings, numbers, booleans, declared missing-value atoms, `\$` references, `~` references, and statement boundaries. It does not infer an omitted parameter, select an ontology, or reinterpret a malformed line. The resolver then binds every path to a locked operator contract. The schema validator checks parameter legality and cardinality. The type checker checks values and references. The context checker checks world and profile compatibility. The provenance checker confirms required source links.

Validation cannot establish semantic fidelity by itself. A syntactically valid representation may still misread the source. The toolchain therefore needs contrastive and round-trip tests. A controlled renderer produces a plain-language account from the candidate without consulting the original wording. Reviewers compare the material commitments. Paraphrase tests compile meaning-preserving variants and compare canonical structures. Semantic mutation tests alter negation, quantifier scope, participant order, source mode, time, exception, or modality and check that the appropriate node changes while unrelated nodes remain stable.

Ambiguity is represented rather than solved invisibly. When a phrase permits two consequence-relevant attachments, reference assignments, scopes, or causal readings, the compiler creates alternatives under an interpretation set. A model can rank them, and a user can select or reject them, but ranking alone does not authorise collapse. A downstream query may return a conclusion that holds under every represented alternative, one that holds under at least one, one that holds under none, or an unknown status because the set is incomplete.

Grounding is layered. Lexical grounding proposes operator paths and symbols. Referential grounding links mentions to entities or preserves distinct candidates. Ontological grounding selects types and relations. Temporal grounding locates states and events. Evidential grounding connects propositions to observations, reports, data, or assumptions. Normative grounding identifies issuer, mandate, jurisdiction, and effective time. Each layer produces its own diagnostics so that a source mismatch is not reported generically as a reasoning failure.

Coverage is part of negative reasoning. Failure to retrieve a prior publication does not establish that none exists; absence of a row in an open dataset does not establish non-existence; a bounded email export does not establish that no message was sent outside the export. A coverage object records the source set, revision, retrieval method, exclusions, temporal interval, schema, and closure policy. A query that requests absence or completeness must cite a compatible coverage object or return a weaker status.

Identity compilation requires special restraint. Repeated labels, titles, aliases, versions, and descriptions are not merged solely by string similarity. Candidate identity links name their criterion and evidence. Strict identity licenses substitution only under a declared profile. Process continuity, lineage, material overlap, legal continuity, and functional equivalence remain separate relations. A translator that cannot determine the intended relation should preserve alternatives or emit an unresolved identity obligation.

The trusted computing base should be small relative to the overall system. It includes canonical source handling, path and version resolution, schema validation, type checking, profile selection, verified or well-tested lowering components, interpreter receipts, result-envelope validation, and source-linked diagnostics. Models, retrieval systems, heuristic ontology matchers, approximate memories, and prose renderers remain outside that boundary. They can propose useful work without being allowed to redefine accepted semantics silently.

Human review is not a single approval button. A domain expert may judge whether a measurement or causal concept is appropriate. A source specialist may verify a quotation and reference. A methodologist may review a statistical or formal lowering. An authorised institutional actor may approve an operational effect. The representation records who accepted which commitment under which role. This permits selective responsibility rather than asking one reviewer to certify the entire artefact.

Correction must be local. When a reviewer changes an evidence mode, reference link, boundary, unit, or operator path, the compiler creates a revised node and preserves unaffected identities. The dependency graph identifies which validations, derivations, queries, and rendered claims require reconsideration. Full recompilation remains available as a reference, but local invalidation makes long-lived scientific objects economically maintainable.

The final renderer is constrained by a claim-licensing relation. It receives accepted result objects, proof or evidence links, contexts, conditions, and unresolved limitations. It may choose wording and explanatory order, but it may not turn possibility into actuality, conditional support into unconditional fact, conflict into consensus, or a report into direct observation. Each generated sentence maps back to the nodes that license it. This does not make generated prose infallible; it makes overstatement detectable.

The complete translation record should retain the source span, candidate generation metadata, active libraries, model identifier, prompt or instruction digest, alternatives, deterministic diagnostics, reviewer changes, and final accepted nodes. Privacy and intellectual-property constraints may limit what can be stored, but the limits themselves should be recorded. A published semantic artefact that omits its translation history may remain useful, yet it offers a weaker basis for audit and reproducibility.

The research question is whether this intermediate step improves outcomes over simpler alternatives. Direct prompting may be sufficient for low-consequence summarisation. A JSON schema may be sufficient for extraction. A conventional compiler may be superior in a narrow language. SLEngish earns its cost only where reusable semantic libraries, cross-task queries, explicit alternatives, provenance, and local correction provide measurable value.

Validation layer	Question answered
Lexical structure	Does the source conform to the one-shape syntax without silent repair?
Path resolution	Does every operator path resolve to a locked and versioned contract?
Schema	Are required parameters present and are illegal or repeated parameters rejected?
Type	Do values, node references, and returned objects satisfy declared types and cardinalities?
Context	Are worlds, jurisdictions, scenarios, and interpretation profiles combined only through declared bridges?
Grounding	Can every accepted symbol and semantic commitment be traced to source material, data, a definition, or an explicit assumption?
Coverage	Are negative and completeness claims licensed by the examined domain and closure policy?
Translation fidelity	Do round-trip, contrastive, paraphrase, mutation, and human review tests support the source-to-node mapping?
Execution receipt	Did the selected interpreter apply the named semantics, bounds, versions, and assumptions?
Realisation licence	Does every sentence in the final explanation remain within the accepted result envelope?

Interpreter Families, Generated Code, and Circuit Runtimes

SLEngish is designed to be interpreted, but it is not defined by one interpreter. The surface representation provides typed operators, values, references, contexts, rules, queries, and dependencies. A backend adapter

may lower a compatible fragment to Datalog, Answer Set Programming, a description-logic reasoner, SPARQL, SQL, an SMT or CSP solver, a temporal engine, a theorem prover, a simulation framework, conventional generated code, or another domain method. The adapter must state which semantics it preserves and which constructs it rejects or approximates.

Backend neutrality is not the absence of semantics. Every operator contract names compatible profiles and expected outputs. A relational closure operator may have a Datalog interpretation. A defeasible rule may require an ASP profile. A quantity constraint may lower to SMT with units normalised beforehand. A temporal process may use interval algebra, event calculus, automata, or model checking. A statistical claim may invoke an analysis package whose result is evidence rather than logical entailment. The common result envelope records status, context, assumptions, bounds, provenance, and an engine-specific receipt.

Generated conventional code is a legitimate interpretation route. An operator family may compile into a typed host-language library, a database query plan, a workflow definition, a test suite, or a simulation configuration. Generated code must remain subordinate to the semantic source: versions, inputs, environment, side effects, outputs, and validation tests are recorded. The existence of code does not prove that the generated procedure faithfully represents the source; the compiler contract and tests address that question.

Dynamic SOP Lang circuits provide one execution substrate for SLEngish. The circuit research programme studies typed nodes, explicit dependencies, composable structures, bounded execution, provenance-bearing connections, validation, and interpreter composition (Axiologic Research 2026). These properties align naturally with SLEngish because SSA identities and named references already expose dependency structure. Circuit execution can therefore be evaluated alongside rule engines, graph stores, solvers, temporal systems, generated code, and domain-specific interpreters.

A source-language boundary allows execution technologies to evolve independently. Circuit schedulers, activation rules, memory systems, dynamic expansion, and resource control can change behind an adapter while SLEngish operator contracts remain stable. The same SLEngish package can also target a different runtime when its contracts map more directly to another formalism. Backend comparisons can therefore measure correctness, performance, interoperability, and review cost while holding the source representation constant.

A circuit backend gives operational meaning to SLEngish dependencies. ``$`` references can become dataflow edges; ``~`` references can become proposition or provenance edges; imports can become module dependencies; rules and queries can generate demand; node hashes can support caching; declared purity and monotonicity can support incremental or reordered evaluation. These mappings define an adapter contract and can be tested without fixing one scheduler or memory architecture in the language specification.

Other backends may interpret the same structure differently. A graph database can store nodes and indexed roles without treating them as scheduled computation. A Datalog compiler can materialise compatible relations and recursive rules. An SMT compiler can generate variables and constraints for a bounded query. A theorem-prover adapter can generate declarations and proof obligations. A code generator can map operators to host-language calls. A hybrid runtime can route different subgraphs to specialised methods while retaining one evidence and dependency record.

Heterogeneity requires explicit conversion. Moving from a probabilistic estimate to a Boolean threshold, from an OWL class assertion to a closed-world database fact, or from an approximate retrieval score to an accepted premise changes guarantees. Conversion operators should appear in the graph and state information loss, threshold, calibration, and authority. A common envelope must not disguise incompatible semantics as if every node participated in one uniform proof system.

Effects require a separate authorisation layer. A SLEngish proposition may describe, recommend, require, or hypothesise an action without causing the action to occur. An operational request can be interpreted only by a runtime with the relevant capability, policy, identity, budget, privacy permission, and approval. This prevents a sentence such as ‘the sample should be heated’ from becoming a laboratory command merely because it has a syntactically valid representation.

Execution should be epoch-based. An epoch binds the source revision, operator locks, compiler versions, active contexts, selected profiles, interpreter versions, policies, and captured external inputs. A later epoch can reuse a cached result only when the operator contracts and invalidation rules justify reuse. Full replay remains the reference for validating incremental execution. Historical results retain the environment and assumptions under which they were produced.

A backend must be able to stop with a structured status. Unsupported operator, unresolved type, incomplete coverage, ambiguous grounding, unavailable engine, resource exhaustion, policy denial, inconsistent assumptions, and failed certificate are distinct outcomes. Returning unknown with a reason is preferable to asking an LLM to improvise across an unsupported gap. The result envelope can distinguish entailed, refuted, possible, necessary, conflicted, unsupported, bounded, timed out, and unknown without pretending that all statuses arise from the same logic.

Approximate retrieval and high-dimensional representations may assist operator selection or source discovery, but they do not define accepted consequence. Similarity can nominate a candidate path, related passage, or possible analogy. The candidate returns to the typed and source-grounded pipeline. If lexical search, graph indexes, embeddings, vector-symbolic methods, or a language model are used, the system should compare recall, semantic precision, latency, memory, provenance quality, and downstream false assurance. The language itself remains independent of the winning retrieval method.

The engine router should be driven primarily by declared contracts. An LLM can propose a plan, but a deterministic registry should check that every operator supports the selected lowering and that the requested guarantee is available. The router records why a backend was selected, what fragments were not handled, what conversions occurred, and which certificates or test results were returned. An unsupported request remains an explicit obligation rather than being silently delegated to free-form generation.

Scientific work combines exact proof, statistical estimation, simulation, measurement, causal modelling, database retrieval, interpretation, and institutional judgement. SLEngish represents their common semantic dependencies while preserving the boundaries among their execution semantics. A result therefore records the operator contracts, conversions, assumptions, evidence, and backend receipts that license it.

The same representation supports several levels of execution. It can serve as a typed annotation format, semantic diff layer, compilation target for domain tools, query-oriented research object, or interchange representation between LLMs and existing systems. Where a general runtime is useful, the dependency graph can coordinate multiple interpreters. This makes runtime choice an empirical engineering variable rather than a property encoded into the source grammar.

The backend-agnostic position is therefore an empirical commitment to modularity. It can be falsified. If most operators require backend-specific surface syntax, a common source may become an unhelpful abstraction. If adapters introduce unacceptable semantic loss, direct authoring in established formalisms may be better. If the dependency structure adds little to correction or reuse, a simpler frame representation may suffice. The programme should preserve the evidence required to reach any of these conclusions.

Interpreter family	Appropriate SLEngish contribution and required caution
Datalog or deductive database	Supports typed facts, recursive closure, and explainable rules; requires explicit treatment of negation, stratification, and closure.
Answer Set Programming	Supports defaults, exceptions, competing models, and optimisation; result semantics and priorities must be named.
RDF, SPARQL, and graph systems	Support graph interchange, federation, and query; n-ary events, context, and profile assumptions require explicit modelling.
Description logic or OWL	Supports class and property reasoning within decidable profiles; it should receive only compatible fragments.
SMT, CSP, SAT, and optimisation	Support arithmetic, finite-domain, scheduling, and constraint obligations; units, bounds, and encoding correctness must be recorded.
Temporal engine or model checker	Supports event ordering, persistence, state transitions, and temporal properties; the chosen time and transition semantics must be explicit.
Theorem prover or proof assistant	Supports deductive proof under formal definitions; fidelity of the formalised statement remains a separate obligation.
Generated conventional code	Supports procedures, simulations, tests, and integrations; environment, effects, versions, and semantic tests remain part of the record.
Dynamic SOP Lang circuit runtime	May exploit typed dependencies, bounded execution, caching, and provenance; it is an optional substrate described in a separate research programme.
Human or institutional interpreter	Remains necessary for contested meaning, legitimacy, ethical authority, and domain judgement; the role and decision must be recorded rather than hidden.

CHAPTER 7

Executable Science as a Semantic Build System

From Reproducible Computation to an Executable Argument

Executable science is the broader application that motivates the language, but the phrase must remain narrower than the claim that science can be reduced to software. Scientific work includes concept formation, measurement, modelling, interpretation, criticism, ethical judgement, institutional authority, and revision. The proposed change is that a larger share of the operational and evidential structure should become inspectable and queryable, while unresolved judgement remains explicit rather than being simulated as deterministic output.

The semantic build-system analogy is useful because scientific claims depend on artefacts and assumptions in ways that ordinary documents record only partially. A build system knows which outputs depend on which inputs and what must be recomputed after change. An executable research object should similarly know which conclusions depend on a dataset revision, a calibration, an identity rule, a causal interpretation, a software environment, a reported source, or a reviewer-approved exception. The analogy does not imply that every dependency is mechanical; it requires each dependency to state its kind.

Executable science is often reduced to reproducible code. Reproducible computation is indispensable, but it does not make a scientific argument executable. A notebook can rerun a regression while leaving unclear which sentence the regression supports, which assumptions are required, which uncertainty is statistical rather than structural, which dataset revision was used, and which claims depend on a contested interpretation of identity or causation. The gap is not execution of the calculation; it is execution of the argument's semantic dependencies.

A scientific project can be treated as a semantic build system. Concepts, hypotheses, protocols, instruments, observations, datasets, transformations, models, simulations, results, claims, critiques, and conclusions become typed nodes connected to source material and executable artefacts. A build resolves definitions, checks units and scopes, validates protocol obligations, executes analyses, derives eligible claims, reports contradictions, and materialises proof and provenance graphs. The prose remains a human view over that build rather than being replaced by it.

SLEngish files do not duplicate datasets, code, or workflow specifications. They state the relations among them. A measurement node identifies the variable, entity, unit, uncertainty, instrument, calibration, source field, and time. An analysis node identifies code, environment, inputs, method, parameters, and output. A claim node states which results support it, under which assumptions, population, temporal scope, identity criteria, and causal mode. A review node can challenge one premise without rewriting the original claim.

Claims become build targets. A conclusion depends on observations, statistical analyses, simulations, background theory, methodological assumptions, and interpretive choices. Some dependencies are deductive; some probabilistic; some defeasible; some institutional. The objective is not to force them into one proof calculus. It is to type the dependency and preserve the provenance so that changes propagate appropriately.

When a dataset is corrected, a software dependency changes, a source is retracted, or an ontology is revised, the build graph identifies affected downstream claims. Some conclusions may remain textually identical while their support weakens. Others may need recomputation. Superseded results remain in prior epochs. This is the semantic analogue of incremental compilation and gives scientific revision a more precise memory.

Research objects, FAIR workflows, notebooks, RO-Crate, provenance standards, containers, and workflow languages already solve important parts of this problem. They package data, code, environments, runs, and metadata. SLEngish targets the missing semantic layer of the prose claim and its relation to those artefacts. It should interoperate with mature tooling rather than replace it.

Executable science should not be confused with fully automated science. A machine may establish that a conclusion follows from declared assumptions while experts dispute the assumptions, measurement validity, causal interpretation, or social relevance. Formalisation localises that disagreement; it does not eliminate judgement. The system's success lies partly in making disputes addressable at the node where they arise.

The strongest near-term use cases are bounded: verifying that reported numbers match outputs; checking unit and population consistency; linking claims to exact source spans and analysis runs; exposing unsupported

causal language; tracking retractions; validating protocol conditions; comparing theory versions; and asking structured questions across a research package. These tasks can demonstrate value without claiming machine understanding of all science.

A paper in this architecture has two aligned surfaces. The first is ordinary prose, which preserves rhetoric, explanation, uncertainty, and disciplinary style. The second is an executable semantic package containing the claims selected for formal treatment. Alignment may be paragraph-level, sentence-level, or span-level depending on importance. Readers can stay in prose until they need to inspect evidence, query a claim, compare versions, or reproduce a derivation.

The package begins with a concept layer. Terms are defined, mapped to external vocabularies where useful, and versioned. Identity criteria are explicit for samples, organisms, datasets, models, software versions, institutions, and populations. Units, scales, temporal frames, and missing-value semantics are declared. Competing definitions can coexist in namespaces rather than being collapsed into a supposedly neutral ontology.

The hypothesis layer distinguishes conjecture, operational hypothesis, statistical hypothesis, mechanistic model, and prediction. A hypothesis specifies conditions under which it would be challenged, but executable science should not reduce scientific status to one falsification operator. Replication, calibration, exploratory analysis, model comparison, severe testing, Bayesian updating, and qualitative evidence require different profiles. The semantic package makes the test relation visible rather than imposing one philosophy of science.

The protocol layer connects planned actions to actual execution. Preconditions, interventions, measurement schedules, randomisation, blinding, inclusion criteria, exclusions, stopping rules, and deviations can be represented. External workflow engines and laboratory systems remain operational authorities. SLEngish records the semantic contract and imports receipts. A protocol query can identify which conclusions rely on a deviation or unavailable measurement.

The result layer distinguishes raw observation, processed measurement, statistical estimate, model output, simulation result, and derived quantity. Probability is separated from confidence in the estimate. Statistical uncertainty is separated from structural, measurement, and model uncertainty. Bounds and units are mandatory under scientific profiles. A result without source and method remains a proposition but cannot be promoted to an empirical claim.

The claim layer records modality and scope. An association is not a causal effect. A population average is not an individual prediction. A result inside a simulation is not an observation in the actual world. A bounded verification is not an unbounded theorem. A claim states which transformation licensed it and which assumptions remain external. The renderer uses these distinctions to prevent familiar overstatements.

The discussion layer can represent competing interpretations rather than only the authors' preferred narrative. A reviewer may accept data and analysis while rejecting an identity criterion or causal layer. An alternative model can share evidence but derive different implications. May/must analysis can show which conclusions are robust across the admissible interpretations. The paper becomes capable of exposing its own hinge assumptions.

The publication artefact records a build lock: source digests, library versions, compiler, model profile, engine versions, environment, policies, and reviewer decisions. This does not guarantee that the science is correct. It makes the machine-mediated transformations inspectable and the computational parts replayable. The reproducible unit becomes the argument package, not only the notebook.

SLEngish contributes a human-facing representation for this semantic layer. Existing workflow and research-object standards remain responsible for packaging files, data, code, environments, and runs. Domain repositories remain responsible for authoritative records. Formal systems remain responsible for their own proofs or certificates. SLEngish links these artefacts to propositions, contexts, evidence, interpretations, and queries so that the relation between a calculation and a published claim does not remain an informal caption.

The build graph should distinguish normative from descriptive dependencies. A protocol may require an action without establishing that it occurred. A dataset may contain an observation without licensing a causal conclusion. A reviewer may authorise publication without endorsing every interpretation. A legal permission may allow data use without establishing scientific validity. The common representation is valuable only if these different relations remain typed.

Scientific change is then represented as revision rather than overwrite. A corrected dataset, altered calibration, retracted source, changed code dependency, or new identity criterion creates a new epoch. The system identifies potentially affected claims, but it does not declare all of them false. Some require recomputation, some require reinterpretation, some retain support through independent evidence, and some become unsupported. This distinction is more faithful than treating a scientific article as either permanently valid or globally retracted.

The minimal useful outcome is not an autonomous research system. It is a document whose selected claims can answer bounded questions about source, scope, support, contradiction, dependence, and change. A reader should be able to ask which measurements support a conclusion, which assumptions are external, what would change under a competing identity model, which claims rely on a retracted source, and which parts of the argument cannot yet be represented.

Research object	Semantic build obligation
Concept or definition	Declare version, scope, competing definitions, mappings, and the claims whose meaning depends on the choice.
Observation or measurement	Record target, time, instrument, unit, uncertainty, method, source field, and applicable identity criterion.
Dataset	Record revision, schema, coverage, exclusions, provenance, licence, consent scope, and relation to source observations.
Code or model	Record version, environment, inputs, parameters, assumptions, outputs, tests, and generated or human-authored status.
Analysis	Connect method, data, code, environment, statistical or formal profile, result, and known limitations.
Scientific claim	State modality, population, scale, temporal scope, causal status, evidence, assumptions, and support relation.
Review or challenge	Target a specific node or dependency and record evidence, authority, requested change, response, and unresolved status.
Publication view	Render only licensed claims while retaining links to the maintained semantic package and its locked versions.
Revision	Create successor nodes, preserve history, calculate affected dependencies, and rebuild only where the contracts permit.

Review, Governance, and Bounded Scientific Use

Peer review becomes more precise when criticism can target semantic nodes and dependencies rather than only paragraphs. A reviewer can accept a measurement but reject its unit conversion, accept an association but reject a causal operator, challenge the identity rule used to group samples, mark two sources as non-independent, or request a sensitivity analysis. The review itself becomes a versioned semantic artefact linked to the claim it contests.

Automated review agents can perform bounded checks: verify citations and spans, compare reported values with analysis outputs, detect missing units, identify unsupported modality changes, inspect temporal consistency, test rules, search for contradictory claims, and generate counterexamples under declared models. They should not return a universal quality score. Each finding carries evidence, profile, coverage, and verifier status. Unknown remains acceptable when the required semantic layer has not been materialised.

Rebuttal can also become executable. Authors respond by linking a correction, alternative interpretation, new analysis, or reasoned rejection to the review node. The build shows which downstream claims changed. Disagreement can persist without being erased. A journal or consortium can declare which unresolved issues block publication and which remain visible after publication.

Post-publication correction is a central advantage. Retractions and dataset updates can invalidate dependencies automatically. A literature graph can identify claims downstream of a retracted source without assuming that every dependent claim is false. The status is affected and requires review. Replacement evidence can restore support in a new epoch. This is more faithful than both permanent citation and indiscriminate deletion.

Research compliance benefits from the same structure. Privacy constraints, consent scopes, copyright licences, data-use agreements, jurisdictional rules, and funder policies can be represented as versioned modules and checked against project nodes. The compliance engine remains distinct from scientific truth. A

result can be scientifically credible yet legally unusable, or compliant yet scientifically weak. The language must preserve those axes.

The social risk is bureaucratic capture. Once claims become machine-checkable, institutions may reward what is easy to formalise and penalise work whose value depends on new concepts, qualitative interpretation, or productive ambiguity. Metrics may become targets. A SLEngish profile must therefore expose Goodhart pressure, coverage, and scope rather than presenting formal completeness as epistemic superiority. Executability should expand criticism, not narrow science to what current operators can express.

Another risk is authority laundering. A formal vocabulary can make contested classifications appear natural. A library may encode one legal doctrine, medical taxonomy, or social ontology and then describe outputs as objective. Namespaces, versioning, provenance, governance, and bridge modules are safeguards, but they do not replace political scrutiny. The source of a semantic operator and the community authorised to maintain it should be visible.

The system should support literature and education as well as verification. Controlled smoothing, perspective marking, and deliberate omission can be legitimate in diplomacy, pedagogy, and art. The objective is not to purge ambiguity. It is to detect and govern when ambiguity is being used, and to require stricter profiles where consequences demand them. A language that treats every unformalised sentence as defective would misunderstand both human communication and science.

A credible pilot must be smaller than the vision. It should select one research package with bounded data, a limited ontology, a tractable set of claims, and several genuinely different reasoning needs. Suitable candidates include a computational experiment with a published dataset, a privacy-preserving learning protocol, a systematic review with explicit inclusion criteria, or a formalised software-security study. The package should contain prose, data, code, claims, and at least one contested interpretation.

The pilot begins with manual gold work rather than end-to-end automation. Domain experts define a reference SLEngish package, identify source spans, declare key ambiguities, and specify expected query outcomes. This reference is not assumed metaphysically complete. It is an evaluation instrument. Independent annotators record disagreements so that the benchmark contains legitimate alternatives rather than forcing one hidden interpretation.

The first experiment tests compiler fidelity. LLM, encoder, hybrid, and direct-authoring paths produce candidate packages under the same libraries. Metrics include syntax validity, verb and parameter accuracy, source alignment, assumption invention, ambiguity preservation, correction time, and semantic equivalence on queries. Token count is secondary. A verbose representation is justified only if it reduces material error and review effort.

The second experiment tests execution and update. Queries cover provenance, identity, time, units, causality, and rebuild impact. The system executes through relational, constraint, temporal, and proof profiles where needed. A dataset field and a library definition are then changed. Incremental results are compared with full replay. Every changed claim should be explainable through the dependency graph.

The third experiment compares representation stacks. A competent team implements the same package using RDF/OWL/SHACL/SPARQL/PROV and conventional workflow metadata. Another uses a frame-oriented CNL or structured JSON. Direct LLM reasoning and tool-augmented prompting form additional baselines. The comparison evaluates authoring effort, number of artefacts, semantic locality, interoperability, query accuracy, update trace, false assurance, and expert preference.

The fourth experiment tests the missing-grammar hypothesis. Participants solve source-monitoring, collective-agency, process-identity, causal-depth, or reversibility tasks using ordinary English views or SLEngish-informed views. The objective is not to prove that a language globally improves thought. It is to measure whether selected explicit operators reduce predictable errors enough to justify their verbosity. Negative results should remove or demote profiles.

The fifth experiment tests governance. A red team introduces malicious source instructions, stale world knowledge, incompatible modules, ontology poisoning, ambiguous licences, and effectful requests. The system must preserve source as data, reject unauthorised effects, pin versions, expose conflicts, and produce stopped states. Security and semantic accuracy are reported separately.

Success is bounded. A pilot succeeds when experts can inspect and correct translations, queries execute with traceable evidence, changes rebuild the correct claims, competing interpretations remain visible, and the

package outperforms at least one simpler baseline on a target metric without unacceptable cost. It does not establish general executable language or autonomous science. It establishes that a more ambitious experiment is justified.

Term	Explanation
Research objects	FAIR workflows, RO-Crate, notebooks, containers, and provenance records package computation; SLEngish targets the semantic dependencies of prose claims (Soiland-Reyes et al. 2022; Wilkinson et al. 2025).
Executable review	Claims, objections, rebuttals, corrections, and retractions become source-linked artefacts whose downstream effects can be rebuilt.
Governance risk	Formalisation can amplify bureaucratic preference, Goodhart pressure, and authority laundering; profiles must expose scope and contested definitions.
Bounded scope	The near-term target is semantic audit and executable dependency, not autonomous judgement or complete formalisation of science.

The social risk is not secondary. Formalisation can convert a contestable category into infrastructure. A profile for risk, quality, novelty, eligibility, or responsibility may appear technical while encoding institutional interests. SLEngish should therefore represent definition provenance, affected parties, authority, appeal, dissent, and revision. A transparent ontology can still be unjust, but opacity makes contestation harder.

Domain sensitivity is essential. Formal mathematics can use proof kernels once the theorem is correctly formalised. Software can use tests and static analysis while remaining vulnerable to specification error. Laboratory science depends on calibration and contact with measured reality. Clinical and social research involve consent, distributional effects, confounding, and long validation latency. Historical and interpretive disciplines require source criticism and plural argument rather than one universal execution metric. The common substrate is evidence and dependency, not a uniform standard of truth.

Governance must separate capability from authority. A runtime may be able to execute code, query a database, or recommend a laboratory action without being authorised to do so. A model may identify a likely error without having the standing to retract a claim. A reviewer may request a change without controlling the underlying data. SLEngish can represent these boundaries, while institutional processes determine who may accept, reject, publish, or act.

The practical adoption path should begin with narrow checks that have clear evaluators: claim-to-output consistency, citation-span verification, unit and population checks, protocol conformance, retraction propagation, source provenance, and semantic diff. Broader causal or interpretive reasoning can be added only where the representation and evaluation are adequate. The language should expand through demonstrated correction rather than through catalogue growth alone.

A successful executable-science package will remain readable as ordinary scholarship. The formal companion should not force all readers into the technical layer or turn the paper into a generated report. Its role is to expose what prose cannot maintain reliably at scale: exact dependencies, versions, evidence paths, alternative interpretations, and bounded machine actions.

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SLEngish in Relation to Ontologies and Formal Knowledge Representation

What Existing Standards and Formalisms Already Solve

The comparison belongs after the language proposal because otherwise SLEngish is easily defined only by opposition. RDF, OWL, SHACL, SPARQL, PROV, Common Logic, controlled natural languages, Datalog, Answer Set Programming, constraint solvers, temporal formalisms, and workflow standards each solve substantial problems. The research question is not whether they can represent knowledge. It is whether an additional source-level language improves the passage from natural language to auditable, heterogeneous execution.

RDF provides a general graph data model and a mature ecosystem for identifiers, linked data, federation, and interchange. OWL provides description-logic semantics for classes and properties under defined profiles. SHACL validates graph shapes. SPARQL queries RDF datasets. PROV supplies a standard vocabulary for entities, activities, agents, derivation, attribution, and generation. Common Logic provides a family of first-order representations. These capabilities are baselines and targets for interoperability, not obsolete technologies to dismiss.

Controlled natural languages address a different boundary. Attempto Controlled English, Common Logic Controlled English, PENG, and related systems show that readable restricted language can map deterministically to formal structures. Their trade-offs among precision, expressiveness, naturalness, and simplicity have been studied systematically (Kuhn 2014). SLEngish differs in degree and architecture: it sacrifices more surface naturalness, assumes LLM-assisted translation, exposes SSA identity, and uses open path-addressed operator contracts. Those differences require evidence, not rhetorical claims of novelty.

A new representation language should begin by acknowledging what existing formalisms already do well. Controlled Natural Languages provide readable entry points with explicit interpretation rules. Formal semantics offers principled accounts of quantification, discourse, events, modality, and reference. RDF provides a simple graph interchange model; RDFS and OWL support ontology semantics; SHACL validates graph shapes; SPARQL queries graphs; PROV-O represents entities, activities, agents, and derivations. Common Logic provides a first-order interchange framework. Datalog, Answer Set Programming, temporal calculi, SAT, SMT, CSP, theorem provers, and workflow systems each supply mature execution semantics for particular classes of problems.

The relevant comparison is therefore not expressive possibility in the abstract. Almost any finite structure can be encoded in JSON, RDF, relational tables, or a general programming language. The questions are engineering questions: Which representation is easiest for an LLM to generate correctly? Which keeps an n-ary event, its roles, and its provenance locally inspectable? Which permits stable semantic diffs? Which exposes a dependency graph without a separate reconstruction step? Which makes it cheap to route one claim to OWL and another to SMT while retaining common identity and source links? Which failures are visible to a reviewer rather than buried in adapters?

Traditional CNLs often preserve a substantial amount of English surface grammar. That can improve naturalness and direct authoring, but it requires a grammar sophisticated enough to regulate articles, agreement, relative clauses, quantifier scope, anaphora, and permitted verb frames. A representation that is designed primarily as an LLM compilation target can make a different trade. It can abandon subject–verb–object as its computational substrate, expose roles by name, and tolerate a more visibly formal surface. Human users review the semantics rather than pretending to write ordinary prose without ambiguity.

Formal semantic representations such as Discourse Representation Structures are richer than flat triples and can preserve discourse referents, negation, quantification, and anaphoric structure. Their breadth is valuable for interpretation research. They are not automatically ideal as a stable authoring and execution language for scientific packages. Production systems still need source identities, package versioning, module imports, operator contracts, engine routing, coverage, incremental invalidation, effect policies, and proof receipts. SLEngish treats these engineering obligations as part of the language environment rather than as downstream infrastructure.

The most important principle inherited from the earlier survey is that the semantic pivot should not pretend to be the final logic. It should preserve identities, roles, propositions, contexts, evidence, alternatives, and dependency information sufficiently well that specialised lowerings remain possible. Some questions can be answered relationally over the pivot. Others require a temporal formula, a constraint system, a probabilistic model, a transition system, or a proof assistant. The common language coordinates these methods; it does not erase their differences.

This portfolio approach creates its own obligations. Every lowering must declare which source distinctions it preserves, approximates, rejects, or ignores. A proof in the target formalism is useful only when linked to the compilation assumptions. A contradiction produced by one engine must not be silently combined with a default conclusion from another profile. Certificates require a common envelope without forcing a common proof calculus. SLEngish profiles are intended to make these boundaries explicit and testable.

The research claim is narrow enough to falsify. SLEngish should be retained only if its uniform source form reduces generation errors, adapter complexity, review time, and provenance loss on target workloads. If a conventional CNL yields better fidelity and correction cost, the surface should become more natural. If RDF plus SHACL and a rule engine produces equivalent packages with less complexity, SLEngish may remain only a source syntax or be abandoned. A serious alternative earns its place by measurable workflow properties, not by novelty of notation.

The Semantic Web stack is the strongest baseline because it separates concerns deliberately. RDF offers Web-scale identifiers and a uniform graph model. OWL provides description-logic-based ontology profiles with well-understood entailment. SHACL defines validation over RDF graphs. SPARQL provides query and update. PROV-O supplies reusable provenance concepts. JSON-LD connects graph data to familiar serialisation. These are not failed technologies. They solve interoperability problems that any new system would be foolish to ignore.

The mismatch arises from the target workflow. A scientific claim may involve an n-ary event, source attribution, evidence mode, confidence, probability, temporal validity, measurement uncertainty, a causal interpretation, a validation constraint, a derivation rule, a query, and an execution trace. In an RDF-centred implementation these concerns are normally distributed across event nodes, reification or triple terms, named graphs, shapes, ontology axioms, provenance vocabularies, rule engines, query languages, and workflow systems. The architecture is modular, but each boundary introduces identifiers, mappings, serialisations, version rules, and ownership questions.

An LLM compiler must then generate a coordinated artefact across several languages. A human reviewer must reconstruct one semantic unit from dispersed triples and metadata. A runtime must decide which layer owns negation, closure, defaults, time, conflict, and explanation. SLEngish proposes a syntactically uniform source in which a semantic operator and all its named roles remain local to one node. It can still compile to RDF and related standards; the difference is authoring locality and circuit identity, not a claim that triples lack expressive power.

N-ary events illustrate the point. A transfer may include agent, theme, recipient, origin, destination, instrument, time, purpose, and resulting ownership. RDF represents this correctly by reifying the event as a resource and attaching role triples. SLEngish does essentially the same semantically, but keeps the roles in one canonical node whose SSA identity already functions as a graph vertex and cache key. A reviewer can see the event as one unit. A schema can validate required roles. A circuit sees typed input ports without another graph-construction pass.

The open-world assumption of OWL is invaluable for incomplete distributed knowledge. It is not a universal policy. A scientific dataset may be closed for a declared table and revision. A compliance rule may require defaults and exceptions. A planning query may need negation as failure. A contradiction-tolerant review may need paraconsistent status. A numeric claim may require SMT rather than description-logic reasoning. The SLEngish source must therefore name its profile and coverage rather than allowing one backend's assumptions to become ambient.

Interoperability remains a core requirement. A class hierarchy should be exportable to OWL when it falls inside an OWL profile. Graph-shaped facts should be exportable to RDF. Validation constraints should lower to SHACL where semantics align. Provenance should map to PROV-O. Queries may compile to SPARQL, Datalog, SQL, or another declared interpreter. The compiler must document semantic loss in both directions. The project fails if it reinvents mature standards while offering no measurable advantage at the compilation and execution boundary.

The fairest experiment is package-level. Given the same natural-language research material, one team builds an RDF/OWL/SHACL/SPARQL/PROV package and another builds a SLEngish package with equivalent lowerings. Metrics include compiler validity, human correction time, artefact count, semantic locality, source alignment, query accuracy, update impact, proof trace quality, interoperability, and false assurance. The outcome may support a hybrid in which SLEngish is an authoring and circuit language over standard interchange formats. That would still be a successful result.

Logic-programming and solver traditions provide the operational semantics that SLEngish itself does not attempt to replace. Datalog supports recursive, explainable closure over relations. ASP supports defaults, exceptions, alternative stable models, and optimisation. SAT, SMT, and CSP systems support bounded logical and arithmetic constraints. Temporal logics, event calculi, automata, and model checking address change and behaviour. Theorem provers address deductive proof. The relevant question is how reliably a SLEngish fragment can be lowered to each formalism while preserving source alignment and reporting unsupported content.

Graph and ontology standards also provide governance lessons. Stable identifiers, explicit vocabularies, profile restrictions, validation languages, and federated publication are mature practices. A SLEngish implementation should reuse them where possible. A path-addressed operator may export an RDF identity; a type hierarchy may compile to OWL; constraints may compile to SHACL; provenance may map to PROV; graph queries may compile to SPARQL. The adapter must state whether the mapping is exact, partial, approximate, or one-way.

The source-level proposal is justified only where it improves semantic locality. An n-ary event encoded as several triples is not intrinsically defective, but the author or reviewer may find it easier to inspect one canonical statement containing named roles. A path-based contract can keep schema, tests, rendering, and lowering near the operator definition. SSA identities can support diffing and dependency analysis before graph expansion. These are engineering hypotheses about authoring and maintenance, not claims that graph representation is incapable of expressing the content.

Existing approach	Capability that SLEngish should reuse or match
RDF and linked data	Stable identifiers, graph interchange, federation, established serialisations, and a broad tooling ecosystem.
OWL and description logics	Formal class and property semantics with decidable profiles and mature reasoners.
SHACL	Declarative validation of graph shapes, cardinalities, and constraints.
SPARQL	Standard graph query, federation, filtering, aggregation, and update mechanisms.
PROV	Interoperable provenance for entities, activities, agents, use, generation, derivation, and attribution.
Common Logic	A standard framework for exchange among first-order logic dialects.
Controlled natural languages	Deterministic mappings from constrained readable input to formal semantics.
Datalog and ASP	Rule evaluation, recursion, defaults, exceptions, model generation, and explainability.
SMT, CSP, SAT, and model checking	Efficient decision procedures, witnesses, unsatisfiable cores, and bounded verification.
Workflow and research-object standards	Execution metadata, packaging, environments, artefact lineage, and reproducibility.

The Additional Hypothesis and the Conditions under Which It Should Be Rejected

Abstract interpretation and symbolic execution occupy different places in the architecture. Abstract interpretation was developed to compute safe approximations of program behaviour over domains too large or infinite to enumerate. It replaces concrete states with abstract properties and propagates them through sound transfer functions until a fixpoint or frontier is reached. In language, one established use certifies a neural model's behaviour over a formally defined family of textual perturbations. That guarantee concerns model

stability under the chosen perturbation relation; it does not prove that every perturbation preserves meaning or that the original answer is true.

A second use is more speculative and central here. The concrete domain can be a finite or symbolically represented set of admissible semantic interpretations. The abstraction may record possible referents, intervals, worlds, modalities, or evidence states. A may result holds in at least one represented interpretation. A must result holds in all. Cannot means no represented interpretation supports it. Unknown records insufficient coverage or abstraction precision. This gives a disciplined semantics to ambiguity, but soundness remains conditional on the interpretation set and transfer rules.

Symbolic execution complements this summary by searching for concrete witnesses. An abstract analysis can report that a violation remains possible. A symbolic engine, SMT solver, model checker, or planner can attempt to construct an interpretation and path that realises it. A replayable witness strengthens the finding. An infeasible witness or unsatisfiable core can reveal that the abstraction merged entities, widened time, ignored an exception, or treated a modal claim as actual.

This leads to counterexample-guided semantic refinement. Traditional CEGAR refines a program abstraction after a spurious counterexample. In the language setting, the refinement targets semantic commitments. A conflict is mapped back to source spans and the exact identity link, temporal boundary, quantifier, modality, or ontology choice responsible. An LLM or human revises only that local commitment. The rest of the semantic graph remains stable, and the analysis is replayed. This is more informative than asking a model to reread an entire document and try again.

Different engines contribute different evidence. Datalog provides recursive closure and explainable rules. ASP handles defaults, exceptions, and optimisation. SMT and CSP handle arithmetic, finite structure, and resource constraints. Automata and chart methods handle ordered patterns. Model checking handles temporal behaviour. Symbolic execution handles path witnesses. Theorem provers and proof assistants handle deductive obligations. Statistical engines estimate empirical parameters. A uniform result envelope must record semantics, coverage, assumptions, certificate, and frontier without claiming that these methods are interchangeable.

The engine router is therefore a typed compiler, not an LLM guess at runtime. Verb schemas and module profiles declare compatible lowerings. A query states the guarantee it requests. The planner selects engines under capability and budget contracts. The canonical result cites the source nodes, lowering version, engine, proof or witness, and unresolved assumptions. If no compatible engine exists, the correct output is an unsupported obligation rather than an improvised answer.

The reference architecture should be understood as a compiler and verification toolchain rather than as one agent prompt. Each stage materialises an immutable or content-addressed artefact so that errors can be assigned, replayed, and compared. Learning may improve a producer or theory, but it may not retroactively change the meaning of a published run.

The first artefact is the source package. It contains canonical text or data, revision digest, language, structure, channels, and stable spans. Paragraphs, headings, tables, quotations, equations, code blocks, and external records retain identity. The source package is lexical authority: it states what was present, not what it means. The second artefact is the task contract, which defines the question, source scope, world, jurisdiction, desired guarantee, allowed assumptions, budgets, and output policy. Semantic demand follows from the task.

The third layer is the evidence graph. Deterministic producers add structure and exact observations. Controlled parsers add high-confidence analyses for recognised fragments. LLMs propose entities, events, relations, definitions, and alternatives linked to spans. Retrieval systems add candidate background material with versioned sources. Coverage records state what was examined and what remained unavailable. Nothing becomes accepted merely because it appears in this graph.

The fourth layer is the interpretation set and grounding state. Candidate nodes are typed against selected libraries. Identity links, temporal locations, modalities, quantities, source relations, and contexts are checked. Alternatives remain separate. Human decisions are recorded as commitments rather than erased into the parser output. SLEnglish is the canonical language in which these nodes and commitments are expressed.

The fifth layer is the executable theory. Task-relevant nodes are lowered into one or more formal engines. Query-first materialisation can avoid formalising irrelevant parts of a long document, but it must expose coverage. A relational query, rule closure, temporal property, constraint system, symbolic path problem, or

theorem is generated together with a source map. Compiler loss and unsupported constructs become diagnostics.

The sixth layer is execution and evidence. Engines return bindings, models, unsatisfiable cores, proofs, counterexamples, traces, statistical outputs, or stopped states. Independent checkers replay certificates where possible. Execution-level provenance records which nodes and versions contributed. A result inherits the weakest relevant guarantee: a verified proof over model-proposed premises remains a verified derivation over proposed premises, not an unconditional fact.

The final layer is constrained realisation. A renderer or LLM can produce accessible prose, but every sentence must be licensed by the canonical result. It may not convert conditional into unconditional language, possibility into actuality, conflict into consensus, or missing evidence into evidence of absence. Round-trip tests compare the realisation with the semantic record. The generated explanation is a view, not the authority.

Governance surrounds every stage. Libraries, compilers, prompts, models, engines, policies, and renderers have versions and digests. Learning agents work in staging. Publication requires conformance tests, semantic mutations, benchmark evidence, and explicit approval. Production reads immutable releases. Current-world knowledge packs have dates and jurisdictions. This architecture turns failure into a named artefact—unsupported type, unresolved identity, incomplete coverage, ambiguity, compilation loss, unsatisfiable theory, unavailable engine, or unlicensed sentence—rather than a vague hallucination.

Term	Explanation
Semantic Web standards	RDF, OWL, SHACL, SPARQL, and PROV-O are mature interoperability, reasoning, validation, query, and provenance standards (W3C 2012, 2013, 2017, 2026).
Common Logic	ISO/IEC 24707 supplies a framework for a family of first-order logic languages and remains an important interchange target.
CNL trade-offs	Precision, expressiveness, naturalness, and simplicity cannot all be maximised; SLEngish intentionally sacrifices surface naturalness for regular execution (Kuhn 2014).
Counterexample-guided refinement	The CEGAR pattern motivates local semantic repair from infeasible witnesses and unsatisfiable cores rather than global “try again” prompting.

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Counterexample-guided refinement	The CEGAR pattern motivates local semantic repair from infeasible witnesses and unsatisfiable cores rather than global “try again” prompting.

The additional hypothesis is that a uniform semantic source layer can reduce the cost of translating, reviewing, and revising knowledge that must move among several backends. The same statement envelope can contain an observation, a reported claim, a qualified identity relation, a rule template, a query, or an operational request while its path and profile determine specialised semantics. A human reviewer sees one regular notation; an adapter sees a typed contract; an existing standard remains available as an interchange or reasoning target.

This hypothesis has several failure modes. The operator libraries may fragment into incompatible local dialects. The generic envelope may hide rather than expose backend-specific semantics. Path-addressed files may create maintenance and federation problems. LLM translation may remain too unreliable or too expensive. Human reviewers may prefer forms, diagrams, or ordinary controlled English. RDF and established standards may provide equivalent locality when used well. A code generator or typed host language may prove a simpler target.

A fair comparison therefore uses the same source material and questions. One implementation can use an RDF, OWL, SHACL, SPARQL, and PROV stack; another can use SLEngish with adapters to equivalent components; a third can use direct structured LLM output; a fourth can use a conventional CNL or frame

representation. The study measures valid compilation, semantic correction time, query accuracy, false assurance, artefact count, source traceability, update impact, interoperability, execution cost, and user comprehension.

A likely engineering outcome is hybrid. SLEngish can operate as an authoring and semantic-diff layer over RDF and formal engines, as a compact representation for LLM-to-tool boundaries, or as a specialised representation for evidence and claims while mathematical and workflow artefacts retain their native languages. The comparison therefore measures where SLEngish provides locality, readability, provenance, and compilation advantages and where established standards remain the better representation.

The comparison is technical and empirical. The relevant measures are correctness, locality, compositionality, governance, performance, tooling, migration cost, semantic preservation, and the ability to show how a natural-language commitment became a computational object. RDF, OWL, SHACL, SPARQL, PROV, Common Logic, Datalog, ASP, graph systems, and conventional CNLs provide strong baselines for these measurements.

CHAPTER 9

Research Programme and Evaluation

Research Questions, Baselines, and Measurements

The programme treats syntax, operator granularity, parameter commonality, path organisation, translation strategy, reasoning profiles, and user interfaces as separate hypotheses. Each can be evaluated against strong baselines while the others are held constant. Frozen corpora, canonical fixtures, semantic mutation tests, independent parsers, and backend adapters provide the experimental controls needed to identify which choices add value.

The primary outcome is false assurance: a determinate answer presented with stronger support than the source, coverage, formalisation, or execution justifies. Evaluation therefore rewards justified abstention, preserved ambiguity, explicit unknown states, correct source anchoring, and accurate dependency tracking in addition to task accuracy.

Evaluation should separate source-to-representation fidelity, representation quality, execution correctness, human review economics, and institutional usability. A strong solver cannot compensate for a mistranslation. A faithful graph that no reviewer can understand may not be deployable. An efficient language whose governance allows silent semantic drift is not a scientific infrastructure. Each layer needs its own baselines and measurements.

The first research question concerns semantic coverage. Can open, namespaced verb libraries represent ordinary event structure, scientific claims, and selected typological distinctions without proliferating incompatible microdialects? Coverage should be measured on held-out corpora from science, policy, software, and social analysis. Failures are classified as missing verb, missing parameter, ontology gap, unsupported discourse structure, context gap, or irreducible ambiguity. The registry evolves only from observed pressure.

The second question concerns translation fidelity. Can LLMs compile prose into valid SLEngish while preserving negation, modality, quantifier scope, evidence, identity, time, and participant roles? The relevant metric is not parse success alone. It includes invented assumptions, lost distinctions, source-span accuracy, alternative preservation, correction time, and query equivalence. A stricter language may improve validity while worsening semantic drift; both must be measured.

The third question concerns cognitive utility. Do explicit operators for evidence, clusivity, identity, collective agency, causal depth, reversibility, scale, and uncertainty improve specific reasoning tasks? Experiments should compare plain English, annotated English, SLEngish forms, and graphical views. Effects may be local rather than general. An operator that helps source monitoring but harms comprehension elsewhere belongs in an optional profile, not the universal core.

The fourth question concerns execution architecture. Does the SSA dependency form improve incremental recomputation, provenance, explanation, and multi-engine integration? Benchmarks should include full versus incremental replay, cache validity under contexts, parallel scheduling, recursive rules, alternative worlds, and non-monotonic updates. A correct full replay is the reference evaluator. Performance gains never justify semantic divergence.

The fifth question concerns interoperability. Can SLEngish lower to RDF/OWL/SHACL, Datalog, ASP, SQL/SPARQL, SMT, temporal systems, and workflow tools with explicit preservation reports? Can existing artefacts be imported without semantic laundering? A small number of deep, tested adapters is more persuasive than many superficial ones. The project should publish both successful and failed mappings.

The seventh question concerns social and scientific governance. Can competing libraries coexist without hidden priority? Can operator authority, cultural provenance, jurisdiction, and deprecation be represented? Do users understand guarantee labels and unknown states? Does formalisation create bureaucratic pressure or false objectivity? Technical success without governance usability would not justify deployment in science.

Evaluation must be decomposed because one aggregate score hides where a system failed. Source-anchor accuracy measures whether nodes point to the correct spans. Verb and parameter precision measure structural compilation. Type validity measures conformity to libraries. Identity-link precision and calibration measure referential grounding. Temporal, modal, evidential, and causal accuracy measure specialised semantics.

Candidate-set coverage evaluates whether legitimate alternatives were retained. Query equivalence tests consequence rather than surface similarity.

False assurance receives its own metric. A case counts as false assurance when the system presents a determinate accepted result although a missing source, unresolved ambiguity, invalid grounding, incomplete coverage, unsupported lowering, or solver limitation should have prevented it. Appropriate abstention is rewarded. Coverage, latency, cost, and answer rate remain relevant, but they are reported beside false assurance rather than combined into a flattering number.

Human evaluation should ask narrow questions. Experts judge whether a node preserves the source, whether an identity link is justified, whether an alternative is material, whether a witness is feasible, whether assumptions are complete, and whether a rendered explanation changes modality. They should not be asked only whether an answer looks good. Inter-rater disagreement is data about semantic ambiguity and should not be forced into one gold label without analysis.

Metamorphic tests provide scalable pressure. Adding a negation should invert only relevant polarity. Replacing a direct observation with a report should change evidence mode but not event content. Changing inclusive to exclusive we should alter membership. Splitting one entity into two should invalidate identity-dependent conclusions. Moving a date across a legal threshold should affect only temporally scoped rules. These tests can be generated for every verb contract.

Mutation testing evaluates execution adapters and verifiers. Operators are altered to drop a parameter, swap roles, ignore a context, collapse alternatives, reuse a stale cache, or omit a source. Tests should detect the mutation. A high answer score with a low mutation score indicates decorative verification. Differential execution compares independent implementations or alternative lowerings on the same canonical package.

Security testing treats source and packages as hostile. Prompt injection, Unicode confusables, malicious imports, cyclic dependencies, resource exhaustion, forged provenance, stale signatures, and effect escalation are tested. A semantic node cannot authorise an operation merely by describing it. Capability checks and package signatures are verified outside the LLM. Privacy and copyright rules constrain captured prompts, datasets, and generated artefacts.

Baselines must be strong. They include direct LLM answering, chain-of-thought-equivalent prompting where permitted, tool-augmented LLMs without persistent semantic artefacts, JSON schema outputs, a conventional frame CNL, RDF/OWL/SHACL/PROV, single-formalism translation, and an optional circuit backend without SLEngish. Ablations remove SSA IDs, explicit alternatives, source spans, open verb libraries, profiles, or independent verification to reveal which complexity earns its cost.

Evaluation object	Measures and comparison
Natural-language compilation	Parse validity, operator selection, role accuracy, scope accuracy, identity accuracy, alternative recall, source-span precision, and unsupported-content detection.
Human correction	Time to understand, number of edits, locality of edits, disagreement between reviewers, and stability after correction.
Semantic representation	Canonical stability, type coverage, library fragmentation, path resolution, migration cost, and ability to express required distinctions.
Reasoning and query	Answer accuracy, proof or witness validity, contradiction handling, unknown calibration, coverage compliance, and profile fidelity.
Update behaviour	Affected-node precision, selective rebuilding, equivalence with full replay, stale-cache detection, and version trace quality.
Interoperability	Round-trip preservation with RDF, PROV, rule engines, solvers, databases, code, and research-object formats.
Security and policy	Injection resistance, malicious import handling, capability enforcement, provenance forgery detection, resource limits, privacy, and copyright constraints.
Human-facing output	Claim-licensing compliance, condition preservation, uncertainty preservation, readability, and source traceability.
Overall value	False assurance, correct abstention, review cost, implementation complexity, latency, storage, maintenance, and domain adoption.

Baseline	Reason for inclusion
Direct LLM response	Establishes the value and limitations of the simplest natural-language interface.

Baseline	Reason for inclusion
Tool-using LLM without persistent semantics	Tests whether explicit semantic artefacts add value beyond ordinary tool orchestration.
Typed JSON or function schema	Provides a strong minimal structured-output baseline with low implementation cost.
Conventional frame CNL	Tests whether greater naturalness or subject–verb–object structure improves authoring and review.
RDF, OWL, SHACL, SPARQL, and PROV	Provides the mature graph, ontology, validation, query, and provenance baseline.
Single-formalism translation	Tests whether a direct Datalog, ASP, SMT, temporal, or proof target is preferable in a bounded domain.
Generated conventional code	Tests whether executable source in an established programming language is simpler and more effective.
SLEngish with alternative backends	Separates the value of the source representation from the value of any particular runtime.
Expert-authored formal representation	Provides an upper-bound or diagnostic reference for translation and formalisation quality.

Falsification Criteria and Research Sequence

The programme separates architectural invariants from provisional mechanisms. Invariants are commitments whose removal would produce a different project: one regular outer sentence shape; explicit named roles; stable semantic node identity; proposition separated from commitment; visible context and provenance; deterministic validation before reasoning; profile-declared semantics; preserved alternatives where consequence changes; and constrained realisation. Even these can be falsified in principle, but they define what the current research is trying to test.

The open verb architecture is provisional in its details. The hypothesis is that semantic operators should be first-class, namespaced, typed, and versioned. The number and granularity of operators are empirical. Corpus studies may show that some distinctions belong as parameter values, type refinements, or rendering conventions rather than verbs. Confusion matrices, definition burden, library fragmentation, and migration cost determine whether verbs are merged or split.

The visible SSA syntax is also provisional. Explicit identifiers may improve diffing, caching, provenance, and review, or they may burden readers without sufficient gain. A comparison should test user-visible IDs against compiler-generated IDs, parenthesised frames, indentation, and structured editors. The canonical IR may retain SSA even if the default human view hides it.

Folder-derived namespaces are a pragmatic hypothesis. They make package structure visible and align semantic work with version control, but they impose one primary hierarchy and create refactoring risks. Manifest-only names, content-addressed modules, URIs, and federation schemes should be tested. Stable semantic identity must not depend accidentally on a file move.

The initial reasoning profile is provisional. Horn-style relational rules with explicit negation and four-valued status are attractive for a bounded core, but early use cases may require defaults, temporal reasoning, constraints, or probabilistic models sooner. The source language should remain stable while backends evolve. No backend earns universal status through implementation convenience.

RDF interoperability provides a direct baseline for the representation layer. Comparative experiments can test whether Semantic Web tooling offers equivalent authoring locality, execution, traceability, and governance with lower complexity. The same method applies to Datalog, ASP, JSON-based schemas, frame CNLs, graph models, and direct LLM tool use. Results determine whether SLEngish is best used as a primary source language, a thin authoring notation, or a specialised interchange layer.

A decisive falsification criterion is failure to reduce false assurance or review cost relative to strong baselines. Additional criteria are semantic fragmentation across libraries, poor translation fidelity, excessive authoring cost, unstable operator semantics, and inability to preserve provenance across backend conversions. These criteria provide concrete reasons to simplify, replace, merge, or retire language features as evidence accumulates.

Reversibility requires migration tools and frozen corpora. Every release has a conformance suite and canonical examples. Deprecation retains old readers. Bridges state whether translations are equivalent,

approximate, or lossy. Published scientific packages remain executable under their locked environment. The language can evolve without rewriting history.

Design decision	How it can change without invalidating the whole programme
Visible SSA identities	Compare visible IDs, editor-generated IDs, hidden canonical IDs, frames, and indentation while retaining stable internal identity.
`\$` and `~` reference distinction	Test alternative sigils or typed references; preserve the semantic difference between value flow and node reference.
Arbitrary-depth path namespaces	Compare folder paths with manifest names, content-addressed identities, URI aliases, and federated registries. File moves must not silently change semantic identity.
One operator per semantic distinction	Merge, split, or demote candidates to parameters, enum values, types, or rendering conventions based on corpus evidence and correction cost.
Common parameter registry	Promote only stable cross-domain roles; retain path-qualified local parameters where generalisation would erase meaning.
LLM-first translation	Compare direct expert authoring, grammar-based parsers, encoders, hybrid compilers, and multiple model families.
Interpreter portfolio	Add or remove backends according to task fit, semantic preservation, complexity, cost, and certificate quality.
Four-valued status	Compare paraconsistent, three-valued, provenance-weighted, and profile-specific conflict semantics.
Canonical text surface	Retain the source representation only if it improves review, diffing, portability, and reproducibility relative to structured editors or binary IRs.

Failure condition	Required response
No reduction in false assurance	Narrow or abandon the language layer even if answer accuracy or demonstration quality appears high.
Correction cost exceeds baselines	Simplify the surface, improve tooling, reduce operator granularity, or restrict use to machine-to-machine exchange.
Semantic library fragmentation	Strengthen governance and bridge modules; if interoperability remains poor, prefer established domain formalisms.
Backend-specific leakage dominates	Allow profile-specific syntax or use direct formal languages rather than preserving a misleading common envelope.
Path organisation proves unstable	Move semantic identity to manifests or content addressing while retaining readable aliases.
Round-trip renderings strengthen claims	Restrict generation, improve licensing checks, or require human-authored publication prose.
Security or authority boundaries remain unclear	Prohibit operational effects and retain SLEngish as a declarative audit representation.
Existing standards perform equivalently at lower cost	Use SLEngish only as a thin view or discontinue it.
No domain shows durable value	Publish the negative result and stop standardisation rather than expanding the vocabulary.

Research phase	Bounded outcome and decision
Reference kernel	Implement independent parsers, path resolution, schemas, SSA references, contexts, canonicalisation, source spans, strict rules, queries, and proof-linked diagnostics.
Linguistic and scientific profiles	Test evidence, time, identity, collective agency, causation, units, uncertainty, coverage, and English-construction libraries on curated corpora.
Interpreter adapters	Implement a small set of serious lowerings, document semantic preservation, and compare query-first with broader materialisation.
Human compiler tools	Evaluate natural-language translation, alternatives, local correction, semantic diff, controlled rendering, and expert authoring.
Executable research object	Create one bounded package containing prose, data, code, claims, reviews, revisions, queries, and independent reproduction.
External replication and governance	Require another implementation or team, held-out domains, migration evidence, issue records, and documented failures before stabilising a public standard.

A research corpus should accompany the implementation from the beginning. It should contain ordinary English constructions, typologically motivated distinctions, scientific claims, contradictory reports, incomplete coverage, identity ambiguity, collective attribution, temporal updates, normative rules, and adversarial cases. Every revision of an operator contract runs against the frozen corpus. New cases are added with an explanation of the failure they expose.

Independent implementation is especially important because a language can appear coherent when parser, compiler, runtime, benchmark, and renderer share the same assumptions. Two parsers should agree on canonical structure. Independent adapters should agree where semantic preservation is claimed. A third party should be able to modify a research package and predict which conclusions require rebuilding. Disagreement is diagnostic evidence rather than an inconvenience to hide.

Negative results and migration records are part of the research output. A candidate operator that confuses reviewers, an ontology bridge that loses meaning, or a backend that cannot preserve provenance identifies a design boundary. Conformance tests and migration tools then convert those failures into changes that can be compared on the same corpus.

Conclusion

Natural language is the main medium in which scientific questions, explanations, disagreements, and institutional decisions are expressed. Its economy leaves source, identity, scope, coverage, authority, time, and alternative interpretations implicit when context is sufficient for human readers. Computational use requires those variables to become explicit at the points where they change denotation, inference, or action.

SLEngish provides that explicit boundary. English grammar supplies the baseline structures to normalise; cross-linguistic research supplies additional semantic distinctions; philosophy and systems research refine identity, agency, causation, boundaries, uncertainty, and collective concepts. The resulting distinctions are represented as path-addressed verbs with named parameters, stable SSA identities, contexts, evidence links, rules, and queries. The operator and parameter tables in the appendices are the concrete synthesis of this analysis.

Natural-language translation can be produced by an LLM, a grammar, an expert, or a hybrid compiler and then checked by deterministic parsers, schemas, type systems, source anchors, and semantic tests. Execution can use rules, graphs, ontologies, solvers, temporal models, theorem provers, generated code, simulations, semantic circuits, and specialised scientific tools. Adapters state the fragment they support and the guarantees they preserve.

Executable science emerges when selected claims, evidence, assumptions, transformations, and dependencies can answer bounded questions, expose contradictions, support review, and respond predictably to correction. SLEngish acts as the semantic source layer that connects these objects while keeping their interpretation and provenance inspectable.

The programme evaluates SLEngish by translation fidelity, source traceability, false-assurance rate, correction cost, review effort, interoperability, and execution behaviour. Syntax, operator granularity, parameter registries, namespace conventions, and backend mappings remain measurable design variables. The operator library becomes progressively smaller, clearer, and more useful as comparative experiments identify which distinctions consistently improve representation and reasoning.

APPENDIX A

Core Grammar and Canonicalisation

This appendix states the candidate source-level grammar independently of any runtime. The grammar is intentionally small. Operator files and profiles supply semantics; the parser supplies only a deterministic envelope and explicit references.

Grammar element	Normative description
Document	A sequence of declarations and comments encoded in UTF-8. Canonical source contains only declarations; comments and editor metadata do not affect semantic hashes unless explicitly included by a profile.
Declaration	<code>`@ssa-id operator-path parameter value ... `</code> One declaration creates one SSA node.
SSA identifier	<code>`@`</code> followed by a lowercase kebab-case identifier unique within the declared source scope. Revision creates a new identifier or epoch rather than mutation.
Operator path	A relative slash-separated path of arbitrary depth ending in <code>`.sle`</code> , such as <code>`knowledge/evidence/report.sle`</code> . Resolution uses root manifests, imports, versions, and content digests.
Parameter	A declared role name followed by one value. The operator contract defines legality, type, cardinality, repetition, conditions, and canonical order.
Value reference	<code>`\$name`</code> requests the primary value produced by an earlier node and creates a data dependency.
Node reference	<code>`~name`</code> refers to the semantic node itself and preserves proposition identity, provenance, alternatives, and status.
Atomic value	A declared lowercase symbol, boolean, canonical number, date-time, unit, enum value, context, entity identifier, or registered datatype literal.
Quoted value	A UTF-8 string delimited by double quotation marks with defined escaping. Language, datatype, or source role is stated by parameters or contract.
Missing-value atom	A declared semantic atom such as <code>`unknown`</code> , <code>`not-applicable`</code> , <code>`withheld`</code> , <code>`not-measured`</code> , <code>`conflicted`</code> , or <code>`open-end`</code> ; omission remains distinct.
Terminator	A final period separated by whitespace from the preceding value. The period in <code>`.sle`</code> belongs to the operator path and is not a statement terminator.
Physical line	A declaration may be displayed on one or several physical lines. Canonical exchange uses one normalised token stream per declaration.
Comment	An editor or source profile may support comments, but comments do not introduce executable meaning unless materialised through a declared annotation operator.

Canonicalisation rule	Required behaviour
Whitespace	Use one ASCII space between canonical tokens; normalise line endings; remove non-semantic indentation and trailing whitespace.
Identifiers	Use lowercase kebab-case for local machine identities. Human-readable and multilingual names remain labels.
Operator paths	Resolve root aliases, normalise relative path segments, prohibit ambiguous traversal, and record package digest plus file version. Arbitrary directory depth is allowed.
Parameter order	Serialise parameters in the order declared by the operator contract. Repeated unordered roles sort by canonical identity; ordered roles preserve semantic order.
Numbers and dates	Use locale-independent canonical forms and declared datatypes. Physical quantities retain a separate unit value.
Booleans and missing values	Use only registered canonical atoms; do not equate omission, false, zero, empty, unknown, or not applicable.
Aliases	Resolve aliases before semantic hashing while retaining the authoring alias in source metadata.

Canonicalisation rule	Required behaviour
Versions	Pin imported roots and external schemas through a lock record. A path without a resolved version is not publication-canonical.
Revision	Do not mutate nodes in place. Record supersession, derivation, and source-revision lineage.
Effects	Declarative interpretation is the default. Any external effect requires a separately authorised operational profile and an execution receipt.
Unknown operators	Reject or retain as explicitly opaque according to policy. Never infer a contract from the spelling alone.
Hash scope	Hash the resolved operator identity, canonical parameters and values, selected context/profile identifiers, and any contract-declared semantic metadata.

```

document    = declaration*
declaration = binding SP operator-path (SP parameter SP value)* SP "."
binding     = "@" identifier
operator-path = path-segment ("/" path-segment)* ".sle"
parameter   = identifier | qualified-parameter
value       = atom | string | number | boolean | missing | value-ref | node-ref
value-ref   = "$" identifier
node-ref    = "~" identifier

```

APPENDIX B

Common Parameter Registry

The registry collects parameter names that recur across linguistic, epistemic, scientific, social, temporal, and systemic operators. A common parameter is retained when its semantics are stable enough for type checking and interoperability; specialised roles use path-qualified names. The table provides the shared vocabulary used by the operator contracts in the following appendices.

Term	Explanation
id	Stable local or qualified identity of an entity, node, artefact, or external record.
name	Canonical symbolic name. Labels for readers remain separate and may be multilingual.
label	Human-readable label; never used as the sole machine identity.
description	Natural-language documentation that is informative but not executable unless compiled separately.
type	Declared semantic or data type of a value, entity, proposition, operator, or result.
parent	Immediate supertype, parent context, parent module, or parent artefact as defined by the verb schema.
module	Qualified package module that owns or supplies the referenced object.
version	Semantic version, immutable release identifier, or external artefact version.
namespace	Readable root or relative operator path used for resolution. Canonical semantic identity also records the package digest, file version, and exported contract.
profile	Named reasoning, validation, evidence, identity, scientific, or operational profile.
context	World, discourse, institutional, quoted, hypothetical, or execution context in which the node applies.
world	Explicit possible, actual, fictional, simulated, counterfactual, or reported world.
jurisdiction	Legal, policy, organisational, or community authority under which a rule or claim applies.
language	Human language or controlled-language variant used by labels, source material, or rendering.
source	Evidence source, originating entity, starting point of a relation, or source artefact according to schema.
source-span	Exact source location supporting or giving rise to the node.
source-depth	Number and structure of testimonial or derivational hops between claim and primary source.
content-digest	Cryptographic or content-addressed digest of a canonical artefact.
valid-time	Interval during which the represented fact, rule, state, licence, or authority is valid.
transaction-time	Interval during which the record exists in the knowledge system.
time	Instant, interval, relative temporal expression, or temporal node associated with the proposition.
start	Beginning instant or boundary of an interval, process, validity period, or action.
end	Ending instant or boundary; 'open-end' may denote an intentionally unclosed interval.
duration	Measured or symbolic length of an event, process, state, or validity period.
frequency	Rate or count of repetitions under an explicit temporal or event scope.
interval	Named temporal, numeric, spatial, or uncertainty interval.
scale	Micro, meso, macro, spatial, temporal, organisational, or measurement scale at which a claim is made.
agent	Intentional or conventionally acting participant responsible for an event role.
actor	Participant that performs a role without presupposing full intention or moral agency.
causer	Participant, event, condition, or structure occupying a causal source role.
causee	Participant caused or enabled to act by a causative construction.

Term	Explanation
patient	Participant undergoing an event or being materially affected.
theme	Participant moved, transferred, perceived, communicated, or otherwise central to an event.
recipient	Participant receiving an object, information, role, permission, or effect.
beneficiary	Participant for whose benefit or detriment an event is performed.
experiencer	Participant undergoing a perception, emotion, cognition, or internal state.
stimulus	Entity or event giving rise to an experience or perception.
instrument	Tool, method, channel, or artefact used to perform an event.
origin	Initial place, owner, state, version, or source endpoint.
destination	Final place, owner, state, version, or target endpoint.
path	Ordered route, derivation chain, causal chain, or transition sequence.
location	Spatial, virtual, organisational, or document location.
possessor	Entity participating in a possession, control, stewardship, kin, or inalienable relation.
possessed	Entity, role, resource, body part, right, or relation associated with a possessor.
whole	Entity treated as the whole in a mereological relation.
part	Component, portion, member, ingredient, phase, or layer related to a whole.
member	Individual or subcollective participating in a collective.
collective	Group, swarm, organisation, institution, population, or aggregate.
speaker	Current discourse producer or quoted speaker.
addressee	Current discourse recipient or quoted addressee.
holder	Agent holding a belief, obligation, permission, role, right, attitude, or perspective.
anchor	Reference point for spatial, temporal, discourse, social, or perspectival interpretation.
perspective	Explicit viewpoint, logophoric centre, information state, or evaluative frame.
source-agent	Person, organisation, model, sensor, or process responsible for producing evidence.
target	Endpoint, compared entity, affected proposition, goal, or object selected by the verb schema.
proposition	Reference to semantic content without necessarily committing to its truth.
premise	Proposition used as an input to a derivation, argument, or analysis.
conclusion	Proposition produced or supported by a derivation, argument, or analysis.
pattern	Proposition template, graph pattern, event pattern, or constraint pattern containing variables.
basis	Reason, criterion, model, observation, rule, definition, or similarity dimension supporting a relation.
mode	Verb-specific semantic subtype when separate qualified verbs would be unnecessarily granular.
polarity	Positive or explicit negative status; omission never means negative.
degree	Scalar, ordinal, fuzzy, or graded strength under a declared scale.
strength	Causal, evidential, modal, normative, or statistical strength as defined by a profile.
confidence	Confidence in an estimate, mapping, claim, source, or compilation—not the event probability itself.
probability	Probability assigned to an event, hypothesis, outcome, or model under an explicit interpretation.
uncertainty	Uncertainty value or structured uncertainty object, including measurement and model uncertainty.
unit	Declared unit of measurement; required by scientific profiles for physical quantities.
value	Literal, symbol, entity, collection, or computed output carried by the node.
from	Initial value or state in a change relation.
to	Final value or state in a change relation.
min	Lower numeric or ordered bound.
max	Upper numeric or ordered bound.

Term	Explanation
threshold	Boundary whose approach or crossing changes status, behaviour, obligation, or phase.
classifier	Counting or ontological classifier applied to a quantity or referent.
cardinality	Exact, minimum, maximum, interval, or named number category.
quantity	Count, magnitude, ratio, amount, or quantity object.
distribution	Statistical, spatial, temporal, responsibility, or participant distribution.
aspect	Internal temporal contour such as perfective, imperfective, progressive, habitual, telic, or iterative.
tense	Temporal relation between event time, reference time, and speech or evaluation time.
remoteness	Distance category such as immediate, recent, remote, historical, near-future, or remote-future.
realis-status	Actual, possible, intended, hypothetical, counterfactual, or otherwise irrealis status.
evidence-mode	Observed, sensed, measured, reported, inferred, remembered, simulated, generated, or unknown source mode.
mirativity	Expected, unexpected, counterexpected, surprising, or model-revising information status.
identity-mode	Strict, material, structural, processual, causal, genealogical, informational, functional, role, or legal continuity.
agency-mode	Individual, delegated, distributed, emergent, centralised, coordinated, or fictionalised agency.
decision-mode	Vote, command, consensus, market outcome, delegated decision, algorithmic selection, or other decision architecture.
causal-layer	Immediate, contributory, enabling, structural, mediated, preventive, triggering, or feedback causal level.
reversibility	Reversible, costly-to-reverse, practically irreversible, or physically irreversible status.
effect-order	Direct, second-order, third-order, cascade, or recursively propagated consequence.
feedback	Positive, negative, delayed, stabilising, destabilising, or mixed feedback regime.
boundary	Sharp, fuzzy, operational, disputed, dynamic, permeable, or scale-dependent boundary status.
mandate	Rule, vote, delegation, charter, consent, or authority that licences representation or action.
coordination	None, local, stigmergic, negotiated, centralised, distributed, synchronised, or emergent coordination.
control	Capacity to select, prevent, redirect, repair, or authorise an outcome.
responsibility	Typed allocation of causal, role, control, knowledge, benefit, or remedial responsibility.
owner	Entity governing a module, artefact, resource, definition, or right.
authority	Institution, person, standard, contract, or capability that validates or permits an action.
method	Measurement, analysis, inference, retrieval, experimental, or compilation method.
dataset	Versioned dataset, table, stream, corpus, or data slice used or produced.
model	Scientific, statistical, causal, simulation, machine-learning, or semantic model.
code	Versioned executable artefact or source module referenced by a scientific or operational node.
environment	Container, hardware, library lock, solver configuration, or execution environment.
query	Reference to the query for which a result, proof, diagnostic, or explanation is produced.
return	Variable, field, proposition, proof component, or aggregate requested by a query.
order	Deterministic ordering criterion for query or execution output.
limit	Result, time, memory, search, token, or iteration limit.
witness	Replayable example, model, path, trace, or counterexample supporting an execution result.
proof	Proof object, certificate, derivation, or checkable receipt.
status	Accepted, proposed, derived, rejected, unknown, conflicted, superseded, or other governed state.

Term	Explanation
coverage	Scope, completeness policy, revision, exclusions, and method supporting positive or negative conclusions.
missing-reason	Unknown, not measured, withheld, not applicable, unsupported, unavailable, or conflicted reason.
capability	Explicit permission required for an operational interpreter or external effect.
budget	Time, memory, cost, search, token, energy, or external-service allowance.
determinism	Deterministic, seeded, stochastic, nondeterministic, or externally contingent execution contract.
monotonicity	Monotonic, antimonotonic, non-monotonic, unknown, or context-dependent semantic behaviour.
effect	Pure, state-read, state-write, file, network, tool, notification, or other operational effect class.
cache	Cacheability and invalidation contract for a node or operator.
severity	Informational, warning, error, blocking, or domain-specific priority of a diagnostic or constraint.
priority	Rule, default, scheduling, review, or conflict-resolution priority.
when	Premise or condition that activates a rule, constraint, decision, or query.
unless	Exception or defeater that blocks a default, rule, constraint, or action.
then	Conclusion, effect, or output activated by a rule.
alternative	Materially distinct semantic candidate retained without premature selection.
assumption	Explicit premise introduced by a model, human, method, or profile rather than directly supported by the source.
alias	Local import or symbol alias resolved to a qualified canonical identity before hashing.
guarantee	Declared guarantee envelope such as syntactic, source-grounded, model-relative, bounded, proved, or replay-verified.
output-type	Declared type of the value or semantic node produced by a registered verb.
required	List or schema object describing parameters that every use of a verb must provide.
optional	List or schema object describing parameters that a verb may accept without changing its identity.
replacement	New node, value, rule, or commitment that supersedes an earlier target.
variable	Typed logical or query variable introduced for binding in patterns and rules.
cause	Event, agent, condition, or proposition presented as the cause of a change under a declared causal profile.
purpose	Goal or intended function attributed to an action without conflating intention with achieved effect.
relation	Declared relation symbol or relation node used as the content of a higher-order construction.
role	Named semantic, social, institutional, discourse, or provenance role.
rule	Registered rule or derivation object used to license an inference.
alternatives	Collection of materially distinct semantic candidates retained for comparison or suspended judgement.
politeness	Register, deference, familiarity, face-management, or conventional politeness value.
distance	Spatial, temporal, social, causal, epistemic, or threshold-relative distance under a named scale.
rank	Social, institutional, taxonomic, preference, or result rank under an explicit ordering.
relationship	Typed interpersonal, kin, institutional, contractual, or discourse relationship.
community	Community, speech group, institution, or governance body relative to which a convention applies.
direction	Absolute, intrinsic, relative, temporal, causal, or scalar direction.
process	Process node or process description relevant to continuation, unboundedness, or transformation.
intention	Declared intention, plan, or goal state of an agent, kept distinct from effect and responsibility.

Term	Explanation
mechanism	Causal, operational, biological, social, or computational mechanism linking source and target.
delay	Temporal lag between cause and effect, signal and response, or action and consequence.
incentive	Reward, penalty, target, metric, or institutional pressure capable of changing behaviour.
result	Result artefact or outcome produced, compared, replicated, or consumed by another node.

APPENDIX C

Candidate Operator Library

The table is the main implementation output of the analysis in the book. It combines ordinary semantic operations with operators motivated by English grammar, cross-linguistic constructions, epistemology, scientific practice, social organisation, and systems reasoning. Each retained path corresponds to a `.sle` contract defining parameters, types, semantic conditions, examples, tests, and compatible interpreters.

Term	Explanation
<code>slenglish/core/import.sle</code>	Loads a versioned module into the current package and binds an optional local alias. Required: module version. Optional: namespace alias content-digest profile. Output: module handle.
<code>slenglish/core/define-verb.sle</code>	Declares a typed semantic operator and its executable contract. Required: name output-type. Optional: module version required optional cardinality determinism monotonicity effect profile description. Output: verb schema.
<code>slenglish/core/define-type.sle</code>	Declares a semantic or data type. Required: name. Optional: parent module version description. Output: type handle.
<code>slenglish/core/define-role.sle</code>	Declares a reusable parameter or semantic role. Required: name. Optional: type cardinality description module version. Output: role handle.
<code>slenglish/core/define-enum.sle</code>	Declares a closed or versioned set of symbolic values. Required: name value. Optional: description module version. Output: enum handle.
<code>slenglish/core/define-unit.sle</code>	Declares a unit, dimension, conversion relation, and authority. Required: name. Optional: type basis authority version description. Output: unit handle.
<code>slenglish/core/alias.sle</code>	Binds a local readable name to a stable qualified symbol. Required: name target. Optional: context version. Output: alias handle.
<code>slenglish/core/entity.sle</code>	Creates or registers a typed entity without asserting further contingent facts. Required: type. Optional: id name label description source content-digest context. Output: entity value.
<code>slenglish/core/variable.sle</code>	Creates a typed variable for patterns, rules, and queries. Required: type. Optional: name description context. Output: variable value.
<code>slenglish/core/context.sle</code>	Creates a world, discourse, legal, quoted, hypothetical, or execution context. Required: name. Optional: parent world jurisdiction profile valid-time source coverage description. Output: context handle.
<code>slenglish/core/assert.sle</code>	Commits a proposition as accepted in a named context. Required: proposition context. Optional: source source-span valid-time authority status. Output: commitment.
<code>slenglish/core/deny.sle</code>	Commits explicit negative polarity for a proposition in a named context. Required: proposition context. Optional: source source-span valid-time authority status. Output: negative commitment.
<code>slenglish/core/claim.sle</code>	Records that a source or agent advances a proposition without automatically accepting it. Required: proposition source. Optional: source-span source-depth evidence-mode confidence valid-time context status. Output: claim.
<code>slenglish/core/hypothesize.sle</code>	Places a proposition in a hypothesis context for testing rather than acceptance. Required: proposition context. Optional: source source-span confidence assumption profile status. Output: hypothesis commitment.
<code>slenglish/core/assume.sle</code>	Introduces an explicit premise for a bounded analysis. Required: proposition context. Optional: source basis valid-time status. Output: assumption commitment.
<code>slenglish/core/quote.sle</code>	Records the exact or controlled content attributed to a source or speaker. Required: proposition source. Optional: source-span speaker addressee time context language. Output: quotation.
<code>slenglish/core/retract.sle</code>	Withdraws a prior commitment while preserving historical provenance. Required: target. Optional: source basis time authority context. Output: retraction.
<code>slenglish/core/supersede.sle</code>	Declares that a new node replaces an earlier node for specified uses. Required: target replacement. Optional: basis time context version. Output: supersession relation.
<code>slenglish/core/rule.sle</code>	Declares a strict, default, causal, temporal, probabilistic, or profile-specific derivation. Required: when then profile. Optional: unless priority context source description. Output: rule handle.
<code>slenglish/core/constraint.sle</code>	Declares a validation, consistency, cardinality, numeric, temporal, or policy obligation. Required: when profile. Optional: unless severity context source description. Output: constraint handle.

Term	Explanation
<code>slenglish/core/query.sle</code>	Creates a first-class query with explicit context, guarantee, coverage, and return contract. Required: pattern return. Optional: context profile coverage guarantee order limit budget description. Output: query handle.
<code>slenglish/core/invoke.sle</code>	Requests execution of a registered operational circuit under explicit capability. Required: target capability. Optional: value context budget environment effect source. Output: invocation request.
<code>slenglish/core/explain.sle</code>	Requests a source-linked explanation of a result, proof, conflict, or dependency. Required: target. Optional: query profile limit language perspective. Output: explanation request.
<code>slenglish/core/diagnose.sle</code>	Creates a typed diagnostic for ambiguity, loss, invalidity, conflict, or unsupported semantics. Required: target status. Optional: source-span severity basis alternative description. Output: diagnostic.
<code>knowledge/logic/and.sle</code>	Constructs a conjunction whose operands must be interpreted under the same named profile. Required: proposition proposition. Optional: context order. Output: conjunctive proposition.
<code>knowledge/logic/or.sle</code>	Constructs an inclusive disjunction unless a profile declares another mode. Required: proposition proposition. Optional: context mode order. Output: disjunctive proposition.
<code>knowledge/logic/not.sle</code>	Constructs explicit logical negation rather than absence or failure to derive. Required: proposition. Optional: context profile. Output: negative proposition.
<code>knowledge/logic/implies.sle</code>	Constructs material, strict, or profile-specific implication without itself asserting either side. Required: premise conclusion. Optional: profile context. Output: implication proposition.
<code>knowledge/logic/equivalent.sle</code>	Declares equivalence under a named semantics and scope. Required: source target. Optional: profile context basis strength. Output: equivalence proposition.
<code>knowledge/logic/each.sle</code>	Binds universal quantification over a typed domain. Required: variable type. Optional: when context profile. Output: quantified proposition.
<code>knowledge/logic/some.sle</code>	Binds existential quantification over a typed domain. Required: variable type. Optional: when context profile. Output: quantified proposition.
<code>knowledge/logic/none.sle</code>	States that no member of a typed, covered domain satisfies a pattern. Required: variable type pattern. Optional: coverage context profile. Output: negative quantified proposition.
<code>knowledge/logic/exactly.sle</code>	Constrains the number of matches to an exact cardinality. Required: pattern cardinality. Optional: coverage context profile. Output: cardinality constraint.
<code>knowledge/logic/at-least.sle</code>	Constrains the number or degree of matches to a lower bound. Required: pattern min. Optional: coverage context profile unit. Output: lower-bound constraint.
<code>knowledge/logic/at-most.sle</code>	Constrains the number or degree of matches to an upper bound. Required: pattern max. Optional: coverage context profile unit. Output: upper-bound constraint.
<code>knowledge/entity/classify.sle</code>	Places an entity in a type without claiming strict identity with the type. Required: source type. Optional: context valid-time basis source-span. Output: classification proposition.
<code>knowledge/entity/exist.sle</code>	States existence in an explicit world, interval, or model. Required: source context. Optional: world valid-time evidence-mode source-span. Output: existence proposition.
<code>knowledge/entity/equal.sle</code>	States strict substitutive identity under an explicitly named profile. Required: source target. Optional: context basis valid-time. Output: identity proposition.
<code>knowledge/entity/different.sle</code>	States non-identity under an explicitly named profile. Required: source target. Optional: context basis valid-time. Output: difference proposition.
<code>knowledge/entity/name.sle</code>	Associates an entity with a label, identifier, or name in a language and scope. Required: source label. Optional: language context valid-time authority. Output: naming proposition.
<code>knowledge/property/state.sle</code>	Attributes a typed state or property value to an entity. Required: source value. Optional: type unit valid-time context basis. Output: state proposition.
<code>knowledge/property/change.sle</code>	Represents change of a property between explicit states or values. Required: source from to. Optional: time duration cause context. Output: change proposition.
<code>knowledge/relation/relate.sle</code>	Constructs a declared relation when no more specialised verb is available. Required: source target type. Optional: context valid-time basis degree. Output: relation proposition.

Term	Explanation
knowledge/relation/compare.sle	Compares two values or entities along an explicit dimension and comparator. Required: source target basis. Optional: mode value unit context. Output: comparison proposition.
knowledge/relation/depend-on.sle	Represents semantic, causal, build, resource, or logical dependency. Required: source target. Optional: mode strength context valid-time. Output: dependency proposition.
knowledge/relation/associate.sle	Represents an explicitly non-causal association with method and strength. Required: source target. Optional: method basis strength confidence context dataset. Output: association proposition.
language/event/occur.sle	Creates an event with explicit participants, time, and context. Required: type. Optional: agent actor patient theme location time duration context source-span. Output: event proposition.
language/event/act.sle	Represents an action by an agent or actor toward an optional target. Required: agent. Optional: target patient theme instrument purpose time location context. Output: action proposition.
language/event/transfer.sle	Represents transfer of a theme between origin and destination or recipient. Required: theme. Optional: agent origin destination recipient instrument time purpose context. Output: transfer event.
language/event/move.sle	Represents spatial, virtual, organisational, or state-space movement. Required: theme origin destination. Optional: agent path instrument time duration context. Output: motion event.
language/event/create.sle	Represents bringing an entity, record, artefact, or state into existence. Required: agent target. Optional: instrument method time context source. Output: creation event.
language/event/destroy.sle	Represents termination or destruction with explicit patient and causal context. Required: agent patient. Optional: instrument method time context reversibility. Output: destruction event.
language/event/communicate.sle	Represents transmission of semantic content between agents or systems. Required: source-agent recipient proposition. Optional: instrument language time context evidence-mode. Output: communication event.
language/event/perceive.sle	Represents perception by an experiencer through an optional sense or instrument. Required: experiencer stimulus. Optional: instrument mode time location context. Output: perception event.
language/event/experience.sle	Represents an internal state or experience without imposing external observability. Required: experiencer value. Optional: stimulus degree time context. Output: experience event.
language/event/change.sle	Represents a transition of an entity, relation, or system between states. Required: target from to. Optional: causer time duration reversibility context. Output: change event.
language/event/repeat.sle	Represents repeated realisations of an event under frequency and distribution. Required: proposition frequency. Optional: interval distribution context. Output: pluractional event.
language/event/distribute.sle	Distributes an action or relation across members, times, or objects. Required: proposition distribution. Optional: collective member interval context. Output: distributed event.
language/event/reciprocate.sle	Represents mutually directed instances of a relation among participants. Required: relation collective. Optional: member time context. Output: reciprocal proposition.
language/event/reflexive.sle	Represents a participant filling two co-referential semantic roles. Required: agent. Optional: patient theme role context. Output: reflexive event.
language/event/causative.sle	Adds an explicit causer and causee to an event structure. Required: causer proposition. Optional: causee instrument causal-layer time context. Output: causative event.
language/event/applicative.sle	Promotes a beneficiary, instrument, location, or recipient into a core event role. Required: proposition target mode. Optional: context. Output: applicative event.
language/event/antipassive.sle	Suppresses or backgrounds a patient while foregrounding an agent and activity. Required: proposition agent. Optional: patient aspect context. Output: antipassive event.
language/event/middle.sle	Represents an event whose affected participant is central without a fully external agent. Required: patient proposition. Optional: agent aspect context. Output: middle event.
language/event/passive.sle	Represents a valency view that backgrounds or omits the agent while retaining event identity. Required: proposition patient. Optional: agent context evidence-mode. Output: passive event view.
language/event/incorporate.sle	Represents semantic incorporation of a noun-like role into an event concept. Required: proposition theme. Optional: mode context. Output: incorporated event.

Term	Explanation
language/event/serialise.sle	Represents a tightly linked sequence of events sharing participants or temporal structure. Required: proposition proposition. Optional: agent theme order time context. Output: serial-event structure.
language/time/locate.sle	Locates an event, state, claim, or rule at an instant or interval. Required: target time. Optional: context valid-time transaction-time remoteness. Output: temporal proposition.
language/time/before.sle	Orders one temporal entity strictly before another. Required: source target. Optional: context basis. Output: temporal relation.
language/time/after.sle	Orders one temporal entity strictly after another. Required: source target. Optional: context basis. Output: temporal relation.
language/time/overlap.sle	States that two intervals overlap under an interval semantics. Required: source target. Optional: degree context. Output: temporal relation.
language/time/during.sle	Places one event or interval wholly or partly inside another. Required: source target. Optional: mode context. Output: temporal relation.
language/time/begin.sle	Marks the inception boundary of a state, process, or validity interval. Required: target start. Optional: context evidence-mode. Output: boundary event.
language/time/end.sle	Marks the termination boundary of a state, process, or validity interval. Required: target end. Optional: context evidence-mode. Output: boundary event.
language/time/continue.sle	States temporal continuation across an interval without requiring material identity. Required: target interval. Optional: identity-mode context evidence-mode. Output: continuation proposition.
language/time/habitual.sle	Classifies a proposition as characteristically repeated rather than a single occurrence. Required: proposition. Optional: frequency interval context. Output: habitual proposition.
language/time/progressive.sle	Presents an event from within its unfolding, without asserting completion. Required: proposition. Optional: time context. Output: progressive proposition.
language/time/perfective.sle	Presents an event as a bounded whole under a named viewpoint. Required: proposition. Optional: time context. Output: perfective proposition.
language/time/imperfective.sle	Presents internal duration, habit, or incompleteness without a closed boundary. Required: proposition. Optional: time aspect context. Output: imperfective proposition.
language/time/perfect.sle	Relates a prior event or state to a later reference time through current relevance. Required: proposition time. Optional: anchor context. Output: perfect proposition.
language/time/prospective.sle	Represents a future-oriented event relative to a reference state. Required: proposition time. Optional: anchor confidence context. Output: prospective proposition.
language/time/remote-past.sle	Places a proposition in a culturally or operationally remote past category. Required: proposition. Optional: time remoteness context. Output: remote-past proposition.
language/time/recent-past.sle	Places a proposition in a recent-past category. Required: proposition. Optional: time remoteness context. Output: recent-past proposition.
language/time/near-future.sle	Places a proposition in a near-future category. Required: proposition. Optional: time remoteness confidence context. Output: near-future proposition.
language/time/remote-future.sle	Places a proposition in a remote-future category. Required: proposition. Optional: time remoteness confidence context. Output: remote-future proposition.
language/time/iterative.sle	Represents discrete repetition of a proposition. Required: proposition cardinality. Optional: frequency interval context. Output: iterative proposition.
language/time/durative.sle	Marks extended duration as semantically relevant. Required: proposition duration. Optional: interval context. Output: durative proposition.
language/time/punctual.sle	Marks an event as point-like relative to the chosen scale. Required: proposition time. Optional: scale context. Output: punctual proposition.
language/time/telic.sle	Marks an event as oriented toward an inherent endpoint. Required: proposition target. Optional: time context. Output: telic proposition.
language/time/atelic.sle	Marks an event as lacking an inherent endpoint under the selected description. Required: proposition. Optional: time context. Output: atelic proposition.
knowledge/evidence/observe.sle	Links a proposition to direct visual or multimodal observation. Required: proposition source-agent. Optional: time location source-span confidence context. Output: observational evidence.

Term	Explanation
knowledge/evidence/sense.sle	Links a proposition to direct sensory evidence with an explicit sensory mode. Required: proposition source-agent mode. Optional: time location confidence context. Output: sensory evidence.
knowledge/evidence/measure.sle	Links a proposition or value to an instrumented measurement. Required: proposition source-agent instrument value unit. Optional: uncertainty method time dataset source-span confidence context. Output: measurement evidence.
knowledge/evidence/report.sle	Links a proposition to testimony, publication, message, or other report. Required: proposition source. Optional: source-agent source-span source-depth time confidence context. Output: report evidence.
knowledge/evidence/quote.sle	Links a proposition to exact or controlled quoted language. Required: proposition source source-span. Optional: speaker addressee language time context. Output: quotation evidence.
knowledge/evidence/infer.sle	Links a proposition to an abductive, statistical, or defeasible inference from premises. Required: proposition premise. Optional: method model confidence context. Output: inferential evidence.
knowledge/evidence/deduce.sle	Links a proposition to a deductive derivation and optional proof. Required: proposition premise. Optional: rule proof profile context. Output: deductive evidence.
knowledge/evidence/abduce.sle	Records a best-explanation hypothesis rather than an established consequence. Required: proposition premise. Optional: model alternatives confidence context. Output: abductive evidence.
knowledge/evidence/remember.sle	Links a proposition to personal or institutional memory with time and confidence. Required: proposition source-agent. Optional: time source-depth confidence context. Output: memory evidence.
knowledge/evidence/simulate.sle	Links a proposition to a simulation run and its model assumptions. Required: proposition model. Optional: code environment dataset method uncertainty context. Output: simulation evidence.
knowledge/evidence/generate.sle	Links content to a generative model or algorithm without treating generation as truth. Required: proposition source-agent model. Optional: method confidence source-span context. Output: generated evidence status.
knowledge/evidence/trace.sle	Links a proposition to an executable trace, log, or provenance chain. Required: proposition source. Optional: path content-digest environment time context. Output: trace evidence.
knowledge/epistemic/know.sle	Attributes knowledge to a holder under a named epistemic profile. Required: holder proposition. Optional: basis source time context. Output: knowledge attitude.
knowledge/epistemic/believe.sle	Attributes a belief without asserting its content in the enclosing context. Required: holder proposition. Optional: confidence basis time context. Output: belief attitude.
knowledge/epistemic/suspect.sle	Attributes a low-commitment belief or suspicion. Required: holder proposition. Optional: confidence basis time context. Output: suspicion attitude.
knowledge/epistemic/doubt.sle	Attributes doubt toward a proposition. Required: holder proposition. Optional: degree basis time context. Output: doubt attitude.
knowledge/epistemic/suspend.sle	Records suspension of judgement between competing propositions. Required: holder proposition. Optional: alternative basis time context. Output: suspended attitude.
knowledge/epistemic/revise.sle	Records a change in epistemic commitment with explicit earlier and later states. Required: holder target replacement. Optional: basis time context. Output: belief-revision event.
knowledge/epistemic/conflict.sle	Records mutually incompatible supported propositions without explosion. Required: proposition proposition. Optional: source context severity. Output: epistemic conflict.
knowledge/epistemic/unknown.sle	States that a value or proposition is not known under current coverage. Required: target. Optional: missing-reason coverage context. Output: unknown status.
knowledge/epistemic/unknowable.sle	States principled or model-relative non-knowability with explicit basis. Required: target basis. Optional: profile context. Output: unknowability claim.
knowledge/epistemic/discover.sle	Records transition from unknown or unaccepted to supported. Required: holder proposition. Optional: evidence-mode time source context. Output: discovery event.
knowledge/epistemic/expect.sle	Records an expectation before evidence arrives. Required: holder proposition. Optional: probability confidence model time context. Output: expectation attitude.
knowledge/epistemic/counterexpect.sle	Marks evidence as contrary to an explicit prior expectation. Required: holder proposition premise. Optional: degree time context. Output: counterexpectation relation.

Term	Explanation
knowledge/epistemic/surprise.sle	Marks information as producing a model-relevant expectation update. Required: holder proposition. Optional: mirativity degree model time context. Output: surprise event.
knowledge/modality/possible.sle	Places a proposition in at least one admissible world or interpretation. Required: proposition context. Optional: world basis profile. Output: possibility proposition.
knowledge/modality/necessary.sle	Places a proposition in every admissible world or interpretation under a profile. Required: proposition context. Optional: world coverage basis profile. Output: necessity proposition.
knowledge/modality/impossible.sle	States that no admissible world or interpretation satisfies a proposition. Required: proposition context. Optional: coverage basis profile. Output: impossibility proposition.
knowledge/modality/contingent.sle	States that a proposition holds in some but not all admissible worlds. Required: proposition context. Optional: coverage profile. Output: contingency proposition.
knowledge/modality/counterfactual.sle	Creates a context contrary to accepted actuality and records its antecedent. Required: proposition context. Optional: assumption world basis. Output: counterfactual proposition.
knowledge/modality/realis.sle	Marks a proposition as presented as actual or realised in the named context. Required: proposition context. Optional: time evidence-mode. Output: realis proposition.
knowledge/modality/irrealis.sle	Marks a proposition as intended, desired, hypothetical, conditional, or otherwise non-actual. Required: proposition context. Optional: mode time holder. Output: irrealis proposition.
social/deontic/require.sle	Creates an obligation with holder, authority, conditions, and validity. Required: holder proposition authority. Optional: when unless jurisdiction valid-time severity context. Output: obligation.
social/deontic/permit.sle	Creates an explicit permission. Required: holder proposition authority. Optional: when unless jurisdiction valid-time context. Output: permission.
social/deontic/forbid.sle	Creates a prohibition with explicit scope and authority. Required: holder proposition authority. Optional: when unless jurisdiction valid-time severity context. Output: prohibition.
social/deontic/recommend.sle	Creates a non-binding recommendation with rationale and target. Required: holder proposition source. Optional: basis strength valid-time context. Output: recommendation.
social/deontic/obligate.sle	Assigns responsibility for satisfying an existing obligation. Required: holder target. Optional: authority mandate valid-time context. Output: assignment proposition.
language/speech-act/ask.sle	Represents a question act by a speaker to an addressee. Required: speaker addressee proposition. Optional: time context language. Output: speech event.
language/speech-act/command.sle	Represents an authoritative directive without automatically granting capability. Required: speaker addressee proposition. Optional: authority mandate time context. Output: speech event.
language/speech-act/promise.sle	Represents a commitment by a speaker to a future proposition. Required: speaker addressee proposition. Optional: time valid-time context. Output: promise event.
language/speech-act/request.sle	Represents a non-authoritative request. Required: speaker addressee proposition. Optional: time politeness context. Output: request event.
language/speech-act/warn.sle	Represents a warning about risk, consequence, or obligation. Required: speaker addressee proposition. Optional: basis severity time context. Output: warning event.
language/speech-act/declare.sle	Represents an institutional or performative declaration under authority. Required: speaker proposition authority. Optional: jurisdiction valid-time time context. Output: declaration event.
language/speech-act/concede.sle	Records acceptance of a proposition within an argument while retaining other disagreement. Required: speaker proposition. Optional: addressee basis time context. Output: concession event.
language/reference/refer.sle	Creates a discourse or document reference to an entity under an explicit resolution status. Required: source target. Optional: context source-span confidence perspective. Output: reference proposition.
language/reference/demonstrate.sle	Constructs a demonstrative reference relative to speaker, addressee, and spatial or discourse anchor. Required: source anchor. Optional: speaker addressee mode distance context. Output: demonstrative reference.
language/reference/anaphor.sle	Links a later expression to an earlier discourse antecedent. Required: source target. Optional: context source-span confidence. Output: anaphoric relation.

Term	Explanation
language/reference/cataphor.sle	Links an earlier expression to a later discourse referent. Required: source target. Optional: context source-span confidence. Output: cataphoric relation.
language/reference/logophor.sle	Links a pronoun-like element to the perspective holder of reported thought or speech. Required: source holder. Optional: perspective context source-span. Output: logophoric relation.
language/reference/proximate.sle	Marks a third-person referent as discourse-proximate or central. Required: source. Optional: anchor perspective context. Output: proximate status.
language/reference/obviative.sle	Marks a third-person referent as less central relative to a proximate anchor. Required: source anchor. Optional: perspective context. Output: obviative status.
language/reference/inclusive-we.sle	Constructs a first-person plural set that necessarily includes speaker and addressee. Required: speaker addressee. Optional: member collective context. Output: participant-set value.
language/reference/exclusive-we.sle	Constructs a first-person plural set that includes speaker and excludes addressee. Required: speaker addressee. Optional: member collective context. Output: participant-set value.
language/reference/speaker.sle	Returns or declares the speaker of a context. Required: context speaker. Optional: none. Output: speaker value.
language/reference/addressee.sle	Returns or declares the addressee set of a context. Required: context addressee. Optional: none. Output: addressee value.
language/number/singular.sle	Marks a referent as one individuated unit under a classifier or identity criterion. Required: source. Optional: classifier context. Output: number proposition.
language/number/dual.sle	Marks exactly two participants or units as a grammatical or semantic category. Required: source. Optional: classifier context. Output: number proposition.
language/number/trial.sle	Marks exactly three participants or units as a grammatical or semantic category. Required: source. Optional: classifier context. Output: number proposition.
language/number/paucal.sle	Marks a culturally or profile-defined small plurality rather than an exact count. Required: source. Optional: min max classifier context. Output: number proposition.
language/number/plural.sle	Marks a plurality without committing to exact cardinality. Required: source. Optional: min classifier context. Output: number proposition.
language/number/associative.sle	Constructs an individual plus contextually associated others, such as "Alice and her group." Required: source. Optional: member context basis. Output: associative collective.
language/number/collective.sle	Treats multiple members as a collective unit for a named predicate. Required: collective member. Optional: basis context. Output: collective-number proposition.
language/number/distributive.sle	Distributes a proposition separately across members or units. Required: proposition member. Optional: collective distribution context. Output: distributed proposition.
language/possession/alienable.sle	Represents ownership or association that can normally be transferred or severed. Required: possessor possessed. Optional: valid-time authority context. Output: possession proposition.
language/possession/inalienable.sle	Represents a constitutive, kin, body-part, or culturally non-transferable relation. Required: possessor possessed. Optional: mode valid-time context. Output: inalienable relation.
language/possession/ownership.sle	Represents legal or normative ownership under explicit jurisdiction and time. Required: owner possessed. Optional: authority jurisdiction valid-time source context. Output: ownership proposition.
language/possession/kin.sle	Represents a kinship relation rather than generic ownership. Required: possessor possessed type. Optional: valid-time context source. Output: kin relation.
language/possession/body-part.sle	Represents a body-part relation with identity and temporal scope. Required: whole part. Optional: valid-time identity-mode context. Output: body-part relation.
language/possession/temporary.sle	Represents temporary custody, use, or control without ownership. Required: possessor possessed. Optional: start end authority context. Output: temporary possession.
language/possession/stewardship.sle	Represents responsibility to maintain or govern a resource without full ownership. Required: possessor possessed. Optional: authority responsibility valid-time context. Output: stewardship relation.
knowledge/mereology/part-of.sle	Represents structural part-whole relation. Required: part whole. Optional: boundary valid-time context. Output: mereological proposition.

Term	Explanation
knowledge/mereology/member-of.sle	Represents membership in a group without implying material constitution. Required: member collective. Optional: valid-time mandate context. Output: membership proposition.
knowledge/mereology/portion-of.sle	Represents a quantity or portion of a mass or resource. Required: part whole. Optional: quantity unit context. Output: portion relation.
knowledge/mereology/ingredient-of.sle	Represents compositional inclusion as an ingredient. Required: part whole. Optional: quantity unit valid-time context. Output: ingredient relation.
knowledge/mereology/phase-of.sle	Represents a temporal or process phase of a persisting entity or process. Required: part whole. Optional: start end identity-mode context. Output: phase relation.
knowledge/mereology/layer-of.sle	Represents a spatial, conceptual, organisational, or system layer. Required: part whole. Optional: scale boundary context. Output: layer relation.
knowledge/mereology/constituted-by.sle	Represents constitution without strict identity. Required: whole part. Optional: basis identity-mode valid-time context. Output: constitution proposition.
knowledge/mereology/boundary-of.sle	Represents an operational, physical, legal, or fuzzy boundary of an entity. Required: part whole. Optional: boundary scale valid-time context. Output: boundary relation.
language/space/locate.sle	Locates an entity relative to a place, coordinate system, or frame. Required: source location. Optional: anchor time context scale. Output: spatial proposition.
language/space/north-of.sle	Places one entity north of another in an absolute geographic frame. Required: source target. Optional: anchor scale context. Output: spatial relation.
language/space/south-of.sle	Places one entity south of another in an absolute geographic frame. Required: source target. Optional: anchor scale context. Output: spatial relation.
language/space/east-of.sle	Places one entity east of another in an absolute geographic frame. Required: source target. Optional: anchor scale context. Output: spatial relation.
language/space/west-of.sle	Places one entity west of another in an absolute geographic frame. Required: source target. Optional: anchor scale context. Output: spatial relation.
language/space/uphill.sle	Places or directs an entity uphill relative to terrain. Required: source anchor. Optional: degree path context. Output: spatial relation.
language/space/downhill.sle	Places or directs an entity downhill relative to terrain. Required: source anchor. Optional: degree path context. Output: spatial relation.
language/space/upriver.sle	Places or directs an entity toward a river source. Required: source anchor. Optional: path context. Output: spatial relation.
language/space/downriver.sle	Places or directs an entity toward a river mouth or flow direction. Required: source anchor. Optional: path context. Output: spatial relation.
language/space/seaward.sle	Places or directs an entity toward the sea. Required: source anchor. Optional: path context. Output: spatial relation.
language/space/landward.sle	Places or directs an entity away from the sea and toward land. Required: source anchor. Optional: path context. Output: spatial relation.
language/space/inside.sle	Places an entity inside a boundary or container. Required: source target. Optional: boundary time context. Output: spatial relation.
language/space/outside.sle	Places an entity outside a boundary or container. Required: source target. Optional: boundary time context. Output: spatial relation.
language/space/near.sle	Represents context- and scale-dependent proximity. Required: source target. Optional: degree scale anchor context. Output: spatial relation.
language/space/far.sle	Represents context- and scale-dependent distance. Required: source target. Optional: degree scale anchor context. Output: spatial relation.
language/space/toward.sle	Represents directed orientation or motion toward a target. Required: source target. Optional: path degree time context. Output: direction relation.
language/space/away.sle	Represents directed orientation or motion away from a target. Required: source target. Optional: path degree time context. Output: direction relation.
language/space/left-of.sle	Places one entity left of another relative to an explicit viewer or frame. Required: source target. Optional: anchor perspective context. Output: spatial relation.
language/space/right-of.sle	Places one entity right of another relative to an explicit viewer or frame. Required: source target. Optional: anchor perspective context. Output: spatial relation.

Term	Explanation
language/space/front-of.sle	Places one entity in front of another under an intrinsic or relative frame. Required: source target. Optional: anchor perspective context. Output: spatial relation.
language/space/behind.sle	Places one entity behind another under an intrinsic or relative frame. Required: source target. Optional: anchor perspective context. Output: spatial relation.
language/space/above.sle	Places one entity above another in a gravity, intrinsic, or diagram frame. Required: source target. Optional: anchor scale context. Output: spatial relation.
language/space/below.sle	Places one entity below another in a gravity, intrinsic, or diagram frame. Required: source target. Optional: anchor scale context. Output: spatial relation.
language/space/through.sle	Represents a path through a region or boundary. Required: source target path. Optional: time context. Output: path relation.
language/space/from.sle	Represents a source endpoint in a motion or orientation relation. Required: source origin. Optional: time context. Output: origin relation.
language/space/to.sle	Represents a destination endpoint in a motion or orientation relation. Required: source destination. Optional: time context. Output: destination relation.
language/discourse/same-reference.sle	States that central participants of linked clauses are co-referential. Required: source target. Optional: anchor context source-span. Output: switch-reference relation.
language/discourse/different-reference.sle	States that central participants of linked clauses are distinct. Required: source target. Optional: anchor context source-span. Output: switch-reference relation.
language/discourse/continue-topic.sle	Maintains the current discourse topic across a unit boundary. Required: source target. Optional: context perspective. Output: topic relation.
language/discourse/switch-topic.sle	Marks a change of discourse topic. Required: source target. Optional: context perspective. Output: topic relation.
language/discourse/foreground.sle	Marks a proposition as foregrounded in discourse or event structure. Required: proposition. Optional: context perspective. Output: discourse status.
language/discourse/background.sle	Marks a proposition as background, circumstance, or supporting frame. Required: proposition. Optional: context perspective. Output: discourse status.
language/discourse/focus.sle	Marks a constituent or proposition as informational focus. Required: target. Optional: alternative context perspective. Output: focus status.
language/discourse/contrast.sle	Places a target in an explicit contrast set. Required: target alternative. Optional: basis context perspective. Output: contrast relation.
language/discourse/presuppose.sle	Records content treated as backgrounded commitment by an utterance. Required: proposition source. Optional: context speaker source-span. Output: presupposition.
language/discourse/implicate.sle	Records defeasible implicated content rather than asserted content. Required: proposition source. Optional: basis confidence context. Output: implicature.
language/discourse/quote-direct.sle	Represents direct quotation with speaker and exact source span. Required: speaker proposition source-span. Optional: addressee language time context. Output: direct quotation.
language/discourse/report-indirect.sle	Represents reported discourse under an explicit perspective anchor. Required: speaker proposition. Optional: source addressee perspective time context. Output: indirect report.
language/perspective/anchor.sle	Creates a perspective centre for evaluation, reference, or report. Required: holder context. Optional: time source-span. Output: perspective handle.
language/perspective/logophoric.sle	Evaluates a referent or proposition relative to a reported consciousness. Required: holder target. Optional: context source-span. Output: logophoric relation.
language/perspective/egophoric.sle	Marks privileged personal authority for a state, intention, or action. Required: holder proposition. Optional: evidence-mode context. Output: egophoric status.
language/perspective/source.sle	Sets the epistemic source whose information state controls interpretation. Required: source-agent context. Optional: time source-span. Output: source perspective.
language/perspective/self.sle	Represents self-oriented reference relative to an explicit holder. Required: holder target. Optional: context. Output: self relation.
language/perspective/pivot.sle	Selects the privileged participant for inverse, empathy, or reference computations. Required: anchor context. Optional: basis. Output: perspective pivot.

Term	Explanation
language/perspective/empathy.sle	Marks relative discourse empathy or alignment without asserting moral approval. Required: holder target. Optional: degree context. Output: empathy relation.
social/relation/address-deferential.sle	Marks a deferential form of address and its social relation. Required: speaker addressee. Optional: rank context language. Output: address event.
social/relation/address-familiar.sle	Marks a familiar or intimate form of address. Required: speaker addressee. Optional: relationship context language. Output: address event.
social/relation/honor.sle	Elevates the social status of a referent in the current utterance or context. Required: speaker target. Optional: rank context language. Output: honorific relation.
social/relation/humble-self.sle	Lowers or backgrounds the speaker's status relative to another. Required: speaker target. Optional: rank context language. Output: humilific relation.
social/relation/elevate-other.sle	Marks respectful elevation of another participant. Required: speaker target. Optional: rank context language. Output: honorific relation.
social/relation/kinship.sle	Represents socially recognised kin relation under a community schema. Required: source target type. Optional: community context valid-time. Output: social relation.
social/relation/rank.sle	Represents rank or hierarchy within an institution or context. Required: source value. Optional: authority valid-time context. Output: rank proposition.
social/relation/role.sle	Represents occupancy of a social or institutional role. Required: holder value. Optional: collective authority mandate valid-time context. Output: role proposition.
social/relation/mandate.sle	Represents authority to act or speak for a collective. Required: holder collective mandate. Optional: authority valid-time context. Output: mandate proposition.
social/relation/authorize.sle	Grants a capability or permission under an authority. Required: authority holder capability. Optional: valid-time jurisdiction context. Output: authorization proposition.
social/relation/represent.sle	States that an agent represents a collective or constituency. Required: agent collective. Optional: mandate authority valid-time context. Output: representation proposition.
social/relation/include.sle	Includes a participant in a group, pronoun set, process, or decision scope. Required: collective member. Optional: basis valid-time context. Output: inclusion proposition.
social/relation/exclude.sle	Excludes a participant from a group, pronoun set, process, or decision scope. Required: collective member. Optional: basis valid-time context. Output: exclusion proposition.
social/relation/taboo.sle	Marks a term, action, or referent as restricted under a community and context. Required: target authority. Optional: language valid-time context severity. Output: taboo status.
social/relation/register.sle	Marks formal, informal, ceremonial, technical, or other register. Required: target mode. Optional: language community context. Output: register status.
social/relation/politeness.sle	Represents a politeness strategy and relation without reducing it to truth conditions. Required: speaker addressee mode. Optional: degree rank context language. Output: politeness relation.
science/quantity/count.sle	Counts individuated units under an explicit classifier and coverage. Required: source cardinality. Optional: classifier coverage context. Output: count proposition.
science/quantity/measure.sle	Associates a magnitude with a unit, method, and uncertainty. Required: source value unit. Optional: uncertainty method instrument time context. Output: measurement proposition.
science/quantity/classifier.sle	Applies an ontological or grammatical counting classifier to a referent. Required: source classifier. Optional: context language. Output: classifier proposition.
science/quantity/mass.sle	Treats a referent as non-individuated mass under a measurement profile. Required: source. Optional: unit quantity context. Output: mass proposition.
science/quantity/countable.sle	Declares an individuation criterion under which a referent can be counted. Required: source basis. Optional: classifier context. Output: countability proposition.
science/quantity/approximate.sle	Represents an approximate value with tolerance or interval. Required: source value. Optional: min max uncertainty unit context. Output: approximate quantity.

Term	Explanation
science/quantity/exact.sle	Represents an exact value relative to a formal or measurement semantics. Required: source value. Optional: unit method context. Output: exact quantity.
science/quantity/range.sle	Represents a bounded or open numeric interval. Required: min max. Optional: unit uncertainty context. Output: range value.
science/quantity/ratio.sle	Represents a ratio with numerator, denominator, unit policy, and scope. Required: source target value. Optional: unit context method. Output: ratio proposition.
science/quantity/probability.sle	Assigns probability to an event or hypothesis under a model. Required: proposition probability. Optional: model dataset method confidence context. Output: probability proposition.
science/quantity/confidence.sle	Records confidence in a claim, estimate, mapping, or probability assignment. Required: target confidence. Optional: basis method context. Output: confidence proposition.
science/quantity/uncertainty.sle	Represents measurement, statistical, model, structural, or deep uncertainty. Required: target uncertainty. Optional: mode distribution min max confidence context. Output: uncertainty proposition.
science/quantity/zero.sle	Represents exact absence under a declared quantity and coverage semantics. Required: source. Optional: classifier unit coverage context. Output: zero proposition.
science/quantity/finite.sle	States finiteness under a specified domain and cardinality semantics. Required: source. Optional: cardinality basis context. Output: finiteness proposition.
science/quantity/unbounded.sle	States absence of a finite bound without asserting an actual infinite totality. Required: source. Optional: basis direction context. Output: unboundedness proposition.
science/quantity/actual-infinite.sle	Represents a completed infinite totality inside a formal theory. Required: source. Optional: basis model context. Output: formal infinity proposition.
science/quantity/potential-infinite.sle	Represents an indefinitely extendable process whose realised stages remain finite. Required: source. Optional: basis process context. Output: potential-infinity proposition.
science/quantity/outfinit.sle	Represents a process with no internal terminal rule but explicit physical or computational frontiers. Required: source. Optional: basis process budget context. Output: outfinit proposition.
science/quantity/threshold.sle	Represents a quantity's distance from or crossing of a consequential threshold. Required: source threshold. Optional: value unit degree time context. Output: threshold proposition.
knowledge/identity/strict.sle	States substitutive identity under a formal semantics. Required: source target. Optional: basis context valid-time. Output: strict identity.
knowledge/identity/material.sle	States continuity or similarity of material composition. Required: source target. Optional: degree basis valid-time scale context. Output: material-continuity proposition.
knowledge/identity/form.sle	States continuity of form or structure despite possible material change. Required: source target. Optional: degree basis valid-time scale context. Output: form-continuity proposition.
knowledge/identity/pattern.sle	States continuity of an abstract organisational pattern. Required: source target. Optional: degree basis model valid-time context. Output: pattern-continuity proposition.
knowledge/identity/process.sle	States continuity of a process, flow, or dynamically maintained organisation. Required: source target. Optional: degree basis path valid-time boundary scale context. Output: process-continuity proposition.
knowledge/identity/causal.sle	States continuity through a causal chain. Required: source target. Optional: path degree basis valid-time context. Output: causal-continuity proposition.
knowledge/identity/lineage.sle	States genealogical, derivational, version, or descent continuity. Required: source target. Optional: path basis version valid-time context. Output: lineage proposition.
knowledge/identity/informational.sle	States identity or similarity of information content under an encoding profile. Required: source target. Optional: degree basis content-digest context. Output: informational-continuity proposition.
knowledge/identity/functional.sle	States functional equivalence or continuity under specified tests. Required: source target. Optional: degree basis profile valid-time context. Output: functional-continuity proposition.
knowledge/identity/role.sle	States continuity of a role despite change of occupant or material substrate. Required: source target. Optional: basis authority valid-time context. Output: role-continuity proposition.

Term	Explanation
knowledge/identity/copy.sle	Represents a copy relation without collapsing source and copy. Required: source target. Optional: method time content-digest context. Output: copy relation.
knowledge/identity/fork.sle	Represents branching lineage from one predecessor into a distinct continuation. Required: source target. Optional: version time basis context. Output: fork relation.
knowledge/identity/merge.sle	Represents convergence of multiple lineages or states into a new entity. Required: source target. Optional: alternative method time basis context. Output: merge relation.
knowledge/identity/authenticate.sle	Evaluates authenticity against provenance, signature, lineage, or institutional criteria. Required: source basis. Optional: proof authority confidence context. Output: authenticity proposition.
knowledge/identity/provenance.sle	States identity-relevant origin and derivation chain. Required: source target. Optional: path content-digest time context. Output: provenance proposition.
knowledge/identity/boundary.sle	Represents the boundary criterion used to individuate an entity or process. Required: source boundary. Optional: scale basis valid-time context. Output: boundary proposition.
knowledge/identity/continuity.sle	General qualified continuity operator used only when a specialised identity verb is unavailable. Required: source target identity-mode. Optional: degree basis valid-time context. Output: continuity proposition.
knowledge/identity/similarity.sle	Represents graded similarity along named dimensions without implying identity. Required: source target basis. Optional: degree profile context. Output: similarity proposition.
knowledge/identity/replacement.sle	Represents functional, institutional, material, or version replacement. Required: source target. Optional: identity-mode basis time context. Output: replacement relation.
knowledge/identity/version.sle	Links an artefact or entity to a versioned continuation. Required: source target version. Optional: basis time content-digest context. Output: version relation.
social/collective/aggregate.sle	Constructs a set-like collection without collective agency. Required: member. Optional: collective cardinality boundary context. Output: aggregate value.
social/collective/crowd.sle	Constructs a transient co-present collective with limited coordination. Required: member collective. Optional: location time coordination boundary context. Output: crowd value.
social/collective/group.sle	Constructs a recognised group with membership criteria. Required: member collective. Optional: boundary valid-time mandate context. Output: group value.
social/collective/team.sle	Constructs a coordinated group with shared task or goal. Required: member collective. Optional: coordination mandate valid-time context. Output: team value.
social/collective/swarm.sle	Constructs a distributed collective whose behaviour emerges from local interactions. Required: member collective. Optional: coordination feedback boundary time context. Output: swarm value.
social/collective/organization.sle	Constructs an organisation with roles, rules, memory, and decision mechanisms. Required: collective. Optional: member mandate authority decision-mode valid-time context. Output: organisation value.
social/collective/institution.sle	Constructs a persistent rule-governed social entity. Required: collective. Optional: member authority mandate boundary valid-time context. Output: institution value.
social/collective/state.sle	Constructs a political state or government entity under explicit jurisdiction and time. Required: collective jurisdiction. Optional: authority member mandate valid-time context. Output: state value.
social/agency/individual.sle	Attributes action to an individual agent. Required: agent proposition. Optional: control intention time context. Output: agency proposition.
social/agency/delegated.sle	Attributes action through a delegated mandate. Required: agent collective proposition. Optional: mandate authority control time context. Output: delegated-agency proposition.
social/agency/distributed.sle	Attributes outcome to coordinated contributions without one complete controller. Required: collective proposition. Optional: member distribution coordination control time context. Output: distributed-agency proposition.
social/agency/emergent.sle	Represents a collective outcome without central intention or decision. Required: collective proposition. Optional: member coordination feedback time context. Output: emergent-agency proposition.
social/agency/centralized.sle	Represents action controlled by a central decision point. Required: collective proposition. Optional: agent authority decision-mode control time context. Output: centralised-agency proposition.

Term	Explanation
social/agency/coordinate.sle	Represents coordination among agents toward a task or outcome. Required: collective proposition. Optional: member coordination mandate time context. Output: coordination proposition.
social/agency/decide.sle	Represents a decision with agent, procedure, content, and mandate. Required: agent proposition decision-mode. Optional: collective mandate authority time context. Output: decision event.
social/agency/represent.sle	Represents action or speech performed on behalf of a collective. Required: agent collective proposition. Optional: mandate authority valid-time context. Output: representative-agency proposition.
social/responsibility/assign.sle	Assigns a typed responsibility to a holder for an outcome or obligation. Required: holder target responsibility. Optional: basis authority valid-time context. Output: responsibility proposition.
social/responsibility/distribute.sle	Allocates responsibility among multiple holders along explicit dimensions. Required: holder target distribution. Optional: responsibility control basis context. Output: responsibility distribution.
social/responsibility/control.sle	Records degree and kind of control relevant to responsibility. Required: holder target control. Optional: time basis context. Output: control proposition.
social/responsibility/remediate.sle	Assigns capacity or duty to repair, compensate, or mitigate an outcome. Required: holder target. Optional: authority capability valid-time context. Output: remedial responsibility.
social/responsibility/benefit.sle	Records receipt of benefits relevant to responsibility allocation. Required: holder target. Optional: value unit time context. Output: benefit proposition.
social/responsibility/bear-cost.sle	Records who bears a cost, risk, or externality. Required: holder target. Optional: value unit time context. Output: cost-bearing proposition.
knowledge/causation/direct.sle	Represents a direct causal relation under an explicit causal model. Required: source target. Optional: model method strength time context. Output: causal proposition.
knowledge/causation/contributory.sle	Represents a factor that contributes without being sufficient or necessary. Required: source target. Optional: strength model context. Output: causal proposition.
knowledge/causation/enabling.sle	Represents a condition that makes another causal pathway possible. Required: source target. Optional: model time context. Output: enabling proposition.
knowledge/causation/structural.sle	Represents causal influence through rules, incentives, networks, or institutional structure. Required: source target. Optional: model mechanism scale time context. Output: structural-cause proposition.
knowledge/causation/mediated.sle	Represents causation through an explicit intermediary or chain. Required: source target path. Optional: model strength context. Output: mediated-cause proposition.
knowledge/causation/preventive.sle	Represents a factor preventing or reducing an outcome. Required: source target. Optional: strength model time context. Output: preventive-cause proposition.
knowledge/causation/trigger.sle	Represents an event that crosses or activates an already prepared causal condition. Required: source target threshold. Optional: model time context. Output: trigger proposition.
knowledge/causation/mechanism.sle	Represents a mechanistic pathway linking cause and effect. Required: source target path. Optional: model evidence-mode context. Output: mechanism proposition.
knowledge/causation/intervene.sle	Represents a controlled or hypothetical intervention. Required: agent target value. Optional: model time context. Output: intervention event.
knowledge/causation/counterfactually-depend.sle	States counterfactual dependence under a specified model and contrast. Required: source target. Optional: model alternative context. Output: counterfactual-cause proposition.
knowledge/causation/sufficient.sle	States sufficiency only relative to a declared model and conditions. Required: source target. Optional: when model context. Output: sufficiency proposition.
knowledge/causation/necessary.sle	States necessity only relative to a declared model and contrast class. Required: source target. Optional: model alternative context. Output: necessity proposition.
social/decision/intend.sle	Attributes intention toward an outcome or action. Required: holder proposition. Optional: time confidence context. Output: intention attitude.
social/decision/foresee.sle	Attributes advance awareness of a possible effect. Required: holder proposition. Optional: probability confidence time context. Output: foresight attitude.
social/decision/tolerate.sle	Represents acceptance of a foreseen effect without making it the primary intention. Required: holder proposition. Optional: basis time context. Output: toleration attitude.

Term	Explanation
social/decision/accident.sle	Marks an effect as unintended and not reasonably foreseen under a profile. Required: proposition. Optional: holder basis profile time context. Output: accident status.
social/decision/choose.sle	Represents selection among explicit alternatives. Required: holder target. Optional: alternative basis time context. Output: choice event.
social/decision/opportunity-cost.sle	Represents the value of the best relevant forgone alternative. Required: target alternative value. Optional: unit model context. Output: opportunity-cost proposition.
social/decision/option-value.sle	Represents the value of preserving future choice under uncertainty. Required: target value. Optional: alternative probability uncertainty time context. Output: option-value proposition.
knowledge/risk/aleatory.sle	Represents uncertainty treated as stochastic variability under a model. Required: target probability. Optional: distribution model context. Output: aleatory-risk proposition.
knowledge/risk/epistemic.sle	Represents uncertainty arising from limited knowledge or evidence. Required: target uncertainty. Optional: coverage source-depth context. Output: epistemic-risk proposition.
knowledge/risk/deep-uncertainty.sle	Represents disagreement or ignorance about models, probabilities, or outcomes. Required: target. Optional: model alternative uncertainty context. Output: deep-uncertainty proposition.
knowledge/risk/unknown-unknown.sle	Records a recognised frontier where relevant possibilities or questions are not enumerated. Required: target. Optional: basis coverage context. Output: unknown-unknown status.
knowledge/risk/reversible.sle	Marks an action or state as reversible under explicit cost and time conditions. Required: target. Optional: value unit time context. Output: reversibility proposition.
knowledge/risk/irreversible.sle	Marks an action or state as practically or physically irreversible. Required: target. Optional: basis time scale context. Output: irreversibility proposition.
knowledge/risk/lock-in.sle	Represents persistence caused by switching cost, network effects, path dependence, or institutional inertia. Required: target. Optional: basis value unit time context. Output: lock-in proposition.
knowledge/risk/path-dependent.sle	Represents dependence of current state on the particular historical sequence. Required: target path. Optional: basis time context. Output: path-dependence proposition.
knowledge/system/emerge.sle	Represents a system-level property not attributable to any single component in isolation. Required: target part. Optional: basis scale model context. Output: emergence proposition.
knowledge/system/aggregate.sle	Computes or represents a macro property from member-level values under an aggregation rule. Required: part whole method. Optional: value unit scale context. Output: aggregate proposition.
knowledge/system/feedback-positive.sle	Represents a loop whose effects amplify a deviation or process. Required: source target path. Optional: strength delay scale context. Output: positive-feedback proposition.
knowledge/system/feedback-negative.sle	Represents a loop whose effects counteract a deviation or stabilise a variable. Required: source target path. Optional: strength delay scale context. Output: negative-feedback proposition.
knowledge/system/threshold.sle	Represents a system boundary at which behaviour changes discontinuously or materially. Required: target threshold. Optional: value distance model scale time context. Output: system-threshold proposition.
knowledge/system/phase-change.sle	Represents transition between qualitative regimes after a threshold or parameter change. Required: target from to. Optional: threshold model time context. Output: phase-change event.
knowledge/system/scale.sle	Indexes a proposition to the scale at which it is asserted. Required: proposition scale. Optional: basis context. Output: scale-qualified proposition.
knowledge/system/micro.sle	Marks individual or component-level scope. Required: proposition. Optional: scale context. Output: micro-level proposition.
knowledge/system/meso.sle	Marks organisational, network, or intermediate-level scope. Required: proposition. Optional: scale context. Output: meso-level proposition.
knowledge/system/macro.sle	Marks population, institutional, ecosystem, or system-wide scope. Required: proposition. Optional: scale context. Output: macro-level proposition.
knowledge/system/externality.sle	Represents a benefit or cost imposed on an entity outside the primary transaction or decision. Required: source target value. Optional: unit mode time context. Output: externality proposition.
knowledge/system/commons.sle	Represents a shared resource with access, governance, depletion, and benefit relations. Required: target collective. Optional: authority boundary value unit time context. Output: commons proposition.

Term	Explanation
knowledge/system/resilience.sle	Represents ability to preserve function or recover after disturbance. Required: target. Optional: basis degree model time context. Output: resilience proposition.
knowledge/system/antifragile.sle	Represents improvement under a defined distribution of stressors rather than mere recovery. Required: target. Optional: basis degree distribution scale context. Output: antifragility proposition.
knowledge/system/fragile.sle	Represents disproportionate harm under variability or stress. Required: target. Optional: basis degree distribution scale context. Output: fragility proposition.
knowledge/system/cascade.sle	Represents ordered propagation of effects through a system. Required: source target path. Optional: effect-order delay scale context. Output: cascade proposition.
knowledge/system/network-effect.sle	Represents value or behaviour that changes with network participation. Required: target member. Optional: basis value scale time context. Output: network-effect proposition.
knowledge/system/second-order.sle	Marks effects produced by direct effects rather than the initiating action alone. Required: source target. Optional: path model time context. Output: second-order proposition.
knowledge/system/third-order.sle	Marks effects produced through at least two intermediate consequence layers. Required: source target. Optional: path model time context. Output: third-order proposition.
knowledge/system/future-generation.sle	Indexes benefits, burdens, rights, or risks to people not yet existing. Required: target holder. Optional: time value uncertainty context. Output: posterity proposition.
knowledge/system/goodhart-pressure.sle	Represents pressure for a measure to become distorted when used as a target. Required: target. Optional: basis degree incentive time context. Output: Goodhart-pressure proposition.
science/research/hypothesis.sle	Declares a scientific hypothesis and the context in which it is tested. Required: proposition context. Optional: source model confidence status description. Output: hypothesis artefact.
science/research/observation.sle	Records an observation with observer, target, time, and source. Required: source-agent target. Optional: evidence-mode source-span time location context. Output: observation artefact.
science/research/measurement.sle	Records a measured value with unit, uncertainty, instrument, and method. Required: target value unit. Optional: uncertainty instrument method dataset time source-span context. Output: measurement artefact.
science/research/experiment.sle	Declares an experiment with hypothesis, protocol, intervention, and environment. Required: target method. Optional: premise dataset code environment start end context. Output: experiment artefact.
science/research/protocol.sle	Declares a versioned procedure and its obligations. Required: name version. Optional: when then unless authority environment source description. Output: protocol artefact.
science/research/dataset.sle	Registers a dataset with identity, version, schema, scope, and provenance. Required: id version. Optional: source content-digest coverage valid-time transaction-time description. Output: dataset artefact.
science/research/analysis.sle	Declares an analysis linking inputs, method, code, environment, and outputs. Required: method dataset. Optional: code model environment value uncertainty time context. Output: analysis artefact.
science/research/model.sle	Registers a scientific, statistical, or simulation model with assumptions and scope. Required: name version. Optional: premise method code dataset profile description. Output: model artefact.
science/research/simulation.sle	Declares a simulation run under a model, parameters, and environment. Required: model environment. Optional: dataset code value uncertainty time context. Output: simulation artefact.
science/research/result.sle	Records an output with method, uncertainty, and source artefacts. Required: value. Optional: unit uncertainty method dataset model code context. Output: result artefact.
science/research/claim.sle	Declares a scientific claim with scope, modality, and support relation. Required: proposition. Optional: source source-span model dataset confidence valid-time context. Output: scientific claim.
science/research/support.sle	Links evidence or results to a claim under a named method. Required: source target. Optional: basis strength confidence context. Output: support relation.
science/research/refute.sle	Links evidence or a counterexample to a challenged claim. Required: source target. Optional: basis witness strength context. Output: refutation relation.
science/research/replicate.sle	Records a replication attempt and its relation to an earlier experiment or result. Required: source target. Optional: method dataset environment result time context. Output: replication relation.

Term	Explanation
science/research/review.sle	Creates a review artefact targeting a claim, method, dataset, or package. Required: source-agent target. Optional: status severity proposition source-span time context. Output: review artefact.
science/research/challenge.sle	Records a specific objection to a premise, inference, identity, scope, or evidence relation. Required: source-agent target proposition. Optional: basis source-span severity context. Output: challenge artefact.
science/research/retract.sle	Records scientific withdrawal of a claim or artefact with reasons and authority. Required: target authority. Optional: basis time source context. Output: scientific retraction.
knowledge/provenance/derive.sle	States that an artefact or proposition was derived from one or more inputs. Required: source target. Optional: method code model time context. Output: derivation relation.
knowledge/provenance/attribute.sle	Attributes an artefact, action, or claim to an agent. Required: source target. Optional: role time authority context. Output: attribution relation.
knowledge/provenance/generate.sle	States that an activity generated an entity or artefact. Required: source target. Optional: time environment context. Output: generation relation.
knowledge/provenance/use.sle	States that an activity or analysis used an entity or artefact. Required: source target. Optional: time version context. Output: usage relation.
knowledge/provenance/quote.sle	Links a quotation or claim to exact source material. Required: source target source-span. Optional: language context. Output: quotation provenance.
knowledge/provenance/cite.sle	Links a claim or document to a bibliographic or data source. Required: source target. Optional: source-span basis context. Output: citation relation.
knowledge/provenance/version.sle	Links artefact versions and their content digests. Required: source target version. Optional: content-digest time context. Output: version-provenance relation.
knowledge/provenance/span.sle	Links a semantic node to an exact lexical, table, figure, code, or data span. Required: source source-span. Optional: content-digest context. Output: span relation.
execution/workflow/depend.sle	Declares that a build target depends on another node or artefact. Required: source target. Optional: mode version context. Output: build dependency.
execution/workflow/build.sle	Requests or records construction of a target from its dependencies. Required: target. Optional: profile environment budget time context. Output: build event.
execution/workflow/invalidate.sle	Marks cached or derived nodes invalid after a dependency change. Required: target source. Optional: basis time context. Output: invalidation event.
execution/workflow/recompute.sle	Requests or records recomputation under a new epoch. Required: target. Optional: source environment budget time context. Output: recomputation event.
execution/reasoning/derive.sle	Records a derived proposition with premises, rule, profile, and proof. Required: conclusion premise. Optional: rule profile proof context. Output: derivation result.
execution/reasoning/entail.sle	States that a target follows under a named formal semantics and coverage. Required: premise conclusion profile. Optional: proof coverage context. Output: entailment result.
execution/reasoning/contradict.sle	States incompatibility between propositions under a named profile. Required: proposition proposition. Optional: proof context severity. Output: contradiction result.
execution/reasoning/may.sle	States that a proposition holds in at least one admissible interpretation. Required: proposition context. Optional: coverage witness profile. Output: may result.
execution/reasoning/must.sle	States that a proposition holds in all admissible interpretations represented by the analysis. Required: proposition context. Optional: coverage proof profile. Output: must result.
execution/reasoning/cannot.sle	States that a proposition holds in no admissible interpretation represented by the analysis. Required: proposition context. Optional: coverage proof profile. Output: cannot result.
execution/reasoning/unknown.sle	Records inability to decide because of unsupported semantics, coverage, budget, or evidence. Required: target. Optional: missing-reason coverage budget context. Output: unknown result.
execution/reasoning/conflicted.sle	Records simultaneous support for a proposition and its explicit negation. Required: proposition. Optional: source alternative context. Output: conflicted result.
execution/reasoning/prove.sle	Requests or records proof under a named calculus or checker. Required: target profile. Optional: premise proof budget context. Output: proof result.

Term	Explanation
execution/reasoning/explain.sle	Produces a source- and proof-linked explanation of a result. Required: target. Optional: query return proof source-span language context. Output: explanation result.
execution/reasoning/witness.sle	Records a replayable model, trace, input, or path supporting possibility or violation. Required: target witness. Optional: environment source-span context. Output: witness result.
execution/reasoning/counterexample.sle	Records a replayable case refuting a universal claim or constraint. Required: target witness. Optional: environment source-span context. Output: counterexample result.
execution/query/whether.sle	Asks whether a proposition is true, false, unknown, or conflicted in a context. Required: pattern context. Optional: profile coverage guarantee return budget. Output: truth-status query.
execution/query/select.sle	Returns bindings satisfying a pattern. Required: pattern return. Optional: context profile coverage guarantee order limit budget. Output: selection query.
execution/query/count.sle	Counts matches under explicit classifier and coverage. Required: pattern. Optional: return context profile coverage guarantee limit. Output: count query.
execution/query/compare.sle	Compares results, models, versions, or scenarios along named dimensions. Required: source target basis. Optional: return context profile guarantee. Output: comparison query.
execution/query/why.sle	Requests derivation, provenance, causal path, or assumption explanation. Required: target. Optional: return context profile guarantee limit. Output: explanation query.
execution/query/what-if.sle	Evaluates a proposition under explicit counterfactual changes. Required: target assumption. Optional: alternative context profile guarantee budget. Output: counterfactual query.
execution/runtime/validate.sle	Runs syntax, schema, type, profile, provenance, and policy validation. Required: target. Optional: profile severity context. Output: validation result.
execution/runtime/reject.sle	Records deterministic rejection and its exact cause. Required: target basis. Optional: severity source-span context. Output: rejection result.
execution/runtime/diagnose.sle	Produces a machine-readable diagnostic for a stopped or degraded execution. Required: target status. Optional: basis severity source-span alternative context. Output: runtime diagnostic.
language/english/clause/declarative.sle	Constructs a proposition presented as a declarative clause without itself asserting that proposition. Required: proposition. Optional: source-span context information-structure. Output: clause object.
language/english/clause/interrogative.sle	Represents an interrogative form and its requested variable, polarity, or alternative set. Required: question. Optional: addressee context source-span. Output: question object.
language/english/clause/imperative.sle	Represents an imperative form while keeping command force distinct from operational authority. Required: action addressee. Optional: speaker strength context source-span. Output: directive object.
language/english/voice/active.sle	Records an active-voice surface alignment where relevant to source analysis. Required: proposition. Optional: subject agent source-span. Output: voice annotation.
language/english/voice/passive.sle	Normalises passive alignment while recording whether the agent is expressed, unknown, backgrounded, or strategically omitted. Required: proposition patient. Optional: agent omission-reason source-span. Output: voice-qualified proposition.
language/english/aspect/progressive.sle	Presents an event from within its unfolding interval. Required: event reference-time. Optional: boundedness interruption context. Output: aspect-qualified proposition.
language/english/aspect/perfect.sle	Relates a prior situation to a later reference point without assuming completion or current relevance beyond the selected profile. Required: event reference-time. Optional: result-state context. Output: aspect-qualified proposition.
language/english/aspect/habitual.sle	Represents a recurrent disposition or pattern rather than one actual occurrence. Required: event-pattern holder. Optional: frequency interval exceptions context. Output: habitual proposition.
language/english/tense/past.sle	Places reference time before the evaluation time under an explicit temporal profile. Required: proposition evaluation-time. Optional: remoteness interval context. Output: tense-qualified proposition.
language/english/tense/non-past.sle	Records English non-past morphology while requiring another operator or context to distinguish present, generic, scheduled, or future interpretation. Required: proposition. Optional: evaluation-time interpretation context. Output: tense annotation.

Term	Explanation
language/english/modality/modal.sle	Disambiguates an English modal auxiliary into epistemic, deontic, dynamic, predictive, or logical mode. Required: proposition mode. Optional: holder authority degree source-span context. Output: modal proposition.
language/english/negation/sentential.sle	Represents sentence negation with explicit structural scope. Required: proposition. Optional: scope focus context source-span. Output: negative proposition.
language/english/negation/not-all.sle	Represents narrow negation of a universal claim rather than universal negation. Required: quantified-proposition. Optional: context source-span. Output: scope-qualified negative proposition.
language/english/reference/article-indefinite.sle	Records an indefinite noun phrase and its introduction of a discourse referent or generic reading. Required: type. Optional: variable quantity genericity source-span context. Output: reference object.
language/english/reference/article-definite.sle	Records a definite description with its identification basis rather than assuming uniqueness globally. Required: description. Optional: referent familiarity uniqueness-scope source-span context. Output: reference object.
language/english/reference/pronoun.sle	Represents a pronoun and candidate antecedents, agreement features, discourse role, and confidence. Required: source-span. Optional: antecedent alternative person number gender perspective context. Output: reference candidate.
language/english/reference/expletive.sle	Marks syntactic subjects such as weather it or existential there that do not denote ordinary semantic agents. Required: form clause. Optional: source-span context. Output: expletive annotation.
language/english/discourse/ellipsis.sle	Records omitted recoverable material and candidate reconstructions. Required: source-span alternative. Optional: antecedent context confidence. Output: ellipsis interpretation.
language/english/discourse/presuppose.sle	Records content conventionally treated as backgrounded or already accommodated without automatically asserting it in every context. Required: proposition trigger. Optional: holder context source-span. Output: presupposition object.
language/english/discourse/implicate.sle	Records a defeasible pragmatic enrichment distinct from literal content. Required: proposition basis. Optional: speaker addressee context confidence cancellable source-span. Output: implicature candidate.
language/english/information/topic.sle	Marks the discourse topic associated with a proposition. Required: target proposition. Optional: contrast context source-span. Output: information-structure annotation.
language/english/information/focus.sle	Marks focused material and the relevant alternative set. Required: target proposition. Optional: alternatives contrast context source-span. Output: focus annotation.
language/english/information/cleft.sle	Represents cleft constructions with explicit focus and presupposed background. Required: focus background. Optional: source-span context. Output: information-structure object.
language/english/coordination/and.sle	Represents coordination while allowing the relation to be specified as addition, sequence, result, contrast, or joint participation. Required: proposition proposition. Optional: relation order context source-span. Output: coordinated proposition.
language/english/coordination/or.sle	Represents inclusive or exclusive alternatives with explicit scope and exhaustivity. Required: proposition proposition. Optional: mode exhaustive context source-span. Output: disjunctive proposition.
language/english/subordination/conditional.sle	Represents condition and consequent with explicit modality, counterfactual status, and temporal relation. Required: condition consequence. Optional: mode world context source-span. Output: conditional proposition.
language/english/subordination/causal.sle	Records that an English because or related clause presents an explanatory or causal relation, without deciding the causal profile automatically. Required: source target. Optional: causal-layer evidence context source-span. Output: causal candidate.
language/english/subordination/concessive.sle	Represents a concession relation between an expectation and a proposition. Required: expectation proposition. Optional: holder degree context source-span. Output: concessive proposition.
language/english/report/direct-speech.sle	Preserves quoted wording, speaker, addressee, time, and source span without asserting quoted content. Required: content speaker. Optional: addressee time source context. Output: quotation object.
language/english/report/indirect-speech.sle	Represents reported content with tense, person, and perspective adjustments kept distinct from acceptance. Required: proposition source-agent. Optional: perspective time source-depth context source-span. Output: report object.

Term	Explanation
language/english/nominalisation/event.sle	Expands an event nominalisation into a candidate event frame and records omitted roles. Required: nominal source-span. Optional: agent patient theme time cause alternatives context. Output: event candidate.
language/english/compound/noun-noun.sle	Represents the underspecified semantic relation inside an English noun compound and preserves alternatives when consequence changes. Required: modifier head. Optional: relation alternative context source-span. Output: compound interpretation.
language/english/comparison/comparative.sle	Represents comparison along an explicit dimension, standard, and degree. Required: source target basis. Optional: degree unit context source-span. Output: comparison proposition.
language/english/comparison/superlative.sle	Represents a maximum or minimum relative to an explicit comparison set and coverage. Required: source basis comparison-set. Optional: order coverage context source-span. Output: superlative proposition.
language/english/possession/genitive.sle	Treats the English genitive as an underspecified relation requiring ownership, kinship, authorship, part, control, association, or another declared interpretation. Required: possessor possessed. Optional: relation alternatives context source-span. Output: possession candidate.
language/english/phrasal-verb.sle	Resolves a verb-particle construction to a declared semantic operator while preserving literal alternatives where relevant. Required: verb particle source-span. Optional: operator alternative context. Output: lexical-semantic candidate.
language/english/light-verb.sle	Represents constructions such as take a walk or make a decision through an event operator rather than treating the light verb as the principal action. Required: verb nominal. Optional: agent roles context source-span. Output: event candidate.
language/english/control/subject.sle	Records the understood subject of a non-finite complement when controlled by a matrix argument. Required: controller embedded-proposition. Optional: relation context source-span. Output: control relation.
language/english/raising.sle	Records a raising construction in which a surface subject is not assigned its semantic role by the matrix predicate. Required: source embedded-proposition. Optional: evidential-mode context source-span. Output: raising analysis.
language/english/attachment/alternative.sle	Preserves materially distinct prepositional, relative-clause, or modifier attachments as explicit alternatives. Required: source-span alternative. Optional: ranking confidence context. Output: interpretation set.

APPENDIX D

English and Cross-Linguistic Constructions as Executable Operators

This table records how the English baseline and the cross-linguistic survey are translated into executable distinctions. Each row identifies a grammatical or conceptual phenomenon and the SLEngish operator pattern that can represent it. The mapping is refined through grammar consultation, corpus analysis, speaker or specialist expertise where relevant, translation tests, and reasoning tasks.

Term	Explanation
language/english/clause/declarative	English baseline mapping. Constructs a proposition presented as a declarative clause without itself asserting that proposition. Required: proposition. Optional: source-span context information-structure. Output: clause object.
language/english/clause/interrogative	English baseline mapping. Represents an interrogative form and its requested variable, polarity, or alternative set. Required: question. Optional: addressee context source-span. Output: question object.
language/english/clause/imperative	English baseline mapping. Represents an imperative form while keeping command force distinct from operational authority. Required: action addressee. Optional: speaker strength context source-span. Output: directive object.
language/english/voice/active	English baseline mapping. Records an active-voice surface alignment where relevant to source analysis. Required: proposition. Optional: subject agent source-span. Output: voice annotation.
language/english/voice/passive	English baseline mapping. Normalises passive alignment while recording whether the agent is expressed, unknown, backgrounded, or strategically omitted. Required: proposition patient. Optional: agent omission-reason source-span. Output: voice-qualified proposition.
language/english/aspect/progressive	English baseline mapping. Presents an event from within its unfolding interval. Required: event reference-time. Optional: boundedness interruption context. Output: aspect-qualified proposition.
language/english/aspect/perfect	English baseline mapping. Relates a prior situation to a later reference point without assuming completion or current relevance beyond the selected profile. Required: event reference-time. Optional: result-state context. Output: aspect-qualified proposition.
language/english/aspect/habitual	English baseline mapping. Represents a recurrent disposition or pattern rather than one actual occurrence. Required: event-pattern holder. Optional: frequency interval exceptions context. Output: habitual proposition.
language/english/tense/past	English baseline mapping. Places reference time before the evaluation time under an explicit temporal profile. Required: proposition evaluation-time. Optional: remoteness interval context. Output: tense-qualified proposition.
language/english/tense/non-past	English baseline mapping. Records English non-past morphology while requiring another operator or context to distinguish present, generic, scheduled, or future interpretation. Required: proposition. Optional: evaluation-time interpretation context. Output: tense annotation.
language/english/modality/modal	English baseline mapping. Disambiguates an English modal auxiliary into epistemic, deontic, dynamic, predictive, or logical mode. Required: proposition mode. Optional: holder authority degree source-span context. Output: modal proposition.
language/english/negation/sentential	English baseline mapping. Represents sentence negation with explicit structural scope. Required: proposition. Optional: scope focus context source-span. Output: negative proposition.
language/english/negation/not-all	English baseline mapping. Represents narrow negation of a universal claim rather than universal negation. Required: quantified-proposition. Optional: context source-span. Output: scope-qualified negative proposition.
language/english/reference/article-indefinite	English baseline mapping. Records an indefinite noun phrase and its introduction of a discourse referent or generic reading. Required: type. Optional: variable quantity genericity source-span context. Output: reference object.
language/english/reference/article-definite	English baseline mapping. Records a definite description with its identification basis rather than assuming uniqueness globally. Required: description. Optional: referent familiarity uniqueness-scope source-span context. Output: reference object.

Term	Explanation
language/english/reference/pronoun	English baseline mapping. Represents a pronoun and candidate antecedents, agreement features, discourse role, and confidence. Required: source-span. Optional: antecedent alternative person number gender perspective context. Output: reference candidate.
language/english/reference/expletive	English baseline mapping. Marks syntactic subjects such as weather it or existential there that do not denote ordinary semantic agents. Required: form clause. Optional: source-span context. Output: expletive annotation.
language/english/discourse/ellipsis	English baseline mapping. Records omitted recoverable material and candidate reconstructions. Required: source-span alternative. Optional: antecedent context confidence. Output: ellipsis interpretation.
language/english/discourse/presuppose	English baseline mapping. Records content conventionally treated as backgrounded or already accommodated without automatically asserting it in every context. Required: proposition trigger. Optional: holder context source-span. Output: presupposition object.
language/english/discourse/implicate	English baseline mapping. Records a defeasible pragmatic enrichment distinct from literal content. Required: proposition basis. Optional: speaker addressee context confidence cancellable source-span. Output: implicature candidate.
language/english/information/topic	English baseline mapping. Marks the discourse topic associated with a proposition. Required: target proposition. Optional: contrast context source-span. Output: information-structure annotation.
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language/english/coordination/and	English baseline mapping. Represents coordination while allowing the relation to be specified as addition, sequence, result, contrast, or joint participation. Required: proposition proposition. Optional: relation order context source-span. Output: coordinated proposition.
language/english/coordination/or	English baseline mapping. Represents inclusive or exclusive alternatives with explicit scope and exhaustivity. Required: proposition proposition. Optional: mode exhaustive context source-span. Output: disjunctive proposition.
language/english/subordination/conditional	English baseline mapping. Represents condition and consequent with explicit modality, counterfactual status, and temporal relation. Required: condition consequence. Optional: mode world context source-span. Output: conditional proposition.
language/english/subordination/causal	English baseline mapping. Records that an English because or related clause presents an explanatory or causal relation, without deciding the causal profile automatically. Required: source target. Optional: causal-layer evidence context source-span. Output: causal candidate.
language/english/subordination/concessive	English baseline mapping. Represents a concession relation between an expectation and a proposition. Required: expectation proposition. Optional: holder degree context source-span. Output: concessive proposition.
language/english/report/direct-speech	English baseline mapping. Preserves quoted wording, speaker, addressee, time, and source span without asserting quoted content. Required: content speaker. Optional: addressee time source context. Output: quotation object.
language/english/report/indirect-speech	English baseline mapping. Represents reported content with tense, person, and perspective adjustments kept distinct from acceptance. Required: proposition source-agent. Optional: perspective time source-depth context source-span. Output: report object.
language/english/nominalisation/event	English baseline mapping. Expands an event nominalisation into a candidate event frame and records omitted roles. Required: nominal source-span. Optional: agent patient theme time cause alternatives context. Output: event candidate.
language/english/compound/noun-noun	English baseline mapping. Represents the underspecified semantic relation inside an English noun compound and preserves alternatives when consequence changes. Required: modifier head. Optional: relation alternative context source-span. Output: compound interpretation.
language/english/comparison/comparative	English baseline mapping. Represents comparison along an explicit dimension, standard, and degree. Required: source target basis. Optional: degree unit context source-span. Output: comparison proposition.

Term	Explanation
language/english/comparison/superlative	English baseline mapping. Represents a maximum or minimum relative to an explicit comparison set and coverage. Required: source basis comparison-set. Optional: order coverage context source-span. Output: superlative proposition.
language/english/possession/genitive	English baseline mapping. Treats the English genitive as an underspecified relation requiring ownership, kinship, authorship, part, control, association, or another declared interpretation. Required: possessor possessed. Optional: relation alternatives context source-span. Output: possession candidate.
language/english/phrasal-verb	English baseline mapping. Resolves a verb-particle construction to a declared semantic operator while preserving literal alternatives where relevant. Required: verb particle source-span. Optional: operator alternative context. Output: lexical-semantic candidate.
language/english/light-verb	English baseline mapping. Represents constructions such as take a walk or make a decision through an event operator rather than treating the light verb as the principal action. Required: verb nominal. Optional: agent roles context source-span. Output: event candidate.
language/english/control/subject	English baseline mapping. Records the understood subject of a non-finite complement when controlled by a matrix argument. Required: controller embedded-proposition. Optional: relation context source-span. Output: control relation.
language/english/raising	English baseline mapping. Records a raising construction in which a surface subject is not assigned its semantic role by the matrix predicate. Required: source embedded-proposition. Optional: evidential-mode context source-span. Output: raising analysis.
language/english/attachment/alternative	English baseline mapping. Preserves materially distinct prepositional, relative-clause, or modifier attachments as explicit alternatives. Required: source-span alternative. Optional: ranking confidence context. Output: interpretation set.
Grammatical evidentiality	Languages with grammatical evidentials require or strongly conventionalise distinctions such as direct observation, report, and inference. Representative scholarship includes Aikhenvald and the WALS evidentiality maps. SLEngish mapping: knowledge/evidence/observe.sle, knowledge/evidence/report.sle, knowledge/evidence/infer.sle, knowledge/evidence/remember.sle, knowledge/evidence/measure.sle; profiles may require evidence-mode and source-depth.
Mirativity	Mirative systems mark information as unexpected, newly integrated, or contrary to a model. The category is debated and often overlaps evidentiality. SLEngish mapping: knowledge/epistemic/expect.sle, knowledge/epistemic/counterexpect.sle, knowledge/epistemic/surprise.sle with mirativity, holder, prior model, and degree.
Inclusive versus exclusive first-person plural	Many languages distinguish a “we” that includes the addressee from one that excludes them. English normally leaves this optional. SLEngish mapping: language/reference/inclusive-we.sle and language/reference/exclusive-we.sle, with explicit speaker, addressee, members, and collective scope.
Dual, trial, paucal, and associative number	Languages may grammatically distinguish exactly two, exactly three, a small plurality, or an individual plus associates. SLEngish mapping: language/number/dual.sle, language/number/trial.sle, language/number/paucal.sle, and language/number/associative.sle; classifiers and coverage specify what is counted.
Numeral classifiers	Classifier languages require counting through a classifying unit that can encode shape, animacy, arrangement, or conventional individuation. SLEngish mapping: science/quantity/classifier.sle and science/quantity/count.sle with classifier, identity criterion, and coverage.
Alienable and inalienable possession	Possessive systems often distinguish transferable ownership from kin, body-part, constitutive, or otherwise inalienable relations. SLEngish mapping: language/possession/alienable.sle, language/possession/inalienable.sle, language/possession/kin.sle, language/possession/body-part.sle, and language/possession/stewardship.sle.
Absolute spatial frames	Languages documented by Levinson and others may habitually use cardinal, uphill/downhill, riverine, or seaward/landward frames rather than body-relative left/right. SLEngish mapping: language/space/north-of.sle, uphill, upriver, seaward, and related operators with explicit anchor and scale.
Remoteness tense	Some tense systems distinguish recent and remote pasts or near and remote futures. SLEngish mapping: language/time/recent-past.sle, language/time/remote-past.sle, language/time/near-future.sle, language/time/remote-future.sle, with interval definitions supplied by a context profile.

Term	Explanation
Rich aspect systems	Perfective, imperfective, progressive, habitual, perfect, prospective, iterative, durative, punctual, telic, and atelic distinctions expose event contour rather than only chronological location. SLEngish mapping: the corresponding language/time/* operators.
Switch-reference	Clause-chaining systems can mark whether adjacent clauses share a central participant. SLEngish mapping: language/discourse/same-reference.sle and language/discourse/different-reference.sle with clause nodes, participant anchors, and source spans.
Logophoricity	Logophoric systems identify forms interpreted relative to the consciousness or speech perspective of a reported individual. SLEngish mapping: language/reference/logophor.sle and language/perspective/logophoric.sle with explicit holder and context.
Egophoricity	Some systems mark privileged personal authority or involvement, especially for one's own states and intentional actions. SLEngish mapping: language/perspective/egophoric.sle with holder, proposition, evidence mode, and context; it does not automatically imply truth.
Obviation and proximate status	Algonquian and other discourse systems can distinguish a more central third person from less central third persons. SLEngish mapping: language/reference/proximate.sle and language/reference/obviative.sle, used to stabilise reference and perspective.
Direct-inverse alignment	Direct-inverse systems mark whether an event follows or reverses a person/animacy hierarchy. SLEngish mapping: language/perspective/pivot.sle plus explicit event roles; experimental operator libraries may add alignment/direct.sle and alignment/inverse.sle if corpus evidence justifies them.
Applicative constructions	Applicatives promote beneficiaries, instruments, locations, or recipients into core event structure. SLEngish mapping: language/event/applicative.sle or specialised event verbs with explicit promoted role.
Antipassive constructions	Antipassives foreground agent and activity while reducing patient prominence or definiteness. SLEngish mapping: language/event/antipassive.sle with explicit underlying proposition, agent, optional patient, and aspect.
Causative constructions	Languages grammaticalise adding a causer and causee in varied ways. SLEngish mapping: language/event/causative.sle and the knowledge/causation/* library; the representation must distinguish causing an event from merely asserting causal relation.
Middle and reflexive voice	Middle and reflexive systems alter participant structure and affectedness. SLEngish mapping: language/event/middle.sle and language/event/reflexive.sle with explicit co-reference or agent suppression.
Pluractionality	Some languages mark event plurality, distribution, repetition, or participant plurality morphologically. SLEngish mapping: language/event/repeat.sle, language/event/distribute.sle, language/time/iterative.sle, and language/number/distributive.sle.
Serial verb constructions	Serial verbs encode tightly linked actions, paths, aspects, or participant sharing without ordinary coordination. SLEngish mapping: language/event/serialise.sle with ordered proposition nodes and declared shared roles.
Medial verbs and clause chaining	Clause-chaining languages use medial forms to mark non-final clauses, participant continuity, temporal relations, and discourse dependency. SLEngish mapping: discourse switch-reference operators plus time and workflow dependencies.
Realis and irrealis	Languages may mark whether an event is presented as actual, intended, hypothetical, feared, possible, or counterfactual. SLEngish mapping: knowledge/modality/realis.sle, knowledge/modality/irrealis.sle, knowledge/modality/counterfactual.sle, and explicit contexts.
Quotative and reportative constructions	Many languages distinguish exact quotation, report, hearsay, and speaker perspective. SLEngish mapping: language/discourse/quote-direct.sle, language/discourse/report-indirect.sle, knowledge/evidence/quote.sle, and knowledge/evidence/report.sle.
Honorific and humilific systems	Speech may grammatically encode relative rank, respect, familiarity, or self-lowering. SLEngish mapping: social/relation/honor.sle, social/relation/humble-self.sle, social/relation/address-deferential.sle, and social/relation/address-familiar.sle.
Taboo and avoidance registers	Some communities maintain special vocabularies or registers for particular kin relations, rituals, or contexts. SLEngish mapping: social/relation/taboo.sle and social/relation/register.sle with community authority and validity scope.

Term	Explanation
Noun classes and animacy hierarchies	Languages classify nouns by gender, animacy, shape, social status, or other semantic and grammatical criteria. SLEnglish mapping: typed entity classifications and explicit classifier profiles; no one hierarchy is assumed universal.
Ideophones	Ideophones can depict sensory scenes, movement, texture, intensity, or manner with high expressive precision. SLEnglish should retain the original form as data and may map interpretable dimensions through perception or event operators without claiming exhaustive translation.
Olfactory lexicons	Research on languages such as Jahai and Maniq shows that odours can be named with greater abstraction than English stereotypes suggest. SLEnglish mapping: knowledge/evidence/sense.sle with olfactory mode plus domain smell operators, while culturally specific lexemes remain versioned concepts.
Relational nouns	Some languages lexicalise relations that English expresses as possessed nouns or loose prepositions. SLEnglish mapping: dedicated relation or possession verbs rather than generic ownership.
Body-part incorporation	Incorporation can encode recurrent participant structure inside the verb. SLEnglish mapping: language/event/incorporate.sle plus language/possession/body-part.sle and explicit affected participant.
Distributive versus collective predication	A plural proposition may apply to each member, to the group jointly, or to a subgroup. SLEnglish mapping: language/number/distributive.sle, language/number/collective.sle, social/collective/* constructors, and explicit quantifier scope.
Inclusive imperative and hortative	Languages can distinguish commands including speaker, addressee, or third parties. SLEnglish mapping: participant-set constructors combined with language/speech-act/command.sle or a hortative profile.
Evidential source depth	Natural languages rarely grammaticalise an arbitrary number of testimony hops, but the distinction is crucial for science and online information. SLEnglish invention: source-depth and knowledge/provenance/derive.sle chains attached to evidence verbs.
Process identity	No single grammatical category universally solves the identity of clouds, rivers, flames, organisms, or institutions. SLEnglish conceptual invention: knowledge/identity/process.sle with causal path, boundary grade, scale, interval, and degree.
Collective agency architecture	Natural languages readily personify states, firms, markets, and societies. SLEnglish conceptual invention: collective constructors and social/agency/delegated.sle, distributed, emergent, or centralized to distinguish decision from outcome.
Responsibility vectors	Ordinary grammar compresses causation, knowledge, control, benefit, mandate, and remediation into “responsible.” SLEnglish conceptual invention: social/responsibility/* operators that allocate these dimensions separately.
Reversibility grade	Ordinary verbs rarely mark whether an action can be reversed cheaply, only at high cost, or not at all. SLEnglish conceptual invention: knowledge/risk/reversible.sle, knowledge/risk/irreversible.sle, and reversibility parameters.
Causal depth	Everyday causal language mixes trigger, direct mechanism, enabling condition, structural cause, and feedback. SLEnglish mapping: distinct knowledge/causation/* verbs and causal-layer.
Scale address	A proposition may be true at individual, organisational, population, or ecosystem scale and misleading elsewhere. SLEnglish conceptual invention: knowledge/system/scale.sle, micro, meso, and macro.
Boundary grade	Clouds, communities, species, and institutions may have fuzzy, operational, dynamic, or contested boundaries. SLEnglish conceptual invention: knowledge/identity/boundary.sle and the boundary parameter.
Unknown versus unknowable	English can express the distinction lexically but rarely requires it. SLEnglish mapping: knowledge/epistemic/unknown.sle, knowledge/epistemic/unknowable.sle, coverage, and missing-reason.
Absence versus non-existence	Missing records, zero measured quantity, unavailable evidence, and non-existence are different. SLEnglish mapping: explicit missing-value atoms, science/quantity/zero.sle, knowledge/entity/exist.sle, and coverage-sensitive knowledge/logic/none.sle.
Goodhart pressure	Metrics used as targets can change the system they describe. SLEnglish conceptual invention: knowledge/system/goodhart-pressure.sle linked to incentive, target, and observed distortion.
Posterity reference	Future people cannot participate in current pronoun systems or contracts but may bear effects. SLEnglish conceptual invention: knowledge/system/future-generation.sle with time, uncertainty, benefit, and burden.

Term	Explanation
Option value and opportunity cost	Ordinary language often hides the value of retaining alternatives and the best forgone path. SLEngish mapping: social/decision/option-value.sle and social/decision/opportunity-cost.sle.
Unknown unknowns	A system may know that its current question set is incomplete without enumerating what is missing. SLEngish conceptual invention: knowledge/risk/unknown-unknown.sle with an explicit frontier and coverage statement.

APPENDIX E

Worked SLEngish Programmes

The fixtures show how evidence, clusivity, process identity, collective agency, measurement, ambiguity, rules, and systemic effects use the same outer syntax. The examples are illustrative contracts whose exact paths and parameters remain subject to conformance testing.

Scenario	Canonical SLEngish fixture
Evidential source chain	<pre>@rain language/event/occur.sle type rain location zurich time "2026-08-22T07:00+02:00" . @witness slenglish/core/entity.sle type person label "Witness 4" . @report knowledge/evidence/report.sle proposition ~rain source-agent \$witness source local-message source-depth 2 confidence 0.72 . @claim slenglish/core/claim.sle proposition ~rain source ~report context reported-world . @q execution/query/whether.sle pattern ~rain context actual-world coverage unknown return status .</pre>
Inclusive and exclusive political "we"	<pre>@citizens social/collective/group.sle collective citizens member electorate boundary legal-electorate . @government social/collective/organization.sle collective government authority constitution . @we-exclusive language/reference/exclusive-we.sle speaker government-spokesperson addressee citizens member \$government . @decision social/agency/decide.sle agent \$government proposition tax-increase decision-mode cabinet-vote mandate statute-17 . @utterance language/event/communicate.sle source-agent government-spokesperson recipient \$citizens proposition ~decision . @diag slenglish/core/diagnose.sle target ~utterance status misleading basis pronoun-scope alternative ~we-exclusive .</pre>
Identity of a cloud	<pre>@cloud-t1 slenglish/core/entity.sle type cloud-state label "Cloud at t1" . @cloud-t2 slenglish/core/entity.sle type cloud-state label "Cloud at t2" . @flow knowledge/identity/process.sle source \$cloud-t1 target \$cloud-t2 basis causal-flow boundary fuzzy scale mesoscale degree 0.84 valid-time interval-17 . @matter knowledge/identity/material.sle source \$cloud-t1 target \$cloud-t2 basis water-molecule-overlap degree 0.21 valid-time interval-17 . @q execution/query/compare.sle source ~flow target ~matter basis continuity-dimension return degree .</pre>
Collective decision versus emergent outcome	<pre>@board social/collective/team.sle collective acme-board member board-members mandate corporate-charter . @close social/agency/decide.sle agent \$board proposition close-plant decision-mode centralized-vote mandate charter-clause-8 . @market social/collective/aggregate.sle collective commodity-market member market-participants . @rise social/agency/emergent.sle collective \$market proposition price-increase coordination uncoordinated feedback positive . @check execution/query/whether.sle pattern ~rise context actual-world return status .</pre>
Scientific measurement and claim	<pre>@sample slenglish/core/entity.sle type material-sample label "Sample 17" . @measure science/research/measurement.sle target \$sample value 12.41 unit millisiemens uncertainty 0.08 instrument probe-4 method calibrated-conductivity time "2026-08-21T16:11Z" dataset dataset-17 . @threshold science/quantity/exact.sle source conductivity-limit value 12.0 unit millisiemens . @comparison knowledge/relation/compare.sle source ~measure target ~threshold basis measured-value mode greater-than . @claim science/research/claim.sle proposition ~comparison source analysis-script-4 confidence 0.94 context experiment-17 . @q execution/query/why.sle target ~claim return proof context experiment-17 .</pre>
Explicit ambiguity	<pre>@alice slenglish/core/entity.sle type person label "Alice" . @mary slenglish/core/entity.sle type person label "Mary" . @scope slenglish/core/entity.sle type telescope label "Telescope 1" . @alt-1 language/event/perceive.sle experiencer \$alice stimulus \$mary instrument \$scope . @associated knowledge/relation/associate.sle source \$mary target \$scope basis possession-or-proximity . @alt-2 language/event/perceive.sle experiencer \$alice stimulus ~associated . @fragment slenglish/core/entity.sle type source-fragment label "Alice saw Mary with the telescope" . @diag slenglish/core/diagnose.sle target \$fragment status ambiguous source-span "Alice saw Mary with the telescope" basis attachment alternative ~alt-1 alternative ~alt-2 .</pre>

Scenario	Canonical SLEngish fixture
Strict rule and proof-producing query	@x slenglish/core/variable.sle type person . @y slenglish/core/variable.sle type person . @z slenglish/core/variable.sle type person . @p1 knowledge/relation/relate.sle source \$y target \$x type parent-of . @p2 knowledge/relation/relate.sle source \$z target \$y type parent-of . @g knowledge/relation/relate.sle source \$z target \$x type grandparent-of . @r slenglish/core/rule.sle when ~p1 when ~p2 then ~g profile datalog-strict . @q execution/query/select.sle pattern ~g return \$z context actual-world profile datalog-strict . @proof execution/reasoning/explain.sle target ~q return proof .
Reversibility and second-order effects	@deploy language/event/act.sle agent organisation-a target model-v4 purpose allocation time release-2026-08 . @lock knowledge/risk/lock-in.sle target ~deploy basis integration-cost value 870000 unit euro context deployment-scenario . @bias knowledge/system/second-order.sle source ~deploy target hiring-distribution path decision-feedback-loop context deployment-scenario . @reversal knowledge/risk/reversible.sle target ~deploy value 1200000 unit euro time 180-day context deployment-scenario . @q execution/query/what-if.sle target ~bias assumption disable-model alternative human-review context deployment-scenario .

APPENDIX F

Evaluation Matrix

Term	Explanation
Outer syntax	Compare canonical SLEngish with frame-oriented CNL, JSON/AST, and parenthesised forms. Measure parser size, generation validity, correction time, semantic diff quality, and token cost. Failure response: change the surface while retaining the IR.
Open verb libraries	Use held-out domains and typological constructions. Measure coverage, category confusion, schema burden, library fragmentation, and migration cost. Failure response: merge or split verbs, move distinctions to parameters, or narrow standard libraries.
Parameter registry	Compile corpora of n-ary scientific and institutional claims. Measure awkward encodings, extension frequency, role agreement, and interoperability. Failure response: add namespaces roles, promote proven roles, or abandon premature universal roles.
Visible SSA	Compare explicit and compiler-generated IDs. Measure human load, provenance quality, cache precision, diff stability, and correction locality. Failure response: hide IDs in normal views while retaining canonical SSA internally.
Folder namespaces	Test refactoring, federation, and multi-team packages. Measure rename errors, discoverability, merge conflicts, and stable identity. Failure response: make folders editorial only and use manifest or content-addressed naming.
LLM compiler	Evaluate held-out prose, concepts, negation, scope, identity, evidence, and alternatives. Measure semantic fidelity, invented assumptions, source alignment, abstention, and correction time. Failure response: hybrid parser/encoder, narrower domains, or direct authoring.
RDF-stack comparison	Build equivalent end-to-end research packages. Measure artefact count, authoring and review time, interoperability, validation, query correctness, update impact, and false assurance. Failure response: use SLEngish only as source syntax or abandon runtime primacy.
Missing-grammar profiles	Compare ordinary English, annotated English, SLEngish, and graphical views on source, identity, agency, causality, and systems tasks. Measure target error reduction and cognitive load. Failure response: make markers optional, domain-specific, or remove them.
VSA/HDC	Hold corpus and exact kernel constant while varying retrieval. Measure recall, latency, memory, noise robustness, calibration, and provenance loss. Failure response: remove VSA/HDC from the core and use dense, lexical, or graph retrieval.
Multi-formalism lowering	Compare native and lowered execution across Datalog, RDF/OWL, ASP, SMT, temporal, and proof profiles. Measure semantic preservation, certificate quality, adapter complexity, and cross-engine inconsistency. Failure response: narrow the portfolio.
Incremental circuits	Compare incremental recomputation with full replay under edits, retractions, contexts, and recursion. Measure trace equivalence, invalidation precision, latency, and stale-cache errors. Failure response: restrict incremental execution to proven operator classes.
Executable science	Ask an independent team to reproduce and modify a pilot package. Measure answerable queries, rebuild correctness, proof utility, expert judgement, cost, and false assurance. Failure response: narrow the goal to semantic audit or discontinue the broader claim.
Controlled realisation	Compare templates, constrained LLM rendering, direct generation, and post-hoc checking. Measure modality preservation, omission, unsupported additions, readability, and source traceability. Failure response: use deterministic views for high-assurance output.
Governance	Test competing libraries, cultural provenance, jurisdiction, deprecation, malicious packages, and authorisation. Measure user comprehension, conflict visibility, security outcomes, and authority laundering. Failure response: prevent production deployment until governance is adequate.
Folder-path refactoring test	Move operator files without semantic change and verify that manifest aliases, content identities, migrations, and historical packages remain valid.
Backend neutrality test	Execute equivalent SLEngish fragments through at least two compatible backend families and compare results, receipts, loss reports, and correction effort.
English construction coverage	Measure translation fidelity for tense, aspect, modality, negation, reference, nominalisation, coordination, clause linkage, and information structure.

Term	Explanation
Typological operator review	Require linguistic specialist review for selected evidential, clusivity, reference, possession, space, and discourse operators and document where analytical mappings fail.

APPENDIX G

Critical Reading Map

The reading map relates the programme to linguistic relativity, typology, formal semantics, conceptual engineering, process ontology, collective agency, controlled languages, knowledge representation, formal execution, and reproducible research. It distinguishes what each line of work establishes from the constructive claims proposed here.

Work or research line	Contribution, limitation, and use in this programme
Scope of the reading map	The literature behind this book does not form one unified field. It spans linguistic anthropology, typology, psycholinguistics, neuroscience, philosophy of language, metaphysics, social ontology, translation studies, conceptual engineering, and the history of artificial languages. The following map is therefore organised by intellectual function rather than disciplinary ownership. It identifies what each work makes possible, what it does not establish, and how the present book departs from it.
John A. Lucy: turning Whorf into a research programme	Lucy's <i>Language Diversity and Thought</i> (1992) remains one of the best places to begin because it reconstructs linguistic relativity as a methodological problem rather than a slogan. The book explains why earlier discussion repeatedly oscillated between grand claims and weak evidence, then proposes a comparative strategy: establish a precise grammatical contrast, derive a cognitive contrast from it, and test speakers on tasks that do not merely repeat the wording of their language. His companion study, <i>Grammatical Categories and Cognition</i> , compares English and Yucatec Maya and shows how grammatical treatment of number and nominal reference may organise classification. Lucy contributes the discipline this project needs. A proposed cognitive marker is not justified because it sounds philosophically illuminating. One must specify the contrast, the task, the mechanism, and the alternative explanation. His limitation is historical rather than conceptual: the programme predates much later work in neuroscience, computational modelling, and large-scale typological databases. This book adopts his experimental caution while extending the target from inherited language differences to deliberately designed distinctions.
Stephen C. Levinson: the strongest model of a domain-specific case	<i>Space in Language and Cognition</i> (2003) is a model of how cross-linguistic work can connect grammar, field observation, gesture, memory, and non-verbal reasoning. Levinson distinguishes relative, intrinsic, and absolute frames of reference and documents communities in which speakers routinely use geocentric systems rather than speaker-centred left and right. The importance of the work lies not in the picturesque fact that some people maintain cardinal orientation. It lies in the causal architecture: conversational success repeatedly rewards a spatial coding strategy, which then becomes available in memory, gesture, and navigation. For the present book, spatial framing provides a template for the idea of cognitive interfaces. It shows how a linguistic requirement can train a reusable attentional habit without constituting an impermeable prison. The limitation is equally instructive. Space is an unusually embodied, measurable domain with clear behavioural tasks. Identity, collective agency, and responsibility are socially contested and may not yield effects of comparable size or simplicity.
Alexandra Aikhenvald and the typology of evidence	Aikhenvald's <i>Evidentiality</i> (2004) is the indispensable survey of grammatical systems that mark information source. It shows that languages may require distinctions such as visual, non-visual sensory, inferred, assumed, or reported evidence, and that these systems vary in both structure and discourse use. The work also separates evidentiality from simple certainty: indirect evidence need not be weak evidence, and direct perception need not be reliable. This distinction is foundational for the proposed epistemic checksum. English commonly compresses source, confidence, trust, and truth into vague expressions such as <i>apparently</i> or <i>I think</i> . Aikhenvald shows that source mode is a legitimate grammatical dimension in its own right. Yet typological existence is not evidence of cognitive superiority. A community with obligatory evidentials is not automatically more sceptical, scientific, or truthful. The design opportunity is narrower: source marking may reduce particular provenance failures in settings where claims travel through long human–AI chains.

Work or research line	Contribution, limitation, and use in this programme
Caleb Everett: a broad empirical synthesis, with uneven terrain	Everett's <i>Linguistic Relativity: Evidence Across Languages and Cognitive Domains</i> (2013) surveys research on space, time, quantity, colour, objects and substances, gender, and other domains. Its strength is cumulative: linguistic relativity is not one all-or-nothing hypothesis but a family of local claims supported to different degrees. The book also makes field linguistics central. Without grammars of typologically distant languages, cognitive science repeatedly mistakes properties of familiar European languages for properties of the mind. The breadth is also the danger. Colour boundaries, exact number, spatial frames, grammatical gender, and temporal metaphors involve different mechanisms and evidential standards. They should not be stacked as if each provided an equal vote for a single doctrine. The present book follows Everett in treating linguistic diversity as cognitive evidence, but insists on an epistemic label for every example: attested grammatical contrast, demonstrated behavioural effect, philosophical extrapolation, or new design hypothesis.
Anna Wierzbicka: escaping English as an invisible metalanguage	<i>Imprisoned in English</i> (2013) argues that English has become not only a global medium but an unexamined conceptual standard in the humanities and social sciences. Terms such as mind, emotion, experience, evidence, self, and privacy carry histories that are not neutral windows onto universal reality. Wierzbicka's Natural Semantic Metalanguage programme attempts to analyse meanings through a small set of cross-linguistically grounded semantic primes. The book's most valuable contribution is diagnostic: a scholar can criticise linguistic imperialism while continuing to formulate every comparison in English categories. The proposed escape through universal semantic primes is more controversial. Semantic decomposition may itself import theoretical commitments, and universality is difficult to establish across underdescribed languages. This book accepts the diagnosis without depending on the full NSM programme. Its method is plural: retain several partial metalanguages, expose translation loss, and design terms that declare rather than conceal their assumptions.
Barbara Cassin and the philosophy of untranslatability	The English edition of <i>Dictionary of Untranslatables: A Philosophical Lexicon</i> (Cassin et al. 2014) follows nearly four hundred philosophical, literary, and political terms across languages and intellectual traditions. Its achievement is to show that translation is not the replacement of labels over identical concepts. Words such as <i>logos</i>, <i>Dasein</i>, <i>pravda</i>, <i>stato</i>, and <i>saudade</i> carry networks of contrasts, etymologies, controversies, and institutional uses. An "untranslatable" is not necessarily impossible to explain; it is a site where translation must remain active. This book differs from a dictionary of untranslatables in two ways. First, it is interested less in lexical jewels than in distinctions that grammar makes hard to omit. Second, it moves from historical diagnosis to conceptual design. Still, Cassin's work offers an ethical warning: a new term should not pretend to settle a debate merely by naming it. Good concepts preserve the tensions they clarify.
N. J. Enfield: language as coordination rather than transparent description	In <i>Language vs. Reality</i> (2022), Enfield presents language as an extraordinary tool for social coordination whose efficiency depends on selective compression. Words do not reproduce the world at full resolution. They supply public handles that allow people to align attention, negotiate commitment, tell stories, persuade, and act together. Misfit between word and world is therefore not a defect accidentally added to an otherwise perfect representational system; it is partly the price of usability. This argument prevents the present project from becoming naïvely perfectionist. A language that encoded every relevant dimension in every sentence would be unusable. Better cognitive interfaces must choose where friction is worth imposing. The design goal is not maximum explicitness but strategically priced precision: make a distinction cheap when omission has predictable costs, and leave it optional elsewhere.
Guy Deutscher and John McWhorter: the useful popular opposition	Deutscher's <i>Through the Language Glass</i> (2010) offers an accessible defence of moderate linguistic influence through colour, spatial orientation, and grammatical gender. McWhorter's <i>The Language Hoax</i> (2014) attacks romantic claims that grammar embodies a people's worldview or makes some cultures more perceptive, rational, or profound. Read together, the books define the rhetorical danger zone. Deutscher shows why the topic is too interesting to dismiss; McWhorter shows why attractive anecdotes can outrun evidence. Neither should serve as the primary scientific authority for a research programme. Their value is argumentative. Any public-facing book on language and thought must be able to survive McWhorter's objection: an evidential marker does not make its speakers better epistemologists. It must also preserve Deutscher's insight: habitual requirements can matter even when they do not alter the ultimate limits of thought. The present thesis is deliberately calibrated to occupy that middle ground.

Work or research line	Contribution, limitation, and use in this programme
Blasi and colleagues: English bias as a scientific problem	Blasi, Henrich, Adamou, Kemmerer, and Majid (2022) argue that cognitive science's reliance on English and a narrow range of languages hinders the discovery of both variation and universals. English is typologically unusual in several respects, yet it frequently supplies experimental materials, theoretical categories, and intuitions that are treated as transparent. The problem is not simply missing cultural diversity in samples. It is a feedback loop in which English determines the questions, tasks, and answer formats. This work supports one of the book's strongest claims: linguistic comparison is not decorative anthropology added after a theory has been built. It is a method for debugging theory. The constructive response is not to reverse the hierarchy and romanticise non-European languages. It is to make cross-linguistic robustness a requirement for claims about human cognition and to include speakers and scholars of the languages as intellectual partners rather than sources of data.
Fedorenko, Piantadosi, and Gibson: the communication-first correction	Fedorenko, Piantadosi, and Gibson (2024) synthesise evidence that language is primarily a system for communication and cultural transmission rather than the substrate of all complex thought. Neuropsychological dissociations, brain imaging, and the limited overlap between the language network and domain-general reasoning systems challenge the idea that thought is simply inner language. Their account is a necessary counterweight to any expansive Whorfianism. The present thesis survives this correction because it does not identify language with cognition. Interfaces shape use without constituting the machine. A notation can transform science even though scientists also think visually, spatially, emotionally, and mathematically. The communication-first view may even strengthen the social version of the argument: because language coordinates many minds, small representational defaults can influence institutions and cumulative knowledge even when individual thought remains multimodal.
Gennari and the turn toward mechanism and task	Gennari's 2025 review argues that language effects in higher-order cognition must be situated in task goals, context, and available mechanisms. Linguistic categories may be recruited when a task demands fine discrimination, memory, verbal rehearsal, or social interpretation and may disappear under other conditions. This approach replaces the search for a global verdict with an analysis of when language becomes functionally relevant. That is the proper model for testing SLEngish. Source tags should not be expected to improve every form of reasoning. They may help in delayed summarisation, contested claims, and multi-hop provenance while adding useless friction to direct perception. Identity operators may help in software lineage and institutional succession but not in ordinary object recognition. A design succeeds locally or not at all.
John, Macedo-Rouet, and Gros: current counterevidence matters	A preregistered 2026 study of 1,200 participants across English, French, German, and Mandarin tested whether language-specific temporal metaphors altered strategies for solving arithmetic word problems. The results favoured semantic content over linguistic metaphor: ordinal versus cardinal interpretations were driven by the problem structure rather than by the preferred spatialisation of time in the participants' languages. This finding is valuable precisely because it is negative for a broad relativity claim. It demonstrates why the field must prefer preregistration, multi-language sampling, and task analysis to memorable one-off effects. The present project treats such failures as design information. A linguistic marker will not change reasoning merely because a philosopher can describe a conceptual difference. The marker must be recruited by the task, learned sufficiently, and connected to a useful operation.
WALS and Grambank: a design library with epistemic limits	The World Atlas of Language Structures and Grambank transform typological knowledge into searchable comparative infrastructure. WALS offers curated chapters on evidentiality, clusivity, number, classifiers, tense, possession, and many other dimensions. Grambank aggregates more than four hundred thousand data points across more than 2,400 language varieties and 195 grammatical features. Together they make visible both the breadth of human design and the severe unevenness of documentation. Databases tempt users to treat a feature as a binary property detached from discourse, history, and analysis. "Has evidentiality" can conceal radically different systems; "lacks a feature" may mean that the grammar is incomplete or that analysts disagree. For conceptual invention, databases are hypothesis generators, not substitute grammars. Every promising comparison should return to specialist descriptions and, whenever possible, speakers.

Work or research line	Contribution, limitation, and use in this programme
Nicholson and Dupré: biology after substance ontology	Everything Flows: Towards a Processual Philosophy of Biology (Nicholson and Dupré 2018) argues that organisms and biological systems are better understood as stabilised processes than as substances that merely undergo change. Chapters by DiFrisco and Pradeu address criteria of identity, persistence, and genidentity in biological processes. The volume gives philosophical depth to the intuition behind the cloud, flame, and organism examples: continuity may consist in organised activity, causal dependence, and regulated replacement rather than sameness of matter. The relevance extends beyond biology. Software services, institutions, markets, and AI systems can also persist through controlled transformation. But process language should not become another universal solvent. Some legal and moral practices intentionally track bodies, persons, signatures, ownership, or lineage. The proposal is not to replace object ontology with process ontology; it is to mark which mode of identity a claim invokes.
The Problem of the Many: why a cloud is philosophically serious	Weatherston's contemporary presentation of the Problem of the Many begins with a cloud whose boundary admits many nearly equivalent collections of droplets. If each candidate is equally eligible, ordinary singular reference appears to select one object without a fact that clearly privileges it. The puzzle connects vagueness, composition, identity, and counting. For this book, the cloud is not a decorative metaphor. It exposes a general error: grammar offers crisp singular nouns where the world offers graded boundaries and scale-dependent partitions. A boundary-grade marker would not solve the metaphysics, but it could prevent downstream systems from silently converting operational boundaries into natural ones. This matters in ecology, medical diagnosis, social groups, datasets, and algorithmic categories.
List, Pettit, and Tuomela: groups as agents under explicit criteria	List and Pettit's Group Agency (2011) develops conditions under which organised groups can satisfy standards of agency through decision procedures that generate coherent beliefs and preferences. Tuomela's Social Ontology (2013) distinguishes acting in the I-mode from acting in the we-mode and examines collective commitment and group reasons. These works provide a vocabulary richer than either individualism or loose anthropomorphism. The central design lesson is architectural. A company does not become an agent merely because a sentence places it before a verb. Its agency depends on mechanisms for representation, decision, correction, memory, and authorised action. Conversely, reducing every organisational outcome to isolated individuals can miss structural commitments and emergent constraints. Collective markers should therefore encode the path from members to act, not merely label the size of the subject.
Infinity across mathematics, physics, and metaphysics	The 2025 revision of the Stanford Encyclopedia entry on infinity, by Easwaran, Hájek, Mancosu, and Oppy, treats infinity across mathematics, cosmology, physics, and theology. This breadth is essential because the word infinite conflates several questions: completed mathematical totalities, indefinitely extendable procedures, unbounded magnitudes, singularities in theories, spatial extent, divisibility, and divine attributes. The present book adds outfit as a deliberately engineered concept: a process is outfit when no final operational boundary is specified, although every realised stage remains finite and resource-bounded. The proposal does not compete with mathematical infinity. It isolates a practical form of open-endedness relevant to computation, scientific exploration, institutional evolution, and cultural production. Its value must be judged by whether it prevents invalid movement between "not yet bounded," "too large to exhaust," and "actually infinite."
Cappelen, Chalmers, and the conceptual-engineering turn	Cappelen's Fixing Language (2018), the volume Conceptual Engineering and Conceptual Ethics edited by Burgess, Cappelen, and Plunkett (2020), and Chalmers's account of conceptual engineering shift philosophical attention from analysing inherited concepts to assessing, designing, and implementing better ones. They distinguish diagnosis from construction: showing that a concept is vague, harmful, or theoretically defective does not yet produce a replacement people can use. This literature gives the book its constructive method. A new distinction requires a target function, semantic specification, implementation path, adoption strategy, and evaluation. It also raises an ethical question often missing from engineered languages: who gains authority when a category becomes mandatory? A provenance system can protect inquiry or become surveillance. A responsibility vector can reveal institutional causation or provide bureaucratic theatre. Conceptual engineering must include governance and the right to contest the ontology.

Work or research line	Contribution, limitation, and use in this programme
Maat and the history of philosophical languages	Maat's Philosophical Languages in the Seventeenth Century (2004) reconstructs the projects of Dalgarno, Wilkins, and Leibniz in their intellectual contexts. These schemes sought a systematic language whose symbols reflected an ordered catalogue of reality. Their ingenuity is inseparable from their fragility: once the taxonomy is disputed, the vocabulary becomes a monument to the designer's assumptions. SLEngish therefore rejects the dream of a complete language of things. It proposes small, revisable operators around recurrent reasoning failures. Its units need not encode the essence of a whale, state, or emotion. They can encode relations such as source, lineage, scale, and reversibility while leaving the underlying ontology open. The system should have versions, deprecations, competing profiles, and explicit unresolved categories.
Large language models as a new experimental site	Mahowald and colleagues (2024) distinguish formal linguistic competence from functional competence in large language models. A system can produce well-formed language while lacking reliable world models, grounded goals, causal understanding, or calibrated use of evidence. This distinction mirrors the central argument of the book: richer expression is not automatically better thought. At the same time, language models make conceptual engineering unusually testable. Researchers can insert operators into prompts, intermediate representations, or training data and measure whether models retain provenance, distinguish forks from copies, or assign responsibility more accurately. Models can also act as adaptive interfaces that request a marker only when ambiguity threatens a decision. The danger is institutional scale: an unexamined schema embedded in widely used AI can harden one contested ontology into infrastructure.
What this map leaves open	No existing work combines all four movements attempted here: global typological comparison, cautious cognitive interpretation, metaphysics of identity and collectivity, and experimentally governed conceptual design. That combination is the opportunity, but also the main source of risk. Each field has standards that the others can easily violate. Linguistic categories can be oversimplified, experiments can be overgeneralised, metaphysical distinctions can be presented as facts of nature, and designed concepts can become moralised without evidence. The proper form of synthesis is therefore not a grand unification. It is an interface among disciplines with explicit type checking. Every passage should permit the reader to ask: Is this a fact about a language, a replicated behavioural effect, a philosophical model, or an invention proposed for trial? The book's originality depends less on having unprecedented intuitions than on keeping those modes distinct while making them cooperate.
Controlled Natural Languages and executable semantics	Tobias Kuhn's survey provides the most useful map of the CNL design space and its trade-offs among precision, expressiveness, naturalness, and simplicity (2014). Attempto Controlled English demonstrates deterministic interpretation, discourse representation, querying, and theorem-proving integration. Common Logic Controlled English shows how English-like forms can express first-order semantics and role-based event structures. PENG demonstrates compilation into Answer Set Programming. SLEngish should be evaluated as a deliberately less natural but more circuit-oriented member of this lineage, not as if the problem had no history.
Semantic Web standards as baselines and interoperability targets	RDF, OWL, SHACL, SPARQL, and PROV-O collectively cover graph representation, ontology semantics, validation, querying, and provenance. Their maturity makes them the strongest engineering baseline. The SLEngish hypothesis concerns source uniformity and execution locality, not representational impossibility. A serious implementation must export and import useful subsets, document semantic loss, and avoid proprietary reinvention of standard concepts.
Datalog, ASP, constraints, and temporal reasoning	Datalog provides a clean basis for recursive rule evaluation; Answer Set Programming supports non-monotonic rules, defaults, and optimisation; SMT and CSP systems support arithmetic and finite constraints; event calculi and temporal logics support changing worlds. SLEngish should not conceal these differences behind one vague "reasoning engine." Its profiles are contracts to established semantic families.
Vector Symbolic Architectures and Hyperdimensional Computing	Plate's Holographic Reduced Representations and Kanerva's account of hyperdimensional computing established a family of distributed compositional representations. Later surveys by Kleyko and colleagues systematise architectures, operations, hardware, and applications. Their role here is deliberately bounded: they may provide approximate memory and retrieval over exact semantic graphs. Whether that hybrid yields a net advantage is an experimental question.

Work or research line	Contribution, limitation, and use in this programme
Reproducible workflows and research objects	FAIR workflow research, computational notebooks, provenance standards, and RO-Crate show how data and computations can become portable research objects. They also reveal the remaining separation between executable computation and prose-level claims. Executable science should be understood as an extension of this ecosystem, not a replacement for it.

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